



Future inorganic nutrient and plankton dynamics in a temperate lake

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Abstract:	Lakes are experiencing high temperatures and frequent phytoplankton blooms. These events threaten the health of lake ecosystems. Phytoplankton respond to lake warming in various ways, making it challenging to predict changes in community composition. Although understanding the effects of lake warming on phytoplankton communities is crucial for lake management, these effects are not clear and have been a matter of debate due to contrasting observations. Here we use a data-informed, trait-based model to disentangle and project, based on future warming scenarios, the effects of temperature on the dynamics of inorganic nutrients and plankton. Assuming identical thermal characteristics among phytoplankton size classes, we find that nutrient concentrations and phytoplankton mean cell size increase with rising lake temperature. This is driven by the interplay between reduced growth and nutrient uptake by all phytoplankton under sub-optimal temperatures, resulting in higher summer nutrients and intensified zooplankton grazing on smaller phytoplankton cells. The small phytoplankton typically occurring during winter and spring is replaced by large phytoplankton. In addition, peaks of phytoplankton biomass, typically occurring in early summer, emerge earlier, in early spring. The earlier occurrence of these peaks is caused by a shift in the optimal temperature for phytoplankton growth from spring to late winter. Although, as in all studies of this type, our results are based on various modelling approximations, they highlight the importance of combining observations and trait-based modelling for gaining insights into the future compositions of lake phytoplankton and establish a foundation for examining lake ecosystems in a warming world.



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Scientific Significance Statement Topic

Lake ecosystems are increasingly threatened by high temperatures and frequent phytoplankton blooms. Phytoplankton respond to higher temperatures in various ways, hence producing contrasting observational trends. Using a mathematical model in combination with monitoring data, we disentangle and project, based on different warming scenarios, the effects of temperature on the dynamics of inorganic nutrients and plankton in a Swiss lake, Greifensee. Assuming identical thermal characteristics among phytoplankton size classes, we find that nutrient concentration and phytoplankton mean size both increase with increasing temperature. As the lake warms up, peaks of phytoplankton biomass emerge earlier than under current temperature. Although our results are based on modelling approximations, they highlight the importance of combining observations and trait-based modelling for improving our capacity to design scientifically-sound lake management strategies.

Scientific Significance Statement Outlet

We adopt a mechanistic model, informed by an extensive amount of data, to investigate the effects of temperature on the dynamics of nutrients and plankton at Greifensee. Our study highlights the importance of combining observations and trait-based modelling to advance our understanding of nutrient and plankton dynamics in lakes. This study also provides an open-source modelling tool that allows for examining ecosystem functions in lakes and for experimenting under climate change conditions.

Dear Prof. David Hambright and Dr. Yong Liu,

Please find 'enclosed' our revised manuscript entitled **Future inorganic nutrient and plankton dynamics in a temperate lake**, authored by To, Chakraborty, Acevedo-Trejos, Pomati, and Merico, for your consideration for publication in *Limnology and Oceanography*.

Thank you for the comments and suggestions on our manuscript, which have helped us to improve our work in terms of scientific focus, presentation of methods, and discussion of results. Thus, we have made a number of revisions in response to reviewers' comments and your own editorial concerns. Briefly, we have significantly reduced the length of the manuscript (by four pages), minimized the use of acronyms and abbreviations (i.e., mathematical variables and model parameters), improved the description of the model by highlighting the novel features, and overhauled the Discussion to bring into focus context and relevance of our results.

Regarding novelty, the study presented here constitutes a significant extension of our previous work (To et al. 2024). First, in contrast to our previous work, which was a systematic analysis of multiple environmental scenarios relative to a generic and unspecified lake, we focus here on a real lake, Greifensee (Switzerland), by considering an extensive amount of high-resolution monitoring data collected at Greifensee over several years. Second, we use the calibrated data-informed model to produce ecological projections based on three IPCC-specific climate change scenarios to investigate the impacts of global warming on the steady-state seasonal dynamics of nutrient and plankton at Greifensee. The model we present here is not only completely re-parameterized (with Greifensee-specific data) with respect to the model used in the previous study, it is also substantially extended, for example, in relation to the addition of thermal dependences of major physiological processes (i.e., phytoplankton cellular growth and zooplankton grazing).

We hope that our revisions and detailed responses to reviewers are able to convey more clearly the novelty and relevance of our study and that it can be now considered for publication in *Limnology and Oceanography*.

Sincerely,
Sze-Wing To and co-authors

We are very grateful to all reviewers for their constructive and insightful comments, which have helped us to significantly improve our manuscript both scientifically and in terms of clarity of meaning. Below are our point-by-point responses (black text) to the comments raised (*grey italics*). Revisions to our manuscript are shown here in **smaller blue** text. Please note that the line numbers cited in our replies refer to the revised manuscript with changes tracked (file: 'LO-24-0506_track.pdf'). In that file, added text is shown in **blue** and deleted text is shown as **red strikethrough**.

Response to Reviewer 1:

1. Introduction

- *Overuse of general climate change impacts, I suggest it should have a paragraph more focus on Greifensee-specific challenges, which would strengthen the relevance.*

Thank you for the suggestion. We removed text on the general impacts of climate change from the first and second paragraph and added a few sentences to specify the challenges relevant for Greifensee.

Lines 71-73

'In a number of temperate lakes, contrary to expectations (e.g., Pomati et al. 2020), a clear multi-decadal warming trend and a reduction in nutrient levels were linked to the dominance of large phytoplankton cells.'

Lines 138-144

'Greifensee, a medium-size eutrophic lake near Zurich (Switzerland), experienced multi-decadal warming and re-oligotrophic trends (Pomati et al. 2020; Merz et al. 2023). The composition of the phytoplankton community in the lake was found to be sensitive to long-term changes in thermal and nutrient conditions (Bürgi et al. 2003; Monchamp et al. 2017), making Greifensee a natural laboratory for studying the impacts of changing temperature on abundance and community composition of lake plankton.'

2. Methods

- *The zero-dimensional configuration of the model (slab approximation) simplifies lake dynamics and limits its ability to capture spatial heterogeneity. Vertical stratification (different temperature change at different levels) and lateral transport are critical processes that are overlooked.*

Thank you for pointing out this important aspect. We agree that vertical stratification and lateral transport could be important for some large, deep, and wind-exposed lakes (Imboden and Wüest 1995). However, Greifensee is generally fully mixed in winter and, due to rare high wind events, is well stratified into two stable and homogeneously mixed layers in the spring to summer periods (Mieleiter et al., 2008; Huntscha et al. 2018). Thus, vertical variability and lateral transport are not significant factors at Greifensee. Many studies (e.g., Livingstone 2003; Rempfer et al. 2010; Butcher et al. 2015; Woolway et al. 2021) also show that, in a warming world, lake mixing will be reduced, leading to stagnant water systems. We believe that these aspects make our simplified physical scheme an appropriate first-order approximation approach. We added a few sentences in the Discussion to clarify this aspect.

(Lines 797-805)

'Our results are based on a relatively idealized lake physics, which does not specify vertical stratification and lateral transport processes. We consider that this choice is appropriate for Greifensee, a relatively shallow lake that is typically fully mixed in winter and stratified into two distinct and homogeneously mixed layers in summer (Mieleitner et al. 2008; Huntscha et al. 2018). Many studies (e.g., Livingstone 2003; Rempfer et al. 2010; Butcher et al. 2015; Woolway et al. 2021) also show that, in a warming world, lakes will experience weaker mixing, with higher temperatures potentially structuring the water column into two stable layers, thus, making our simplified physical scheme an appropriate first-order approximation approach.'

- *The exact mechanism for calibrating the Q_{10} factor for zooplankton grazing is not explained clearly.*

Thank you for pointing this out. The Q_{10} factor for zooplankton grazing was calibrated using an observed range of Q_{10} values for maximum ingestion of Flagellates, Ciliates, and Copepods (functional groups abundant at Greifensee) reported by Hansen et al. (1997). We added a sentence detailing this aspect (Lines 297-299) and revised Fig. S5 by including panels E and K showing the results of the calibration.

'The Q_{10} coefficient was determined from a range of values observed by Hansen et al. (1997) for the maximum ingestion rate of Flagellates, Ciliates, and Copepods (functional groups abundant at Greifensee).'

- *Seasonal averaging of input data (e.g., light, temperature) may oversimplify short-term dynamics.*

The time step of our model is actually one day. To calibrate parameters, we compared the model results against three consecutive years of nutrient and biological data, using daily averages over a one-year period. Only for the mixed layer depth we obtained monthly data (from Pomati et al. 2020), which we interpolated to get daily data and match the time step of the model (Lines 337-338). Thus, our model captures the short-term daily dynamics. We added a few sentences in the Methods to specify this aspect.

(Lines 410-414)

'The time step of our model is one day. To calibrate parameters, we compared the model results against three consecutive years of nutrient and biological data, using daily averages over a one-year period (details can be found in Supporting Information S3). The model results, therefore, reflect the daily dynamics of the lake for a "typical" (i.e., average) year under current temperature conditions.'

3. Results

- *The section lacks discussion on interannual variability by the nature of the model configuration; results are presented as steady-state projections without addressing how transient responses might differ.*

As explained in the response above, we focused on the average state of the lake over one-year period or, in other words, on a current (and future) "typical" year. For this, we input all the environmental data to the model as daily averages and reported steady-state results over

a one-year period. We added explanatory text in the Methods (Lines 422-423). In the Discussion, we specified that the topic of inter-annual variability under global warming is outside the scope of our study but that it constitute an interesting line of research for the future (Lines 831-833).

(Lines 422-423)

'Inter-annual variability was not explored as it was beyond the scope of our study.'

(Lines 846-847)

'Studying inter-annual variabilities would be another obvious next step when looking at future projections of lake ecosystems.'

- On line 481 "*A higher T_{opt} or σ_T does not induce as many variations to the model variables; even under RCP 8.5, the greatest relative change is less than 50%.*" why is that? In our model, higher thermal trait values would produce a more heat-tolerant phytoplankton community (see dashed line in the revised Fig. 4), whereas lower thermal trait values produce a less heat-tolerant community (see dotted line in the revised Fig. 4). Under the considered warming scenarios, the phytoplankton community with a higher heat tolerance is less sensitive to higher temperatures, this reduced sensitivity is reflected also in other model variables. We added explanations and revised Fig. 4 to clarify these aspects.

(Lines 591-592)

'... lower values of thermal traits (i.e., lower tolerance to high temperatures, Fig. 4).'

(Lines 597-599)

'That is because higher thermal trait values produce a more heat-tolerant phytoplankton community (Fig. 4).'

- Minor error: line 395-399 should refer to fig. 5 rather than fig. 4
Revised (Lines 506-508).

4. Discussion

- The Discussion is overly reliant on interpreting the model outputs (as the model is overly simplified and results are fairly easy to interpret in this case) without critically addressing the limitations of the assumptions (e.g., simplified stratification dynamics).

Thank you for these valuable suggestions, we revised the Discussion section. Additionally, we included a sensitivity run (Supporting Information S5) that shows the effects of alterations in lake mixing (Fig. S4A) under the most severe climate scenarios (RCP 8.5) on model results. We found that these alterations reduce nutrient concentrations in summer and produce an earlier zooplankton peak (Fig. S4B, S4D). However, many studies (e.g., Livingstone 2003; Rempfer et al. 2010; Butcher et al. 2015; Woolway et al. 2021) show that in a warming world mixing will be reduced, with temperature potentially structuring the water column into two stable and homogeneously mixed layers. This makes our simplified physical scheme an appropriate first-order approximation approach. We added and revised sentences in different parts of the Discussion.

(Lines 660-665)

'This result is based on the assumption that different phytoplankton size classes have identical thermal characteristics (Fig. 2). In nature, species sensitivities to temperature may vary and different phytoplankton can exhibit different growth and nutrient uptake abilities along thermal gradients (Thomas et al. 2015; Bestion et al. 2018). This is an aspect that could generate feedbacks on the nutrient dynamics that are not captured by our model.'

(Lines 668-686)

'The stronger and longer stratification events in a warmer lake (Butcher et al. 2015; Woolway et al. 2021) can further disrupt the nutrient exchanges between epilimnion and hypolimnion and, thus, influence nutrient dynamics (Supporting Information S5, Fig. S4B). Taken together, these results indicate that accounting for the impacts of climate on nutrient regimes would require detailed considerations in relation to physical and biological feedbacks. Our results should be considered as a first order approximation with respect to the impacts that global warming can have on nutrient dynamics and plankton phenology.'

(Lines 797-805)

'Our results are based on a relatively idealized lake physics, which does not specify vertical stratification and lateral transport processes. We consider that this choice is appropriate for Greifensee, a relatively shallow lake that is typically fully mixed in winter and stratified into two distinct and homogeneously mixed layers in summer (Mieleitner et al. 2008; Huntscha et al. 2018). Many studies (e.g., Livingstone 2003; Rempfer et al. 2010; Butcher et al. 2015; Woolway et al. 2021) also show that, in a warming world, lakes will experience weaker mixing, with higher temperatures potentially structuring the water column into two stable layers, thus, making our simplified physical scheme an appropriate first-order approximation approach.'

- *Missing a more "detailed" comparison with similar studies on temperate lakes, which would help contextualize the findings.*

Thank you for pointing this out. We revised the Discussion to better contextualise our results into the current understanding of how lakes may change in a high-temperature world.

(Lines 636-639)

'These patterns are consistent with the general phytoplankton dynamics expected to occur when interactions between inorganic nutrient limitation and grazing pressure are primary driving factors of lake ecosystems (Vanni and Temte 1990; Sommer et al. 2012).'

(Lines 691-706)

'For example, a dynamic diatom-daphnia model showed that, under lake warming, a high phosphorus supply causes earlier diatom spring blooms (Huber et al. 2008). A depth-resolved hydrodynamic model coupled to a simple phytoplankton growth model showed that earlier spring blooms can be caused by changes in lake stratification patterns (Peeters et al. 2007). In our model, the earlier peaks of large phytoplankton are produced by a warmer winter that advances the timing of optimal temperature for the growth of all phytoplankton size classes but intensifies the grazing of zooplankton on small phytoplankton size classes. Thermal characteristics, like optimal temperature for growth, are important factors for driving seasonal succession in culture experiments comprising marine diatoms (Anderson and Rynearson 2020).'

5. Limitations and Future Directions

- *Recommendations for future work are too general. Specific suggestions, such as integrating vertical stratification models or incorporating fish dynamics, would make this section more impactful.*

Good point. We revised a large part of this section to make our suggestions for future work more specific (Lines 832-847).

'Future modelling studies could examine the impacts of temperature on planktonic interactions more realistically by considering size-specific thermal responses (e.g., Chen 2022; Karakuş et al. 2022) or other size-related characteristics, like a more diverse zooplankton community in relation to body size. One could also consider additional temperature dependences on, for example, remineralization processes (Boscolo-Galazzo et al. 2018). Assuming that metabolic and biochemical rates would be enhanced by increasing temperatures, these dependences would produce more accurate dynamics of inorganic nutrients and detritus. Spatially explicit hydrodynamic modelling studies could investigate the extent to which our first-order slab physics approximation is appropriate to capture the future ecological regimes of lakes like Greifensee (e.g., Huntscha et al. 2018). Studying inter-annual variabilities would be another obvious next step when looking at future projections of lake ecosystem.'

- *The role of uncertainty in input parameters are not discussed sufficiently.*

We added new sensitivity tests (Supporting Information S4 and Fig. S3) to evaluate the effects of uncertainties in input parameters. We supplemented a sentence in the Method section to briefly describe this analysis (Lines 414-417) and extended our Discussion to acknowledge the uncertainty involved (Lines 805-808).

(Lines 414-417)

'Additionally, we undertook a sensitivity analysis to evaluate the effects of parameter uncertainty on model results. We addressed those parameters ($n=14$) that could not be directly constrained by the data specifically available for Greifensee and those that were not calibrated (Supporting Information S4).'

(Lines 805-808)

'Also, our results are based on a specific set of model parameters that are informed by lake data. This reduces the level of uncertainty in these parameters, especially those that regulate phytoplankton nutrient uptake and zooplankton grazing (Fig. S3), which are the major determinants of our results.'

6. Conclusion

- *Overgeneralizes the applicability of the model without considering its site-specific calibration for Greifensee.*

We revised the relevant text to highlight the range of validity of our model results (Lines 855-859).

'Acknowledging a number of simplifications and limitations, this study represents a first-order approximation of the ecological responses that Greifensee may exhibit in a warmer world. Most of the model parameters and all environmental forcing are site-specific, thus the validity of our results is limited to Greifensee or to lake systems with characteristics similar to Greifensee.'

Response to Reviewer 2:

Overview and general recommendation:

The manuscript presents a simple process based nutrient, phytoplankton, zooplankton, detritus (NPZD) model, which is used to predict the impact of global warming on the dynamics of nutrients, phytoplankton, and zooplankton. The model is described in detail, implemented in an open source software (python) and all code and data are made available in a Zenodo repository. The topic is of scientific interest and fitting the journals scope. While I like the approach and content I think the way the material is presented could be improved a bit and the discussion would need more self critical evaluation of the results. I think the way the model equations are presented at the moment is hard to read and would profit from some improvement. My main criticism is regarding the discussion and conclusions the authors draw from the results. A (numerical) model is always a simplified representation of the system of interest and I think that even though the used model in this study is very simple that is not a problem. But, I think it is necessary to make the reader more aware of the limitations of this model and the assumptions in the climate scenarios when presenting the results and drawing conclusions from them. Detailed comments are given below.

Thank you for these constructive comments. Concerning the Methods section, we moved most of the mathematical details (which are now presented in a revised form to improve clarity) to the Supporting Information (S1), also in response to editorial requirements. We revised some key aspects of the Methods, especially the ‘*Ecological projections based on climate change scenarios*’, to highlight the assumptions made to produce the future projections. We also revised the Discussion, especially the ‘*Limitations and future directions*’, to highlight the limitations of the model more explicitly and more comprehensively. Please find below our detailed responses to your specific points.

Major comments

- *Model description: I am a bit unhappy with the way the model is presented in the methods. It was quite hard to grasp and there are inconsistencies. On possibility that I personally really like is to use a matrix/vector notation (sometimes also called Petersen matrix wikipedia) to present the system of ODEs. See e.g. Tables 1 – 4 in Omlin et al. (2001) describing the state variables, stoichiometry matrix, and process rates, or Kneis et al. (2017) section 2.1. But I do not think that this is necessarily needed. Just the way it is presented currently was very hard to follow for me.*

To improve clarity, we revised the model description and presented it in a step-by-step manner. Also note that, in response to editorial requirements, the mathematical details common to a previously published model (To et al. 2024) are now in Supporting Information S1. In the main manuscript, we left the description of the novel features of the model (Lines 170-172), they include the temperature dependence of phytoplankton growth and zooplankton grazing, the procedure of model calibration against the data collected at Greifensee, and the procedure followed to produce the ecological projections under future climate change scenarios. We also revised the explanations of the data conversions (Supporting Information S2 and S3) and provided tables, similarly to Omlin et al. (2001), to

document the formulations (Table S1) and the parameters (Table S2) used for these conversions.

In addition there were some inconsistencies and errors in the equations:

- *Eq 5: should read: $(\mu i - \phi i - \lambda i) \cdot A i$ otherwise the growth rate is multiplied with $A i$ squared*
Thank you, we revised this (Supporting Information S1, Eq. S5).

- *Eq 8: $N U(N)$ reads like $N \cdot U(N)$? but it should be $U(N)$ like in Eq 11, right?*
Yes, we revised this (Supporting Information S1, Eq. S10).

- *In lines 155 you write: "A deepening MLD reduces the concentration of the nonmotile variables." this would only be true if $N_0 < N$, but if $N_0 > N$ a deepening will increase the concentration (Eq 4)*

Thank you, you are correct. We did not describe this accurately. We revised our description but due to editorial concerns on the length of the manuscript, this part is moved to Supporting Information S1.

(Supporting Information S1, Lines 32-34)

'A deepening MLD reduces the concentrations of non-motile variables, unless N_0 is larger than N , that is, when the nutrient input from the hypolimnion is greater than the nutrient concentration in the epilimnion.'

- *In Eq. 17 you use $V P$ for the mean biovolume, while in the text and Fig 6. you use $V A$*
Revised (Eq. 3).
- *In Eq. 13 and Eq. 14 it is confusing to use the letter I for ingestion rate as it is also used for irradiation I_z , I would suggest to use a different letter*
Good point. Ingestion rate is now indicated by ψ (Eqs. S12 and S13).

- *I think some of the assumptions and limitations of the model and in the climate scenarios need to be communicated clearer, especially in the discussion and conclusions drawn from the results.*

- *The impact of climate warming on stratification duration and the MLD are ignored.*

Yes, our projections do not include the dynamic impact of warming on the duration of lake stratification. To document the relevance of this omission, we performed additional sensitivity runs (Supporting Information S5) under the most severe warming scenario (RCP 8.5). We altered the duration of mixing (1) by extending the summer stratification (through manipulation of the mixed layer depth) and (2) by reducing the diffusive mixing between epilimnion and hypolimnion. We found that these alterations can affect our projections by reducing nutrient concentrations in summer and by producing an earlier zooplankton peak (Fig. S4B, S4D). We described these sensitivity

tests in the Methods (Line 442-445) and presented the impacts of these alterations on our model results in the Discussion (Lines 666-672).

(Lines 442-445)

'However, our future projections did not include physical feedback that could be induced by increasing temperatures, like changes in lake stratification (but see Supporting Information S5 for a sensitivity run exploring the impact of this omission on our results under scenario RCP 8.5).'

(Lines 668-686)

'The stronger and longer stratification events in a warmer lake (Butcher et al. 2015; Woolway et al. 2021) can further disrupt the nutrient exchanges between epilimnion and hypolimnion and, thus, influence nutrient dynamics (Supporting Information S5, Fig. S4B). Taken together, these results indicate that accounting for the impacts of climate on nutrient regimes would require detailed considerations in relation to physical and biological feedbacks. Our results should be considered as a first order approximation with respect to the impacts that global warming can have on nutrient dynamics and plankton phenology.'

- *Your model has no diffusive exchange between epilimnion and hypolimnion, but you mention this exchange in the discussion (L543–544). It would be good to make it more clear to the reader what you expect according to other studies, if this is implemented in your model, and if not how this limitation might impact the model outcome and conclusions (applies not only to this comment)*

Thank you for the feedback. The simplified stratification dynamics in our model captures vertical mixing through changes in (1) the mixed layer depth and (2) the cross-thermocline diffusive mixing parameter, ω (Eq. S2). Altering these two factors approximates the effects of prolonged lake stratification and reduced mixing. Such changes in lake stratification were suggested to drive earlier phytoplankton blooms in warmer lakes (Peeters et al. 2007; Woolway et al. 2021). To explore the effects of our approximation, we performed additional sensitivity runs (please see Supporting Information S5 and response above).

- *Your model has no loss term for sedimentation for the algae, which would probably increase differences based on size. In the discussion this is not very clear e.g. in lines 590*

Thank you for pointing this out. We added text to clarify that sinking processes were omitted because of the relatively shallow average depth and mixing characteristics of Greifensee.

(Lines 742-745).

'Sinking was not included in our model because Greifensee is characterized by a relatively shallow average depth (18 meters), but this loss process can contribute to varying the size composition of phytoplankton communities (Zohary et al. 2017).'

- *The model has no temperature dependency of the remineralization, which could be especially interesting in the climate scenarios*

Good point, we highlighted such aspect for future investigations in the Discussion (Lines 840-843).

'One could also consider additional temperature dependences on, for example, remineralization processes (Boscolo-Galazzo et al. 2018). Assuming that metabolic and biochemical rates would be enhanced by increasing temperature, these dependences would produce more accurate dynamics of inorganic nutrients and detritus.'

- *All of your algae size classes have the same optimum temperature, this is also the main reason why in the warming climate scenario you get low algae concentrations and high nutrient concentrations at the end of summer. In a real system I would expect that an algae with higher optimum growth temperature would fill this niche e.g. some cyanobacteria. So I do not agree with your conclusions that warming would lead to increased nutrients (L540) but that this is rather a consequence of your model assumptions. In fact, you even mention this in L 546 and have two paragraphs (L595–616 and L641–655) discussing this. So I am not so happy with you mentioning the increase in nutrients under warming temperatures both in the abstract and conclusions without a critical remark.*

Thank you for pointing this out, we revised the relevant text as follows:

(Line 31)

'Assuming identical thermal characteristics among phytoplankton size classes, we find that..'

(Lines 302-303)

'This assumption implies that different phytoplankton and zooplankton size classes have identical thermal responses in relation to, respectively, growth and grazing.'

(Lines 660-665)

'This result is based on the assumption that different phytoplankton size classes have identical thermal characteristics (Fig. 2). In nature, species sensitivities to temperature may vary and different phytoplankton can exhibit different growth and nutrient uptake abilities along thermal gradients (Thomas et al. 2015; Bestion et al. 2018). This is an aspect that could generate feedbacks on the nutrient dynamics that are not captured by our model'.

(Lines 855-857)

'Acknowledging a number of simplifications and limitations, this study represents a first-order approximation of the ecological responses that Greifensee may exhibit in a warmer world.'

- *I had a look at the simulation results in the ncd files in the Zenodo repository and noticed that 5 out of the 10 Phytoplankton groups do not grow at all (both in the standard run and in the projections). From the other five groups only the smallest and largest group have values that are reasonably large, the other groups have values smaller than 1e-50. I think this should be mentioned in the manuscript and especially the conclusions regarding the average bio volume*

and size distribution of the pytoplankton should be critically reevaluated, e.g. lines 336 – 338 “The model parameterization obtained from the standard model run produces model results that reflect current average annual conditions (in terms of nutrient concentrations, plankton biomasses and size structures) at Greifensee” or L515–518

Thank you for performing this check. Such numerical feature, which is very common to plankton ecosystem models, as we showed in our recent work (To et al. 2025), is caused by competitive exclusion (Hardin 1960). Sustaining size diversity for a relatively long time in such models requires the implementation of ad-hoc processes or mechanisms (Flynn, 2005; Anderson, 2005), like trait or niche differentiations (Chen 2022), which were not considered in our study. Despite this limitation, our model constitutes a meaningful approximation as it is capable of capturing, in both magnitude and dynamics, average nutrient, phytoplankton, and zooplankton concentrations as well as mean phytoplankton biovolume (Fig. 5). However, the reviewer is right, we should have mentioned this aspect in the manuscript. We revised accordingly.

(Lines 385-388)

‘We analyzed the phytoplankton size composition in macroecological terms (e.g., Acevedo-Trejos et al. 2018; To et al. 2024), by considering a community-weighted mean cell biovolume, $\overline{V^A}$.’

(Lines 430-433)

‘The model parameterization obtained from the community-integrated outputs of the standard model run produces model results that reflect current average annual conditions (in terms of nutrient concentrations, total phytoplankton and zooplankton biomasses and aggregated phytoplankton size structures) at Greifensee.’

(Lines 810-818)

‘The absence of size-specific temperature dependencies in our model prevents niche differentiation with respect to thermal traits (Chen 2022). This limits the model ability to maintain all 10 phytoplankton size classes over time due to competitive exclusion (Hardin 1960), an aspect common to plankton models (e.g., To et al. 2025) but that does not have an equivalent in nature (the “paradox of the plankton”, Hutchinson 1961). Thus, some of the size classes simulated by our model disappear during the course of the simulations. This, however, does not preclude the model from capturing, both in magnitude and dynamics, nutrient and plankton concentrations, as well as mean phytoplankton biovolume.’

Minor comments

- *The color coding for some Figures is not suited for readers with colour vision deficiencies or when printed in black and white (e.g. Fig 6 + 7). You can check figures using e.g. this webpage: <https://www.color-blindness.com/coblis-color-blindness-simulator/>*

The colors in Figures 4 and 7 are now replaced with line patterns. We modified the colors in Figure 6, now they appear visible to the color blindness simulator.

- *Throughout your manuscript you use “data driven” to describe your model (e.g. in the abstract “Here we use a data-driven, trait-based model to ...”). While I understand what you mean by*

this, the term “data driven” is commonly used to describe statistical or machine learning models and might be confusing to the reader as I would classify your model as process-based or mechanistic.

Thank you for pointing this out. The term “data-driven” is now replaced by ‘data-informed’ through the manuscript.

- *L120 “We aim to understand the direct effect of increased LWST on ...” according to L69 the impact of temperature on zooplankton grazing is a indirect effect, so maybe remove “the direct” in L120*
Done (Line 154).
- *fig 5: maybe add that the data shown is for the epilimnion/MLD*
Done (Line 518-519).
- *Fig S7: I would use “growth rate” instead of growth flux*
Revised.
- *L284 “constrained” this usually means lower and upper boundaries, how is that done for algae?*
The word “constrained” is changed to “calibrated” (Line 348).
- *L22 Sentence is very general, what about lakes at high altitudes or in polar regions and what about oligotrophic lakes*
The sentence is revised as ‘Most lakes on our planet are...’ (Line 52)
- *L139 “The two layers are separated by a seasonally varying mixed layer depth (MLD)” Sounds like there is a third layer but the MLD is the depth of the epilimnion, right?*
Yes, correct. To make this clear, we revised the sentence and added it also to the caption of Figure 1 (Lines 207-208).

‘The depth of the epilimnion reflects the Mixed Layer Depth (MLD)’.

- *L179 & L192 I would not say N_0 is calibrated from observations or did you iteratively run the model and adapt N_0 to improve the outcome? You just set N_0 to the average value from observations, right?*
Yes, the reviewer is correct. We determined the value of N_0 based on average winter observations, when the entire water column is well mixed and nutrient concentrations are homogeneously distributed throughout it. For clarity, we revised the sentences:

(Lines 219-221 and Supporting Information S1, Lines 52-53)

'We determined the value of N_0 based on average winter observations, when the entire water column is well mixed and nutrient concentrations are homogeneously distributed throughout it.'

- L313 "phosphorus-based biomass concentrations" I would suggest to use something like "phosphorus equivalents" instead
 "phosphorus-based biomass concentrations" is changed to "phosphorus-equivalent biomass concentrations" (Line 362)
- L555 "simulation models"
 Thanks for spotting the typo. The sentence is now removed due to the editorial request of shortening the length of the manuscript.
- L601 "A global circulation model resolving .." Is this really a global circulation model that additionally simulates biogeochemical processes or is it a biogeochemical model driven with GCM data?
 The study (Anderson et al. 2024) is a biogeochemical model driven with GCM data. We revised the sentence (Lines 756-759) as:

'An ecosystem model resolving biogeochemical processes, for example, showed that alterations of the Q_{10} factor can markedly change the community composition of phytoplankton under the most severe climate change scenario (Anderson et al. 2024).'

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For Review Only

1 *Research article*

2 **Future inorganic nutrient and plankton dynamics in a temperate lake**

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22 Abstract

23 Lakes are experiencing high temperatures and frequent phytoplankton blooms. These events
24 threaten the health of lake ecosystems. Phytoplankton respond to lake warming in various
25 ways, making it challenging to predict changes in community composition. Although
26 understanding the effects of lake warming on phytoplankton communities is crucial for lake
27 management, these effects are not clear and have been a matter of debate due to contrasting
28 observations. Here we use a data-informed, trait-based model to disentangle and project, based
29 on future warming scenarios, the effects of temperature on the dynamics of inorganic nutrients
30 and plankton. Assuming identical thermal characteristics among phytoplankton size classes,
31 we find that nutrient concentrations and phytoplankton mean cell size increase with rising lake
32 temperature. This is driven by the interplay between reduced growth and nutrient uptake by all
33 phytoplankton under sub-optimal temperatures, resulting in higher summer nutrients and
34 intensified zooplankton grazing on smaller phytoplankton cells. The small phytoplankton
35 typically occurring during winter and spring is replaced by large phytoplankton. In addition,
36 peaks of phytoplankton biomass, typically occurring in early summer, emerge earlier, in early
37 spring. The earlier occurrence of these peaks is caused by a shift in the optimal temperature for
38 phytoplankton growth from spring to late winter. Although, as in all studies of this type, our
39 results are based on various modelling approximations, they highlight the importance of
40 combining observations and trait-based modelling for gaining insights into the future
41 compositions of lake phytoplankton and establish a foundation for examining lake ecosystems
42 in a warming world.

43

44

45 Introduction

46 Most lakes on our planet are under threat by climate change and, specifically, by increasing
47 temperatures (O'Reilly et al. 2015; Woolway et al. 2022). The increase in lake water surface
48 temperature, hereafter water temperature, couples with changes in nutrient conditions in the
49 lake to alter the interactions and compositions of lake communities. This may dampen trophic
50 transfer efficiencies, with serious impacts on the stability and productivity of lake food webs
51 (Strecker et al. 2004; Barneche et al. 2021). Increasing occurrences of anomalous algal blooms
52 (dominated, for example, by cyanobacteria) are observed in eutrophic lakes worldwide,
53 especially in connection to rising temperatures (Paerl and Huisman 2008; Ho et al. 2019).
54 Nevertheless, the mechanistic reasons leading to these blooms are still not clear. In a number
55 of temperate lakes, contrary to expectations (e.g., Pomati et al. 2020), a clear multi-decadal
56 warming trend and a reduction in nutrient levels were linked to the dominance of large
57 phytoplankton cells. Hence, a better understanding of the thermal responses of plankton
58 communities is key to managing lake ecosystems in a warming world.

59 The effect of temperature on lake phytoplankton communities can be of two types: (1)
60 direct effects can alter metabolic processes of cells and (2) indirect effects can reshape the
61 interactions of biotic and abiotic components (Zohary et al. 2021). Elevated temperatures can
62 indirectly affect phytoplankton nutrient uptake by varying autotrophic metabolic rates and
63 nutrient recycling by varying remineralization rates (Vanni 2002; Dory et al. 2024).
64 Additionally, temperature affects the composition of phytoplankton communities indirectly by
65 enhancing zooplankton grazing, which can favour grazer-resistant phytoplankton species
66 (Sommer and Lewandowska 2011; Lüring 2021). The synergistic direct and indirect effects of

temperature can have complex repercussions on phytoplankton communities. These repercussions are frequently observed but they remain difficult to disentangle and predict.

Warming-induced changes in the seasonal succession of phytoplankton have been modelled and observed in different European lakes over the last two decades (reviewed in Winder and Sommer 2012; Vadadi-Fülöp and Hufnagel 2014). Phenological shifts in planktonic communities impact key lake ecosystem functions such as productivity, diversity, and trophic transfer efficiency (Strecker et al. 2004; Woolway et al. 2022). Minimal predator-prey models proposed that earlier onsets can be caused by warming-induced shifts or decoupling of plankton dynamics (Scheffer et al. 2001; De Senerpont Domis et al. 2007a). A competition model based on several phytoplankton species found that warming-induced shifts observed in phytoplankton succession were driven by species-specific responses to temperature change (De Senerpont Domis et al. 2007b). Another way temperature can influence key ecosystem functions is by altering the size composition of phytoplankton, often with opposing effects, especially in the context of temperate lakes. Some mesocosm experiments found that warming favors small phytoplankton cells (Yvon-Durocher et al. 2011; Rasconi et al. 2015), whereas others observed a shift towards large cells (Rüger and Sommer 2012; Yvon-Durocher et al. 2015). Long-term field studies also showed similar contrasting results: high water temperatures fostered both large (Pomati et al. 2020; Dory et al. 2024) and small phytoplankton (Zohary et al. 2017).

The major effects of warming on phytoplankton sizes appear to be driven by temperature-mediated trophic interactions and resource availability (Murphy et al. 2020; Zohary et al. 2021). However, these two drivers are confounded in natural environments and

that limits our ability to make sound predictions on the compositions and dynamics of planktonic communities in lake ecosystems. Mathematical models are useful tools for disentangling the interacting effects of temperature on lake plankton. Recent modelling investigations included more comprehensive ecological interactions between plankton communities and the environment. By considering seasonal variations in light, water temperature, mixed layer depth, and nutrients in a plankton ecosystem model, Cagle and Roelke (2021) and To et al. (2024) found that grazing characteristics of zooplankton regulate the succession of different phytoplankton species and their community composition in temperate lakes. However, such modelling investigations have not been implemented in combination with a high degree of biotic and abiotic data inputs. A data-informed model that is verifiable by in-situ observations can strengthen our understanding on the effects that lake warming can have on composition of phytoplankton communities.

Building on our previous NPZD modelling study (To et al. 2024), we present here an updated data-informed trait-based model to explain plankton dynamics in a specific temperate lake and to explore lake ecosystem responses under different warming scenarios. Greifensee, a medium-size eutrophic lake near Zurich (Switzerland), experienced multi-decadal warming and re-oligotrophic trends (Pomati et al. 2020; Merz et al. 2023). The composition of the phytoplankton community in the lake was found to be sensitive to long-term changes in thermal and nutrient conditions (Bürgi et al. 2003; Monchamp et al. 2017), making Greifensee a natural laboratory for studying the impacts of changing temperature on abundance and community composition of lake plankton. This new model version includes additional physiological details on the thermal dependence of the plankton communities (presented in Methods) and assumes

that temperature impacts phytoplankton growth and zooplankton grazing through data-derived relationships. The physical processes in the model are driven by observed environmental data and the model outputs are calibrated against averaged nutrient concentration, phytoplankton and zooplankton biomasses, and size data collected at Greifensee. The calibrated model is then run into the future to predict the evolution of the ecosystem under projected climate change scenarios. We considered three Representative Concentration Pathways (RCPs), RCP 2.6, RCP 6.0, and RCP 8.5, reflecting low, medium, and high emission scenarios, respectively. We aim to understand the effect of increased water temperature on future nutrient concentrations, plankton biomasses, size composition of the phytoplankton community, and timing of the peaks in phytoplankton biomass.

Methods

Size-based plankton ecosystem model

We consider a size-based plankton model that builds on a previous NPZD-based modelling study (To et al. 2024) but that captures now a real lake, Greifensee. The model consists of: (1) an inorganic nutrient, N (i.e., phosphate, PO_4^{3-}), (2) a community of phytoplankton A_i , subdivided into ten size classes ($i = 1, \dots, 10$), covering the phytoplankton size range typically observed at Greifensee, (3) a community of herbivorous zooplankton Z_j subdivided into two size classes ($j = 1, 2$), and (4) a detritus pool, D (Fig. 1). The phytoplankton community exploits nutrients and light for growth, while water temperature affects the growth of phytoplankton and the grazing of zooplankton. Since phosphate is considered the most limiting inorganic nutrient for most lakes (Schindler 1978), all state variables are expressed in units of

$\mu\text{M P}$. A detailed model description, including equations and formulation of processes, is reported in Supporting Information S1. In the following, we present the novel features of the model, which include the temperature dependence of phytoplankton growth and zooplankton grazing, the model calibration procedure, and the ecological projections under the different warming scenarios.

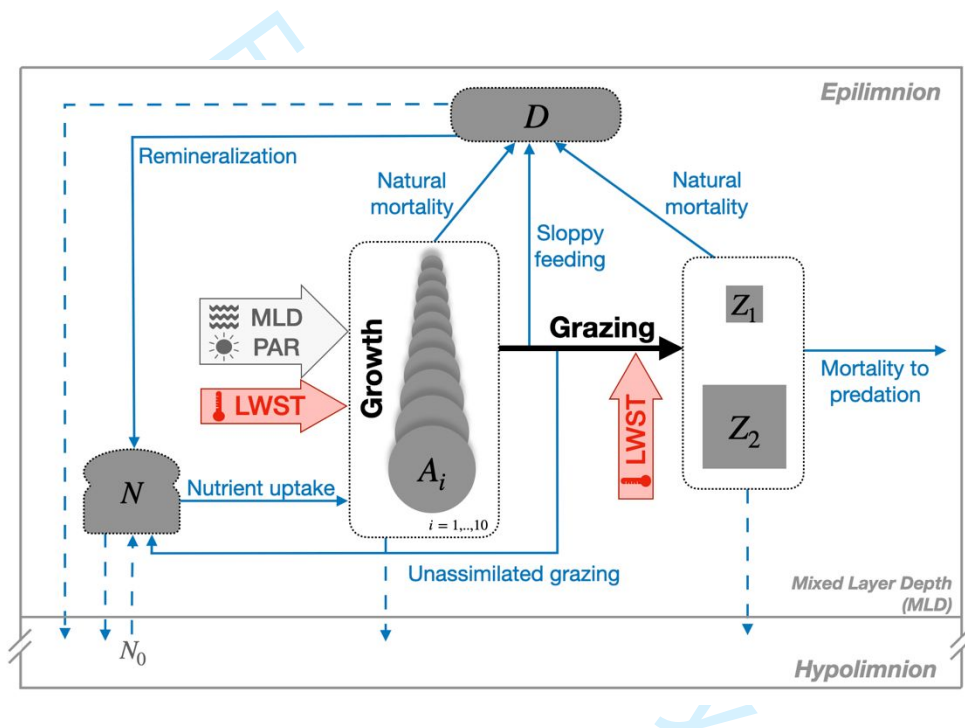


Fig. 1 Schematic of the size-based planktonic model. This model is based on our previous work (To et al. 2024). The new features of this model are highlighted in red. These are the temperature dependences of phytoplankton growth and zooplankton grazing, which are quantified through a unimodal curve and a Q_{10} exponential curve, respectively. The model assumes a zero-dimensional spatial configuration, the so-called slab physics approximation. The epilimnion contains a plankton ecosystem consisting of four components, inorganic nutrient, N , phytoplankton, A_i , zooplankton, Z_j , and detritus, D . Phytoplankton, A_i , is subdivided into i size classes (with $i = 1, \dots, 10$) and zooplankton, Z_j (with $j = 1, 2$), is subdivided into 2 size classes, Z_1 and Z_2 . Phytoplankton growth and zooplankton grazing are size-dependent. The depth of the epilimnion

reflects the Mixed Layer Depth (MLD). The blue arrows indicate ecological and biogeochemical processes (solid lines) and physical processes (dashed lines). Environmental forcing to this model is provided by in situ observations of photosynthetically active radiations (PAR), mixed layer depth (MLD – an indicator of lake mixing), and water temperature (LWST) made at Greifensee. To approximate real-world grazing, the grazed material is split into (1) a portion that is not ingested (due to sloppy feeding) and is immediately lost into detritus, (2) a portion that is ingested and assimilated by zooplankton, and (3) an additional portion that is not assimilated after being ingested and is excreted and quickly remineralised into inorganic nutrients. The loss of zooplankton to higher order predation is density-dependent and formulated as a quadratic function. We assume that the loss of nutrients from the epilimnion is recharged through changes in MLD by a constant nutrient source, N_0 , from the hypolimnion. We determined the value N_0 based on average winter observations, when the entire water column is well mixed and nutrient concentrations are homogeneously distributed throughout it. The detailed model configuration and all equations describing the processes shown in the schematic are reported in the Supporting Information S1.

Temperature dependences

The basic features of the model considered in this study were introduced by To et al. (2024) and are presented in detail in Supporting Information S1. The novel features include the temperature dependences of phytoplankton growth and zooplankton grazing. The temperature-regulated growth of phytoplankton is described by the following function (Norberg 2004):

$$E(T) = e^{0.063T} \cdot \left[1 - \left(\frac{T - T_{opt}}{\sigma_T} \right)^2 \right], \quad (1)$$

with T representing temperature ($^{\circ}\text{C}$). T_{opt} and σ_T representing, respectively, thermal optima and thermal tolerance. The curve is set to reflect the observed range of temperatures, between 3 and 23 $^{\circ}\text{C}$ (Eq. 1 and Fig. 2E). Zooplankton grazing is regulated by temperature through the following function (Fig. 2F):

$$\psi(T) = Q_{10}^{\frac{T-T_{ref}}{10}}. \quad (2)$$

The function scales with the Q_{10} coefficient, with T_{ref} representing the reference temperature at which no factorial change in maximum ingestion rate of zooplankton occurs (set to 15 $^{\circ}\text{C}$, Table 1). The Q_{10} coefficient was determined from a range of values observed by Hansen et al. (1997) for the maximum ingestion rate of Flagellates, Ciliates, and Copepods (functional groups abundant at Greifensee). In our model, the thermal optima (T_{opt}) and the thermal tolerance (σ_T) for phytoplankton growth and the Q_{10} coefficient for zooplankton grazing are assumed constant. This assumption implies that different phytoplankton and zooplankton size classes have identical thermal responses in relation to, respectively, growth and grazing. These temperature dependencies allow us to examine the effects of temperature on phytoplankton growth and zooplankton grazing and to investigate possible changes to lake ecosystem components when temperature increases.

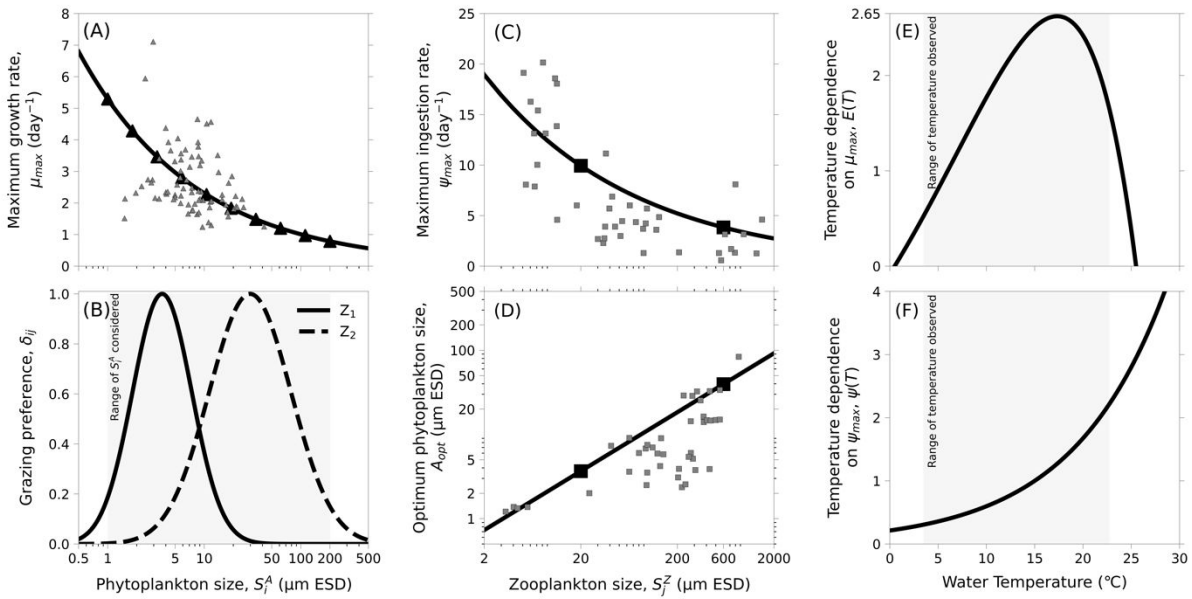


Fig. 2 Model assumptions related to phytoplankton growth and zooplankton grazing. (A) Allometric relationship defining maximum growth rates, μ_{max} , for a given phytoplankton size S_i^A . (B) Size-grazing breadth of the two zooplankton over a range of phytoplankton size classes. Allometric relationships defining (C) maximum ingestion rates, ψ_{max} , and (D) optimal phytoplankton size, A_{opt} , for a given zooplankton size S_j^Z . The community-wide temperature dependence on (E) phytoplankton growth, $E(T)$, and (F) zooplankton grazing, $\psi(T)$. The grey triangles and squares are the data points obtained from literature compilations by Edwards et al. (2012), (A); Hansen et al. (1994), (C); and Hansen et al. (1997), (D). Black triangles and squares represent the specific size values considered in this study for, respectively, phytoplankton and zooplankton. Shaded areas in panel B represent the range of phytoplankton sizes considered in this study, whereas in panels E and F represent the present observed range of water temperatures at Greifensee.

Model calibration

Environmental forcing

196 The model is forced with environmental variables observed at Greifensee, a medium-size
197 (about 8.45 km², with a maximum depth of 33 m) eutrophic lake near Zurich (Switzerland).
198 The environmental variables are: water temperature, Photosynthetically Active Radiation
199 (PAR), and Mixed Layer Depth (MLD). Water temperature is obtained from the CTD profiles
200 as daily average epilimnetic temperature from March 2019 to December 2022. The PAR data
201 is converted from the solar irradiance data collected daily by MeteoSwiss, from a weather
202 station next to Greifensee, for the same period as the water temperature. We retrieved data for
203 mixed layer depth from Pomati et al. (2020), which documented monthly values at Greifensee
204 from 1972 to 2015. The forcing functions for the model are daily values over a one-year period.
205 We used daily means from a three-year dataset (March 2019 to December 2022) for water
206 temperature, 75-percentiles from the same three-year dataset for PAR, and daily values
207 obtained by interpolating the monthly averages for MLD (Fig. 3). Details of the procedures
208 followed for generating the forcing functions are reported in Supporting Information S2 and
209 the environmental data are provided in Supplementary Data.

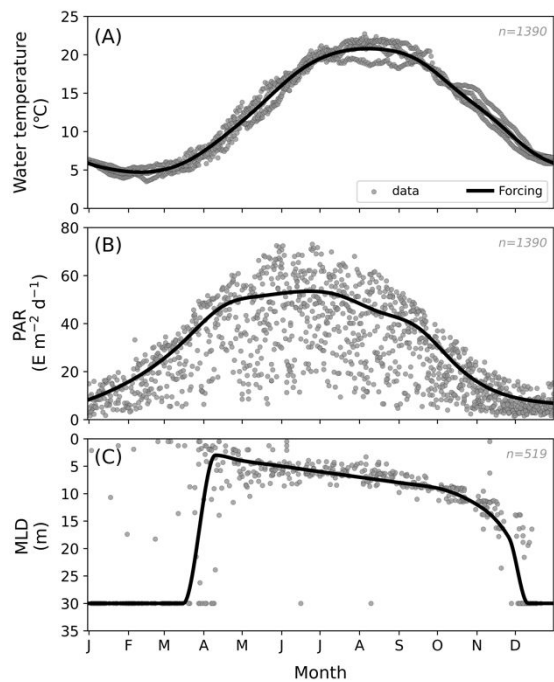


Fig. 3 Forcing functions used in our model. (A) Water temperature ($n = 1390$). (B) Photosynthetically Active Radiation, PAR ($n = 1390$). (C) Mixed Layer Depth, MLD ($n = 519$). A more detailed description of the forcing functions is provided in Supporting Information S2.

Data for model calibration

The model was calibrated by daily mean nutrient concentrations (phosphate) and phytoplankton and zooplankton abundance and size data, provided by the Swiss Federal Institute of Aquatic Science and Technology (Eawag). These data covered the period from March 2019 to December 2022. We selected the six most dominant groups of phytoplankton, namely, chlorophytes, rhodomonas, cryptophyceae, centric diatoms, dinobryon, and uroglena, and two dominant groups of herbivorous zooplankton, namely, ciliates and daphnia. The plankton data were collected at Greifensee using a novel dual-magnification dark-field imaging

microscope deployed at 3 meters deep (Merz et al. 2021). The measured abundance data were in units of ROI/sec (Regions Of Interest per second). This pseudo-unit is a statistical output derived from each detected and processed object by the imaging microscope and is analogous to object counts for both unicellular and colonial phytoplankton and zooplankton individuals (Merz et al. 2021). We converted these data into plankton abundances (individual L^{-1}) based on laboratory-calibrated functions (Fig. S1, Table S1), derived by Merz et al. (2021). The plankton abundances were then converted into phosphorus-equivalent biomass concentrations, following previously reported scaling functions based on the size of phytoplankton and zooplankton (Supporting Information S3).

The size data were collected as bio-area (μm^2) and converted into biovolume, based on an empirical scaling law (Table S1). The median bio-area of the six phytoplankton species we selected ranged from 10^2 to $10^5 \mu m^2$. Our model captures cell size in terms of equivalent spherical diameter, ESD (μm), because it encompasses differences in size and geometric shape of plankton (Jennings and Paslow 1988). Therefore, in the model, we consider a size spectrum ranging from 1 to 200 μm ESD for the phytoplankton community, encompassing 10 size classes separated by logarithmic intervals. For zooplankton, we considered two size classes, Z_1 and Z_2 , of 20 and 600 μm , respectively. We started our simulations with an initial concentration of $6.25e^{-4} \mu M$ P for all planktonic organisms. We analyzed the phytoplankton size composition in macroecological terms (e.g., Acevedo-Trejos et al. 2018; To et al. 2024), by considering a community-weighted mean cell biovolume, $\bar{V}^A$:

$$\overline{V^A} = \frac{\sum_i^n A_i V_i^A}{\sum_i^n A_i} \quad (3)$$

243 where n denotes the total number of size classes, and A_i and V_i^A denote, respectively, total
 244 biomass and biovolume of the i^{th} class of phytoplankton.

245 All model parameters were taken from literature (Table 1). Parameters that regulate
 246 nutrient uptake and light harvesting of phytoplankton and grazing of zooplankton were fine-
 247 tuned manually (i.e., α_{P_I} , K_n , $\alpha_{\psi_{max}}$, $\alpha_{A_{opt}}$, $\square_j^{Z_1}$ and $\square_j^{Z_2}$; within ranges of values reported in
 248 the literature, see Table 1), in order to find best fits (based on visual inspection) between model
 249 results and observations (Fig. S5). This parameter calibration produced what we call a ‘standard
 250 model run’, i.e. a model run that yields the best match with observations. The time step of our
 251 model is one day. To calibrate parameters, we compared the model results against three
 252 consecutive years of nutrient and biological data, using daily averages over a one-year period
 253 (details can be found in Supporting Information S3). The model results, therefore, reflect the
 254 daily dynamics of the lake for a “typical” (i.e., average) year under current temperature
 255 conditions. Additionally, we undertook a sensitivity analysis to evaluate the effects of
 256 parameter uncertainty on model results. We addressed those parameters ($n=14$) that could not
 257 be directly constrained by the data specifically available for Greifensee and those that were not
 258 calibrated (Supporting Information S4). Since we are interested in the average state of the lake
 259 over one year, the model is run to steady-state for 10 years, using the same annual forcing for
 260 every year, and considering only the final year for analysis. Interannual variability was not
 261 explored as it was beyond the scope of our study.

262

263 **Table 1.** List of model parameters with corresponding symbols, standard values, parameter range,
 264 units, and literature sources. When we report a parameter range, the standard value of the parameter was
 265 derived from model calibration.

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/parameter range
Initial slope of P-I curve	α_{P_I}	0.02	0.02 - 0.1	Ein m ⁻² d ⁻¹	Anderson et al. (2015)
Maximum photosynthesis rate	P_{max}	1.1	-	d ⁻¹	To et al. (2024)
Light attenuation coefficient	K_{par}	0.1	0.08 – 0.5	m ⁻¹	Chen et al. (2002); Genkai-Kato et al. (2012)
Thermal tolerance	σ_T	12.5	-	°C	Greifensee data
Thermal optima	T_{opt}	13	-	°C	Greifensee data
Half-saturation constant for nutrient (phosphorus)	K_n	0.06	0.06 -0.1	µM P	Edwards et al. (2012)
Slope of the allometric relationship for maximum growth rate, $\mu_{maxi}(S_i^A)$	$\alpha_{\mu_{max}}$	-0.36	-	-	Edwards et al. (2012)

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/ parameter range
Intercept of the allometric relationship for maximum growth rate, $\mu_{maxi}(S_i^A)$	$\beta_{\mu_{max}}$	$10^{0.69}$	-	d ⁻¹	Edwards et al. (2012)
Cross-thermocline diffusive mixing coefficient	ω	0.1	-	m day ⁻¹	Fasham et al. (1990)
Natural mortality - Phytoplankton	$\square_A$	0.1	-	d ⁻¹	this study
Natural mortality – Zooplankton	$\square_Z$	0.1	-	d ⁻¹	To et al. (2024)
Higher-order mortality rate	η_Z	0.34	-	d ⁻¹	Oschlies & Schartau (2005)
Fraction of grazing not egested	ε	0.69	-	-	Fasham et al. (1990)
Fraction of grazing assimilated to growth	γ	0.75	-	-	Oschlies & Schartau (2005)
Remineralization rate	φ	0.15	-	d ⁻¹	Kerimoglu et al. (2017)
Half-saturation constant to phytoplankton cells	K_A	0.19	-	μM P	Kerimoglu et al. (2017)
Size-grazing breadth - Z_1, Z_2	$\square_j^{Z_1}, \square_j^{Z_2}$	0.435, 0.6	0.2-0.8	μm ESD	Banas (2011)

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/ parameter range
Slope of the allometric relationship for maximum ingestion rate, $\psi_{maxj}(S_j^Z)$	$\alpha_{\psi_{max}}$	-0.28	-0.25 - -0.4	-	Hansen et al. (1997)
Intercept of the allometric relationship for the maximum ingestion rate, $\psi_{maxj}(S_j^Z)$	$\beta_{\psi_{max}}$	23	22-26	d ⁻¹	Hansen et al. (1997)
Q ₁₀ coefficient for the maximum ingestion rate	Q ₁₀	2.8	2.0 - 3.9	-	Data in Hansen et al. (1997)
Reference temperature for the Q ₁₀ coefficient	T_{ref}	15	-	°C	Hansen et al. (1997); Kerimoglu et al. (2017)
Slope of the allometric relationship for optima phytoplankton size, $A_{optj}(S_j^Z)$	$\alpha_{A_{opt}}$	0.7	-	-	Hansen et al. (1997)
Intercept of the allometric relationship for optimal phytoplankton size, $A_{optj}(S_j^Z)$	$\beta_{A_{opt}}$	0.45		µm ESD	Hansen et al. (1997)
Cell sizes – phytoplankton	S_i^A	1-200	-	µm ESD	Greifensee data
Body sizes – zooplankton	S_j^Z	20 (Z ₁) 600 (Z ₂)	-	µm ESD	Greifensee data

Parameter	Symbols	Standard Value	Parameter range	Units	Sources for parameter/parameter range
Initial conditions for nutrient	N_0	0.75	-	$\mu\text{M P}$	Greifensee data
Initial conditions for phytoplankton, zooplankton, and detritus	A_0, Z_0, D_0	6.25e-4	-	$\mu\text{M P}$	this study

Ecological projections based on climate change scenarios

The model parameterization obtained from the community-integrated outputs of the standard model run produces model results that reflect current average annual conditions (in terms of nutrient concentrations, total phytoplankton and zooplankton biomasses and aggregated phytoplankton size structures) at Greifensee. With this parameter configuration, we forced the model with new temperatures, representing three climate change scenarios, based on three Representative Concentration Pathways (RCPs), Table S3, expected by the end of this century (i.e., year 2099) in the region covering Greifensee, between 40° N and 60° N. We considered average and seasonally uniform increases in water temperature by 1.4 °C (RCP 2.6), 2.5 °C (RCP 6.0), and 4.2 °C (RCP 8.5). However, our future projections did not include physical feedback that could be induced by increasing temperatures, like changes in lake stratification (but see Supporting Information S5 for a sensitivity run exploring the impact of this omission on our results under scenario RCP 8.5). Also for these climate change scenarios, we run the model to steady-state for 10 years, using the same annual forcing for every year, and

considering only the final year for analyses. In our analyses, annual seasons are defined as follows: winter, from December to February; spring, from March to May; summer, from June to August; and autumn, from September to November.

Sensitivity analyses on phytoplankton thermal traits

For each temperature scenario, we undertook a series of sensitivity analyses on the thermal optima and the thermal tolerance of the phytoplankton community. These sensitivity analyses examine the effects on the model results (annual medians) under variations of $\pm 15\%$ from standard values (Fig. 4). These thermal traits play a key role in regulating phytoplankton responses to temperature change but are difficult to measure in natural systems and their values can vary within and between species (Boyd et al. 2013; Thomas et al. 2016). The range of variability we considered for these sensitivity analyses covers the values of thermal optima and thermal tolerance observed in freshwater species (Thomas et al. 2016).

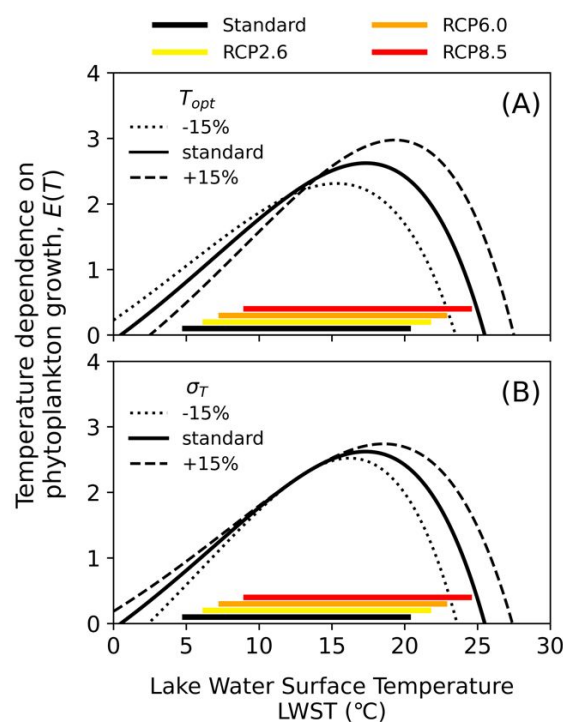


Fig. 4 Temperature dependence curves tested in the sensitivity analyses. These curves result from changes of $\pm 15\%$ in (A) thermal optima, T_{opt} , and (B) thermal tolerance, σ_T , from standard values. The standard values for T_{opt} and σ_T are, respectively, 13 °C and 12.5 °C (black curves). The colored bars represent the annual ranges of water temperatures for various warming scenarios: dark (current average state); yellow (RCP 2.6); orange (RCP 6.0); and red (RCP 8.5).

Results

Current status in the seasonal dynamics of lake ecosystem components

For the model calibration, we focused on the seasonal dynamics of four major variables: (1) nutrient concentration, (2) phytoplankton biomass, (3) zooplankton biomass, and (4) phytoplankton mean biovolume (the latter three are community-integrated). The simulated inorganic nutrient dynamics capture the average annual cycle of epilimnetic phosphorus observed at Greifensee (Fig. 5A). The nutrient concentration is characterized by a steady decline, starting at the beginning of the year and lasting for about three months, it remains low throughout summer and for part of autumn, and it restocks by the end of the year (Fig. 5A).

The simulated plankton dynamics also capture the observed community-integrated biomass qualitatively well (Fig. 5B). The model captures the biomass accumulation of phytoplankton during spring, summer, and autumn, and reproduces correctly the low phytoplankton biomass levels after the summer and autumn peaks (Fig. 5B). However, the model produces a first winter peak that is not shown by the data (Fig. 5B), it underestimates the spring biomass, and it overestimates the summer biomass. The simulated zooplankton

biomass matches well with the autumn and winter observations, despite a slight overestimation for the spring peak (Fig. 5C). A mismatch is noticeable in June and July, when the simulated zooplankton biomass declines due to the dominance of large and less grazable phytoplankton (Fig. 5C and 5D). In summary, despite being an approximation of the real lake in terms of physics and ecology, the model captures several general features of the phytoplankton and zooplankton seasonal cycles; namely the spring, summer, and autumn biomass accumulations of phytoplankton and the spring biomass peaks of zooplankton.

A good match is also obtained between the simulated and observed seasonal patterns in the community-mean biovolume of phytoplankton. In general, both model and data show small mean biovolume in late winter and beginning of spring (Fig. 5D), when smaller cells outcompete larger cells due to weaker grazing on smaller cells (Fig. 5C). Relatively larger phytoplankton cells are present during summer and autumn (Fig. 5D), when zooplankton grazing on smaller cells increases and the larger, less grazable cells gain dominance. Additionally, we find that the phytoplankton size composition is strongly controlled by grazing characteristics, namely, the allometric scaling of grazing (i.e., maximum ingestion rate and optimal phytoplankton size) and the size-grazing breadth of zooplankton (Fig. S3A, S5I, S5J, S5L).

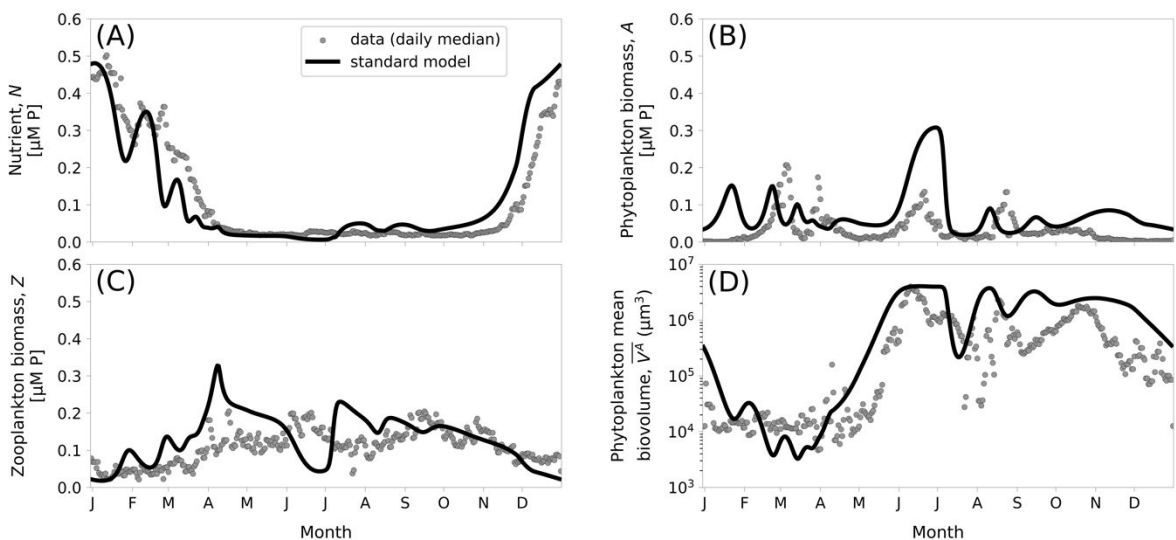


Fig. 5 Results of the standard run showing temporal dynamics of (A) nutrient concentration, (B) phytoplankton community biomass concentration, (C) zooplankton community biomass concentration, and (D) phytoplankton mean biovolume. The grey dots represent the observational data collected in the epilimnion by the underwater imaging device (see details about data transformation in Supporting Information S3) and the black lines represent the model outputs. The model is run to steady-state over 10 years and we show here the results of the 10th year. The observations are median daily values over 3 years of data (from March 2019 to December 2022).

Expected future seasonal dynamics of major variables

Alterations in nutrient dynamics

We find seasonal alterations in the nutrient dynamics under the warming scenarios, with the strongest alteration occurring under RCP 8.5 (water temperature higher by 4.2 °C with respect to current average). The most prominent change in the nutrient concentration occurs in summer and autumn, i.e. from July to October, when there are excessive nutrient accumulations in the

lake, and accumulation intensifies with the warming scenarios (Fig. 6B). Under RCP 8.5, the nutrient accumulation in summer reaches an annual maximum, a pattern that contrasts sharply with the current status (standard run) when the highest nutrient level occurs in winter (Fig. 6B). This summer nutrient peak coincides with a decline in phytoplankton biomass, caused by the unfavorable temperature conditions for growth experienced by the system under RCP 8.5 (Fig. S6A and S7). Warming also causes an increase in the overall annual nutrient concentration in the epilimnion. Under RCP 8.5, the annual median nutrient concentration is about four times higher than that under the current status, i.e. $0.05 \mu\text{M P}$ (current status) versus $0.2 \mu\text{M P}$ (RCP 8.5), Fig. 6C. In summary, our model projections suggest that as the lake warms up, nutrient accumulation increases in the summer months and the lake becomes more eutrophic.

Alterations in plankton phenology

As the lake warms up, the model shows lower winter peaks and earlier summer peaks of phytoplankton (Fig. 6D). The timing of the summer peaks depends on the timing of thermal optima, which shifts with the warming scenarios (Fig. S6C). Under RCP 8.5, the summer phytoplankton peak advances by about two months, from July to May. The earlier arrival of the summer peaks shortens the corresponding period of low phytoplankton biomass after the spring peaks. We also notice an increased number of minor phytoplankton peaks during the year as warming intensifies, with small reductions in annual median phytoplankton biomass concentrations (Fig. 6E). In summary, the increase in water temperature advances the timing of the major phytoplankton peaks and increases the number of minor peaks occurring throughout the year.

The model results also show phenological shifts for zooplankton. The timing of the first zooplankton peak in April remains the same in all climate scenarios, but the second peak moves earlier with the severity of the RCP scenarios (Fig. 6F). Consequently, the period between the first and the second zooplankton peaks shortens by two months under RCP 8.5. Also, the zooplankton biomass in late summer decreases noticeably under RCP 8.5, compared to other scenarios, due to the decline in phytoplankton growth (Fig. S7). The annual median zooplankton biomass is lower in RCP 8.5 compared to other scenarios (Fig. 6G). In summary, major alterations in the phenology of zooplankton include a reduction in the period between the first and the second peak and a reduction in the summer biomass.

Alterations in phytoplankton mean biovolume

An elevated water temperature increases the mean biovolume of phytoplankton and reduces the range of size variations during the year. In particular, as the lake warms up, relatively smaller cells are replaced by larger cells in winter and in spring (Fig. 6H). Under RCP 8.5, the dominance of relatively larger phytoplankton appears already in April, two months earlier than in the current status. In summary, a warmer lake fosters a narrower size diversity of phytoplankton and reduces seasonal variations in mean biovolumes (Fig. 6I).

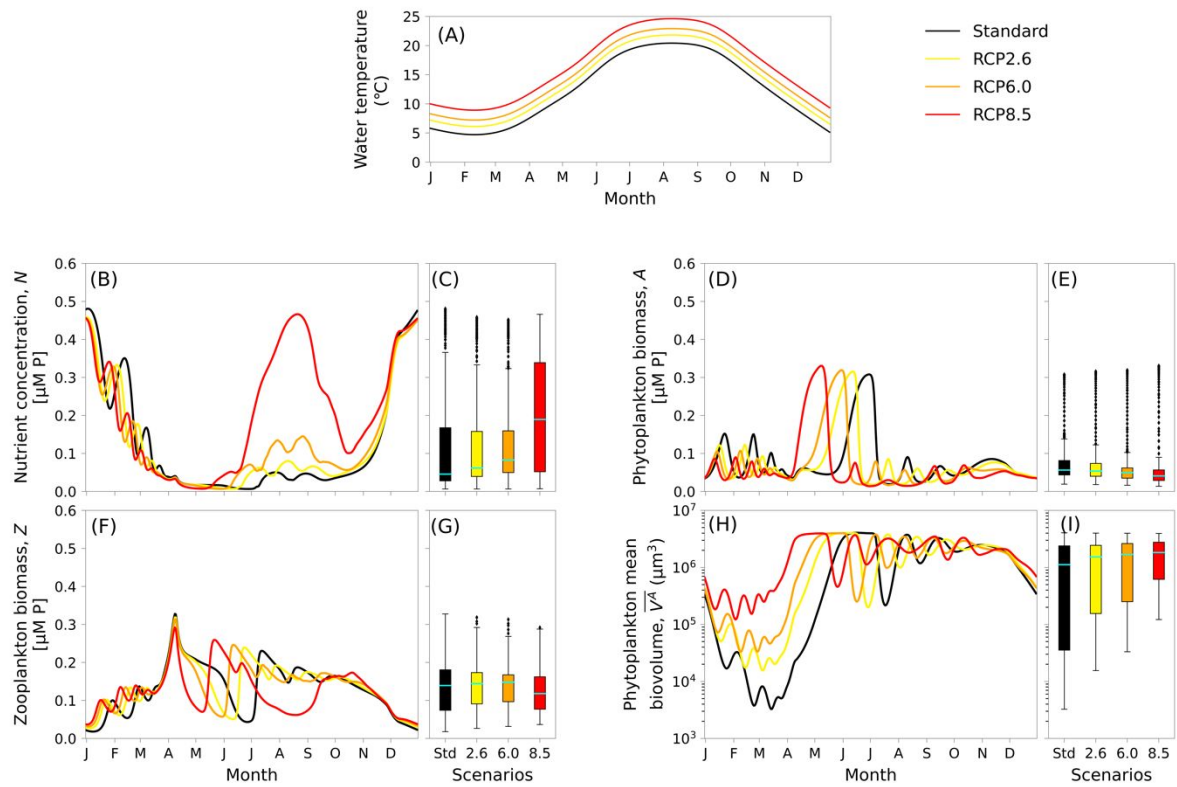


Fig. 6 Results of climate change scenarios compared to the current average status of the lake (standard run, black lines and Fig. 5). The climate change scenarios, RCP 2.6, RCP 6.0, and RCP 8.5 simulate an average and seasonally uniform increase in water temperature by, respectively, 1.4 °C, 2.5 °C, and 4.2 °C (A). The results show temporal dynamics of (B) nutrient concentration in the epilimnion, (D) phytoplankton biomass, (F) zooplankton biomass, and (H) phytoplankton mean biovolume. Boxplots (panels C, E, G, and I) show the overall seasonal variability (width) and median (cyan lines) of the corresponding variables. The model is run to a steady annual cycle over a period of 10 years with repeated annual forcing, the results shown are those of the 10th year.

Sensitivity analysis of future scenarios

We examined the sensitivity of the model results by varying the values of the thermal traits, thermal optima (T_{opt}) and thermal tolerance (σ_T), by $\pm 15\%$ from standard values. The results

show that, under future warming scenarios, the nutrient concentration, among other variables, is most sensitive to changes in the thermal traits. A reduction in either of the thermal traits leads to a higher median nutrient concentration, whereas an increase in either of the thermal traits leads to a lower median nutrient concentration (Fig. 7). These changes are exacerbated by the severity of the warming scenarios. Under RCP 8.5 (Fig. 7C), the median nutrient concentration increases by at least 200% when the phytoplankton community is characterized by lower values of thermal traits (i.e., lower tolerance to higher temperatures, Fig. 4). Such higher nutrient concentrations are caused by reduced growth of phytoplankton, from spring to autumn, due to a less suitable temperature for growth (Fig. S8G, S8I). Additionally, lower T_{opt} or σ_T leads to pronounced reductions in phytoplankton and zooplankton biomasses and in phytoplankton mean biovolume, particularly under RCP 8.5 (Fig. 7C). A higher T_{opt} (or σ_T) does not induce as many variations to the model variables; even under RCP 8.5, the greatest relative change is less than 50%. That is because higher thermal trait values produce a more heat-tolerant phytoplankton community (Fig. 4).

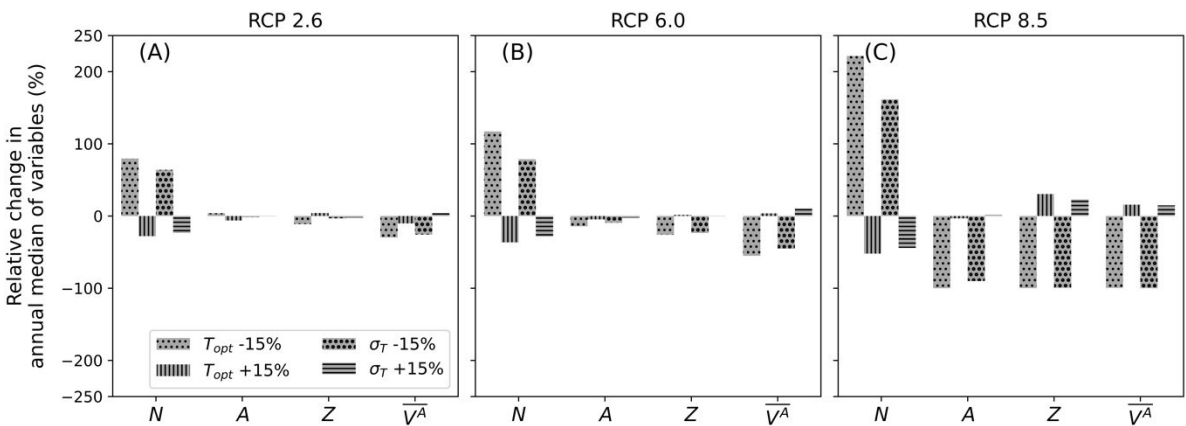


Fig. 7 Sensitivity of model variables (annual medians) caused by $\pm 15\%$ changes from standard values of thermal optima (T_{opt}) or thermal tolerance (σ_T), for the different climate change scenarios: (A) RCP 2.6, (B) RCP 6.0, and (C) RCP 8.5. The model variables are nutrient concentration, N ; phytoplankton biomass, A ; zooplankton biomass, Z ; and phytoplankton mean biovolume, $\bar{V}^A$.

Discussion

Mounting evidence over the last two decades shows rising lake temperatures on a planetary scale, with European temperate lakes expected to experience the fastest warming trend (Woolway et al. 2022). The consequences of rising temperatures on the physiology and ecology of plankton are far from clear. In our model, the growth of phytoplankton depends, among other factors, on a thermal dependence curve characterized by data-constrained (specifically, Greifensee data) optimal temperature for growth and tolerance range; and the grazing of zooplankton depends on a Q_{10} function constrained by literature data. Under future climate change scenarios, our model predicts increased epilimnetic nutrient concentrations, advancements in phytoplankton biomass peaks, and dominance of relatively larger phytoplankton. We also find that a phytoplankton community with lower thermal optima or lower thermal tolerances leads to even higher accumulation of nutrient concentrations due to the less favourable temperature ranges of phytoplankton growth and thus of lower nutrient uptake.

Current seasonal patterns of nutrients and phytoplankton concentrations

435 With respect to the current average state of Greifensee, the seasonal pattern of phosphorus
436 dynamics produced by the model matches well with the observational data and reflects the
437 typical nutrient dynamics of temperate lakes, which are characterized by high nutrient
438 concentrations in winter and low concentrations in summer (Sommer et al. 2012). In temperate
439 lakes, relatively abundant phosphorus at the beginning and at the end of the year coincides with
440 scarce light and less favourable temperatures for growth, resulting in lower growth and uptake
441 rates (Staehr and Sand-Jensen 2007; Sommer et al. 2012). The June-July peaks in the observed
442 phytoplankton and zooplankton biomasses are temporally coincident, whereas the model
443 results show a slight shift, with zooplankton peaking slightly later than phytoplankton. This
444 small delay, typical of predator-prey models, is an expected feature of our model, given its
445 general resource-prey-predator structure. We find that the annual variations in the mean
446 phytoplankton biovolume are characterized by relatively higher abundances of smaller
447 phytoplankton in spring, higher abundances of larger phytoplankton in summer, and higher
448 abundances of intermediate-size phytoplankton in autumn. These patterns are consistent with
449 the general phytoplankton dynamics expected to occur when interactions between inorganic
450 nutrient limitation and grazing pressure are primary driving factors of lake ecosystems (Vanni
451 and Temte 1990; Sommer et al. 2012). Our model further supports findings from previous size-
452 based models (Banas 2011; Taniguchi et al. 2023; To et al. 2024) showing that the biomass of
453 phytoplankton is regulated by variations in resources like light and nutrients, whereas the
454 annual variations in the phytoplankton size composition are strongly controlled by grazing
455 characteristics like the size-feeding breadth of zooplankton. By using appropriate zooplankton
456 size-feeding characteristics under the influence of major physical factors in lakes, we capture
457 and elucidate the phenology of phytoplankton biomass and size composition at Greifensee.

458 *Future increased phosphorus concentration driven by phenological shift of phytoplankton*

459 The ecological projections under future warming scenarios show that epilimnetic phosphorus
460 concentrations increase with the increasing severity of global warming. The higher phosphorus
461 concentration is driven by a shift in phytoplankton phenology, characterized by lower
462 accumulations of phytoplankton biomass in winter and earlier accumulations in spring, causing
463 a reduction of phytoplankton biomass in summer due to sub-optimal temperature conditions
464 for phytoplankton growth. The sub-optimal condition for growth also results from a mismatch
465 between optimal temperature and light conditions, as the optimal temperature for growth occurs
466 earlier than optimal light conditions. This mismatch retains unused phosphorus in the summer
467 epilimnion because of reduced phosphorus uptake by the phytoplankton community. This
468 result is based on the assumption that different phytoplankton size classes have identical
469 thermal characteristics (Fig. 2). In nature, species sensitivities to temperature may vary and
470 different phytoplankton can exhibit different growth and nutrient uptake abilities along thermal
471 gradients (Thomas et al. 2016; Bestion et al. 2018). This is an aspect that could generate
472 feedbacks on the nutrient dynamics that are not captured by our model. Also, higher
473 temperatures lead to stronger lake stratification, which in turn has an impact on competition
474 for nutrients and light with consequences on the composition and dynamics of phytoplankton
475 communities (Tilzer and Goldman 1978; Donis et al. 2021). The stronger and longer
476 stratification events in a warmer lake (Butcher et al. 2015; Woolway et al. 2021) can further
477 disrupt the nutrient exchanges between epilimnion and hypolimnion and, thus, influence
478 nutrient dynamics (Supporting Information S5, Fig. S4B). Taken together, these results
479 indicate that accounting for the impacts of climate on nutrient regimes would require detailed

considerations in relation to physical and biological feedbacks. Our results should be considered as a first order approximation with respect to the impacts that global warming can have on nutrient dynamics and plankton phenology.

Earlier accumulation of phytoplankton biomass driven by thermal optima

Our climate change scenarios produce earlier accumulations of summer phytoplankton biomass. Earlier formation of spring blooms in temperate lakes was detected by previous studies based on simulations and observations. For example, a dynamic diatom-daphnia model showed that, under lake warming, a high phosphorus supply causes earlier diatom spring blooms (Huber et al. 2008). A depth-resolved hydrodynamic model coupled to a simple phytoplankton growth model showed that earlier spring blooms can be caused by changes in lake stratification patterns (Peeters et al. 2007). In our model, the earlier peaks of large phytoplankton are produced by a warmer winter that advances the timing of optimal temperature for the growth of all phytoplankton size classes but intensifies the grazing of zooplankton on small phytoplankton size classes. Thermal characteristics, like optimal temperature for growth, are important factors for driving seasonal succession in culture experiments comprising marine diatoms (Anderson and Rynearson 2020). Although our projections are largely consistent with these studies, a long-term observation conducted in a large Norwegian lake revealed delayed phytoplankton peaks (Moe et al. 2022). Their observation considered taxonomic details of the phytoplankton community and showed that the phenological changes of phytoplankton are contingent to the different succession patterns exhibited by individual taxa. Therefore, taxonomic details related to the different thermal responses in a community are important for understanding specific changes in the phenological

patterns of phytoplankton (Thackeray et al. 2008; Segura et al. 2018). This shows the importance of considering size or taxonomic differences in thermal traits for capturing specific phenological trends.

Changes in phytoplankton mean biovolume under increasing temperature

We find an increase in mean phytoplankton biovolume and a decrease in the range of variability with increasing average temperature. These trends are caused by the disappearance of smaller phytoplankton during spring and the earlier appearance of larger phytoplankton during summer. Our results are driven by thermally-regulated bottom-up (nutrient uptake by phytoplankton) and top-down (zooplankton grazing) processes. As temperature increases, the optimal temperature for phytoplankton growth occurs earlier and the summer temperatures become sub-optimal for growth (Fig. S6C); whereas zooplankton grazing, which targets smaller phytoplankton, intensifies (Fig. S6B). These aspects interact not only to favor phytoplankton cells with intermediate biovolumes over cells with small biovolumes, but also to foster the earlier appearance of large phytoplankton. Intensified grazing on smaller phytoplankton was observed in warmer freshwater environments (Winder and Sommer 2012). Analyses of time-series from eight Swiss lakes revealed a rising proportion of larger phytoplankton cells in association with higher temperatures, higher zooplankton abundances, and lower nutrient levels (Pomati et al. 2020). This multi-decadal dataset showed that larger phytoplankton are found in summer, coinciding with higher grazing pressures compared to winter, suggesting the presence of a grazing-mediated positive correlation between larger phytoplankton and higher temperatures. However, in lakes where zooplankton grazing is weak, there can be different results. In Lake Kinneret (Israel), the prevalence of small phytoplankton

in stratified summer waters (based on an eight-year observation) resulted from a strong selection by sinking processes on the better-suspended small phytoplankton (Zohary et al. 2017). Sinking was not included in our model because Greifensee is characterized by a relatively shallow average depth (18 meters), but this loss process can contribute to varying the size composition of phytoplankton communities (Zohary et al. 2017). In summary, the physical and biological properties of a lake are important for understanding temperature-induced changes in the community composition of phytoplankton. Our findings here emphasize the role played by intensified grazing on the community compositions of phytoplankton in a warmer Greifensee.

Thermal traits affect future nutrient concentrations

The responses of phytoplankton to increasing temperatures are reflected in traits like optimal temperature for growth and thermal tolerance. Our sensitivity analyses suggest that the future projections of nutrient dynamics in the lake depend highly on the optimal temperature for growth and thermal tolerance of phytoplankton, which control their growth and uptake efficiencies under different temperature conditions. Previous modelling studies also showed that the thermal traits of phytoplankton regulate community responses. An ecosystem model resolving biogeochemical processes, for example, showed that alterations of the Q_{10} factor can markedly change the community composition of phytoplankton under the most severe climate change scenario (Anderson et al. 2024). An NPZ model driven by seasonal variations of temperature showed that tackling species differences, especially in terms of thermal optima, is important for estimating the thermal sensitivities of phytoplankton (Chen 2022). In addition, optimal temperatures for phytoplankton growth may co-vary with nutrient and light

availabilities (Edwards et al. 2016; Bestion et al. 2018) in ways that are not captured by our model. We suggest that measuring size-specific thermal trait values would be a crucial asset for modelling studies such as this one. These data would facilitate the formalization of a robust functional relationship between thermal traits and plankton sizes, a fundamental assumption for investigating the effects of temperature on plankton communities and the corresponding ecosystem feedback under climate change scenarios.

Limitations and future directions

Our study investigates the impact of increased water temperature in a specific lake system, relying on specific, data-derived assumptions regarding temperature-dependent phytoplankton growth and zooplankton grazing. Our results are based on a relatively idealized lake physics, which does not specify vertical stratification and lateral transport processes. We consider that this choice is appropriate for Greifensee, a relatively shallow lake that is typically fully mixed in winter and stratified into two distinct and homogeneously mixed layers in summer (Mieleitner et al. 2008; Huntscha et al. 2018). Many studies (e.g., Livingstone 2003; Rempfer et al. 2010; Butcher et al. 2015; Woolway et al. 2021) also show that, in a warming world, lakes will experience weaker mixing, with higher temperatures potentially structuring the water column into two stable layers, thus, making our simplified physical scheme an appropriate first-order approximation approach. Also, our results are based on a specific set of model parameters that are informed by lake data. This reduces the level of uncertainty in these parameters, especially those that regulate phytoplankton nutrient uptake and zooplankton grazing (Fig. S3), which are the major determinants of our results. Additionally, previous data suggested temperature dependences that were not considered in this study like phytoplankton respiration (Robarts and

Zohary 1987) and nutrient remineralization rates (Boscolo-Galazzo et al. 2018). The absence of size-specific temperature dependencies in our model prevents niche differentiation with respect to thermal traits (Chen 2022). This limits the model ability to maintain all 10 phytoplankton size classes over time due to competitive exclusion (Hardin 1960), an aspect common to plankton models (e.g., To et al. 2025) but that does not have an equivalent in nature (the “paradox of the plankton”, Hutchinson 1961). Thus, some of the size classes simulated by our model disappear during the course of the simulations. This, however, does not preclude the model from capturing, both in magnitude and dynamics, nutrient and plankton concentrations, as well as mean phytoplankton biovolume.

Assessing how these limitations impact the community compositions of phytoplankton would be an interesting avenue for future research. Data-derived size-scaling relationships about optimal temperature for growth or thermal tolerance, which we consider crucial to phytoplankton physiology, remain scant. Resolving these scaling relationships can optimize the mechanistic investigations of size-based models under future scenarios (Chen 2022; Litchman 2023), particularly those in relation to the synergistic effects of temperature, nutrients, and zooplankton grazing on phytoplankton communities (Pomati et al. 2020; Dory et al. 2024). Future modelling studies could examine the impacts of temperature on planktonic interactions more realistically by considering size-specific thermal responses (e.g., Chen 2022; Karakuş et al. 2022) or other size-related characteristics, like a more diverse zooplankton community in relation to body size. One could also consider additional temperature dependences on, for example, remineralization processes (Boscolo-Galazzo et al. 2018). Assuming that metabolic and biochemical rates would be enhanced by increasing temperatures,

these dependences would produce more accurate dynamics of inorganic nutrients and detritus. Spatially explicit hydrodynamic modelling studies could investigate the extent to which our first-order slab physics approximation is appropriate to capture the future ecological regimes of lakes like Greifensee (e.g., Huntscha et al. 2018). Studying inter-annual variabilities would be another obvious next step when looking at future projections of lake ecosystem.

Conclusions

By combining a broad range of field observations and data-informed trait-based modelling, our study provides insights into the future dynamics of nutrient, plankton biomasses and community compositions of phytoplankton in a temperate lake under global warming scenarios. Our model predicts three potential consequences in a warmer lake, namely, increased epilimnetic nutrient retention, advancement of phytoplankton blooms, and increased mean phytoplankton biovolume, the latter being brought about by a replacement of small phytoplankton cells with large ones. Acknowledging a number of simplifications and limitations, this study represents a first-order approximation of the ecological responses that Greifensee may exhibit in a warmer world. Most of the model parameters and all environmental forcing are site-specific, thus the validity of our results is limited to Greifensee or to lake systems with characteristics similar to Greifensee. Future investigations of these phenomena, based on alternative sets of assumptions or expanded datasets, would be a valuable effort to verify the likelihood of their occurrence. Our approach emphasizes the multifaceted responses of planktonic communities to global warming, highlighting both challenges and opportunities involved in understanding and managing a lake ecosystem in a changing world. We thus argue that long-term monitoring programs, like those undertaken at Greifensee, in combination with

modelling studies based on temperature dependences on the physiology and ecology of plankton can improve our predictions and enhance our capacity to design effective lake management strategies. Finally, investigating how the changes predicted by our model could impact lake ecosystem functions would help the scientific community to assess the significance that these studies have for lake management and conservation strategies in a warming world.

Acknowledgements

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854

855 **Data Availability Statement**

856 The lake data and model outputs used for this study are available on Zenodo (doi:
857 10.5281/zenodo.11390345). The code to reproduce the analyses of this study is available on
858 Github (<https://github.com/systemsecologygroup/TempSizeMod>).

859

860 **Conflict of Interest**

861 None declared.

Supporting Information for:

Future inorganic nutrient and plankton dynamics in a temperate lake

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For Review Only

Supporting Information S1 – Model configuration and equations

Physical configuration

The lake ecosystem model operates in a zero-dimensional physical configuration that consists of two layers stacked on top of each other, representing the upper epilimnion and the bottom hypolimnion (Fig. 1) and reflecting the so-called slab physics approximation (e.g. Fasham et al. 1990; To et al. 2024). The depth of the epilimnion is known as the mixed layer depth (MLD) and varies seasonally. The epilimnion is biologically active and is assumed to be well-mixed so that all model properties, physical and biological, are considered homogeneously distributed within its volume. The hypolimnion is biologically inactive. In the hypolimnion, a fixed amount of nutrient, N_0 , replenishes through physical mixing the inorganic nutrient pool of the epilimnion, N . Changes in motile (zooplankton) and non-motile (nutrient, detritus, and phytoplankton) model variables, induced by physical processes, can occur due to (1) exchange of fluxes driven by diffusive mixing between the epilimnion and hypolimnion and (2) variations in the MLD. The variation in zooplankton biomass due to changes in the MLD is described by the following term (Evans and Parslow 1985):

$$\dot{Z}_z = \frac{h}{MLD(t)}, \quad (S1)$$

with $h = \frac{dMLD(t)}{dt}$. Here we assume that zooplankton can maintain their position within the upper epilimnion. The shallowing or deepening of the MLD will, respectively, increase or decrease zooplankton concentrations. A shallowing MLD leaves the concentrations of non-motile variables (i.e. nutrient, phytoplankton, and detritus) unchanged because their reduction in number due to detrainment is compensated by the reduction in volume of the epilimnion. A deepening MLD reduces the concentrations of non-motile variables, unless N_0 is larger than N , that is, when the nutrient input from the hypolimnion is greater than the nutrient concentration in the epilimnion. The non-motile variables are also subject to diffusive mixing. The variations in biomasses of non-motile variables due to physical processes can be written as follows (Evans and Parslow 1985):

$$\dot{Z} = \frac{\omega + h^+(t)}{MLD(t)}, \quad (S2)$$

with ω being the diffusive mixing constant and

$$h^+(t) = \max(0, h). \quad (\text{S3})$$

Biological components

The model consists of a set of 14 ordinary differential equations:

$$\frac{dN}{dt} = - \sum_i \mu_i A_i + (1-\alpha) \sum_i \sum_j G_{ij} Z_j + \beta D + \beta N_0 - \beta N \quad (\text{S4})$$

$$\frac{dA_i}{dt} = (\mu_i - \alpha_A - \alpha) A_i - \sum_i \sum_j G_{ij} Z_j \quad (\text{S5})$$

$$\frac{dZ_j}{dt} = \left(\alpha \sum_i \sum_j G_{ij} Z_j - \beta_Z - \alpha_Z - \eta_Z Z_j \right) Z_j \quad (\text{S6})$$

$$\frac{dD}{dt} = \sum_i \beta_P A_i + \sum_j \beta_Z Z_j + (1-\alpha) \sum_i \sum_j G_{ij} Z_j - (\beta + \alpha) D \quad (\text{S7})$$

with $i = 1, \dots, 10$ and $j = 1, 2$. All the terms in the equations are described in the following.

The dynamics of nutrient (Eq. 4) is driven by nutrient lost to phytoplankton growth, $\sum_i \mu_i A_i$, and nutrient replenishment from (1) the portion of grazed materials that is not assimilated by zooplankton, $(1-\alpha) \sum_i \sum_j G_{ij} Z_j$, (2) the remineralization of detritus, βD , and (3) the exchange between the epilimnion and hypolimnion via vertical mixing, $\beta(N_0 - N)$, where N_0 is the nutrient concentration in the hypolimnion. We determined the value N_0 based on average winter observations, when the entire water column is well mixed and nutrient concentrations are homogeneously distributed throughout it.

The temporal dynamics of the i^{th} phytoplankton size class (Eq. 5) is regulated by phytoplankton growth, $\mu_i A_i$, and by several loss processes. The growth rate of phytoplankton, described by,

$$\mu_i = \mu_{\max_i}(S_i^A) \cdot E(T) \cdot U(N) \cdot H(I_Z), \quad (\text{S8})$$

is regulated by a size-dependent maximum growth rate term,

$$\mu_{max_i}(S_i^A) = \beta_{\mu_{max}} \cdot \left(\frac{S_i^A}{1 \mu\text{m}} \right)^{\alpha_{\mu_{max}}} , \quad (\text{S9})$$

with S_i^A representing the size of the i^{th} phytoplankton size class (Fig. 2A), and three environmental factors: temperature T , nutrient N , and light I_z . The temperature-regulated growth term, $E(T)$, following Norberg (2004), a new feature considered in this model, is presented in the main article, section ‘Temperature dependences’. The nutrient uptake term follows a Monod function (Monod 1949),

$$U(N) = \frac{N}{K_N + N} , \quad (\text{S10})$$

with N representing the inorganic nutrient concentration in the lake (Eq. 4) and K_N the half-saturation constant for nutrient.

The light-limited growth term depends on the daily MLD-averaged photosynthetic rate,

$$H(I_z) = \frac{1}{MLD} \int_{z=0}^{z=MLD} P(I_z) dz , \quad (\text{S11})$$

and is calculated by integrating the photosynthesis-irradiance curve (Lewis and Smith, 1983),

$$P(I_z) = P_{max} \frac{\alpha_{P_I} \cdot I_z}{\sqrt{P_{max}^2 + (\alpha_{P_I} I_z)^2}} , \quad (\text{S11a})$$

through depth z (Eq. 12a). In this equation, α_{P_I} is the initial slope of the curve, P_{max} is the maximum photosynthetic rate, and I_z is a function estimating the level of irradiance, I , at depth z , based on the Beer-Lambert’s law:

$$I_z = I_0 \cdot e^{-K_{par} \cdot z} . \quad (\text{S11b})$$

84 K_{par} is the light attenuation coefficient due to water, and I_0 is the irradiance value at surface, i.e., the
 85 observed PAR prescribed as forcing to the model. To optimize numerical performance, $H(I_z)$ is
 86 computed based on an analytical approximation proposed by Anderson et al. (2015):

87

$$H(I_z) = \frac{P_{max}}{K_{par}MLD} \log \left(\frac{\alpha_{P_I} I_0 + \sqrt{P_{max}^2 + (\alpha_{P_I} I_0)^2}}{\alpha_{P_I} I_z + \sqrt{P_{max}^2 + (\alpha_{P_I} I_z)^2}} \right). \quad (S11c)$$

88

89 Phytoplankton is lost due to natural mortality, $\mathbb{D}_A$, mixing, $\mathbb{D}$, and zooplankton grazing $\sum_i \sum_j G_{ij} Z_j$.

90

91 The temporal dynamics for the j^{th} zooplankton size class (Eq. 6) is regulated by net grazing, $\mathbb{D}$
 92 $\sum_i \sum_j G_{ij} Z_j$, which is the portion of grazed phytoplankton successfully ingested and assimilated, and
 93 several loss processes. The grazing rate of zooplankton is determined by,

94

$$G_{ij} = \psi_{maxj}(S_j^Z) \cdot \psi(T) \cdot \left(\frac{\mathbb{D}_{ij}(S_i^A, S_j^Z) A_i}{K_A + \sum_i \mathbb{D}_{ij}(S_i^A, S_j^Z) A_i} \right). \quad (S12)$$

95

96 The first factor describes the size-dependent maximum ingestion rate:

97

$$\psi_{maxj}(S_j^Z) = \beta_{\psi_{max}} \cdot S_j^{Z\alpha_{\psi_{max}}}, \quad (S13)$$

98

99 with S_j^Z representing the size of the j^{th} zooplankton size class (Fig. 2C). The second factor, $\psi(T)$,
 100 describes the temperature dependence of grazing and is presented in detail in the main article, section
 101 ‘Temperature dependences’ (Eq. 13A). The third factor of the grazing formulation comprises a half-
 102 saturation constant for phytoplankton intake, K_A , and the grazing preference for a specific range of
 103 phytoplankton size classes,

104

$$\mathbb{D}_{ij}(S_i^A, S_j^Z) = e^{-\left[\frac{\log_{10}(S_i^A) - \log_{10}(P_{optj}(S_j^Z))}{\mathbb{D}_j} \right]^2}. \quad (S14)$$

105

106 The grazing preference is characterized by a Gaussian curve over a specific range of phytoplankton
 107 size classes. The median and the width of this Gaussian are determined by, respectively, the optimal

108 phytoplankton size, $A_{optj}(S_j^Z)$, and the level of selectivity of the grazer θ_j (Fig. 2B). Grazers of
 109 different sizes are characterized by different optimal phytoplankton sizes following an allometric
 110 scaling as:

$$111 \quad A_{optj}(S_j^Z) = \beta_{P_{opt}} \cdot S_j^{Z^{\alpha_{A_{opt}}}}. \quad (S15)$$

112
 113 This function is empirically derived from the data of Hansen et al. (1997), reflecting the fact that large
 114 zooplankton prefer large phytoplankton cells (Fig. 2D).

115
 116 The loss processes for zooplankton include natural mortality, μ_Z , mixing, μ_Z , and higher order
 117 predation, $\eta_Z Z_j$ (Eq. 6). The portion of grazed phytoplankton that is not ingested due to sloppy feeding,
 118 $(1-\mu)\sum_i \sum_j G_{ij} Z_j$, flows into the detritus pool (Eq. 7). The detritus pool acts as a nutrient recycling
 119 station by remineralizing dead plankton and non-ingested materials from grazing. The portion that is
 120 ingested but not assimilated, $(1-\mu)\sum_i \sum_j G_{ij} Z_j$, is excreted and quickly remineralized into the
 121 nutrient pool (Eq. 4).

122

123

124

125

Supporting Information S2 – Environmental forcing

As forcing to the model, we considered three environmental variables that are specific to Greifensee, namely water temperature, photosynthetically active radiance (PAR), and mixed layer depth (MLD).

129

Data source

These forcing functions are derived from various time-series data, and the data cover different time periods. Water temperatures are obtained from the CTD profiles as daily average epilimnetic temperature from 2019 to 2022, measured by the Swiss Federal Institute of Aquatic Science and Technology, Eawag. The PAR data is converted from solar radiance data collected from a weather station next to Greifensee, measured daily by MeteoSwiss for the same period as the water temperature. The PAR conversion is presented in the next paragraph. Finally, we obtained MLD data from Pomati et al. (2020). These are monthly data collected at Greifensee from 1972 to 2015 (n=519). To match the time step of the model (one day), we obtained daily values by interpolating the monthly averages for MLD (see below).

140

Unit conversion of PAR

PAR values for plankton models are often obtained from light measurements, and light can be expressed as photon flux density (i.e. $\text{Ein m}^{-2} \text{d}^{-1}$ / $\text{mol quanta m}^{-2} \text{s}^{-1}$), or as energy flow density (i.e. W m^{-2}) (Lampert and Sommer, 2007). The photosynthetic processes of modelled phytoplankton are often expressed in $\text{Ein m}^{-2} \text{d}^{-1}$ or $\text{mol quanta m}^{-2} \text{s}^{-1}$, whereas solar irradiance is commonly measured in W m^{-2} . First, we calculate the amount of PAR from the solar irradiance in terms of net short-wave solar irradiance, because not all wavelengths of the light spectrum are usable by photosynthesis. Only wavelengths in the range 0.4-0.7 μm are suitable for most phytoplankton to perform photosynthesis, and PAR is often considered in the range 0.41 to 0.50 of the net short-wave solar irradiance (Howell et al. 1983; Tsubo and Walker 2005). The specific value depends on climatic conditions and differs geographically. We adopt an intermediate value of 0.45 (Table S1, S2). After obtaining PAR in units of W m^{-2} , we transformed these units into $\text{Ein m}^{-2} \text{d}^{-1}$. From Lampert and Sommer (2007), an approximated conversion from W m^{-2} to $\mu\text{Ein m}^{-2} \text{s}^{-1}$ to ranges from 4 to 5. We adopt the value of 5 (Table S1, S2).

155

Derivations of forcing function

157 Based on the monthly values of the three sets of environmental data, we derive corresponding annual
158 forcing functions for water temperature, PAR, and MLD (Fig. 3). The water temperature is obtained
159 from CTD profiles as daily average epilimnetic temperatures from March 2019 to December 2022.
160 After obtaining monthly means from the daily averages over a one-year period, we interpolated the
161 monthly values to derive the forcing function for water temperature. The PAR data exhibits a wide
162 range of variability in the annual composite (Fig. 3), a common feature for irradiance data due to high
163 inter-annual or daily variabilities driven by cloud cover. We obtained the monthly means based on
164 the 75-percentile of the daily averages over a one-year period and interpolated the monthly values to
165 derive the forcing function for PAR (Fig. 3). The MLD data are retrieved from a monthly time-series
166 over 40 years (1972-2015) from Pomati et al. (2020). We manually derived twelve monthly data
167 points that capture the changes in the mixed layer depth and interpolate the monthly data into daily
168 values. The interpolations of the environmental data were performed using piecewise polynomials
169 (i.e. Spline) in the *scipy* package (Python 3.7.x) (Virtanen et al. 2020). The code for generating the
170 function is available on Github (<https://github.com/systemsecologygroup/TempSizeMod>). The
171 environmental data points and the generated forcing functions used as input to the model are shown
172 in Fig. 3.

177 **Supporting Information S3 – Conversions of observations and model results**

178 We used two raw plankton size datasets that are obtained from a field-deployed imaging device dated
179 from March 2019 to December 2022 (available in Supplementary Data,
180 doi:10.5281/zenodo.11390345). The plankton data for Greifensee, collected by the dual-
181 magnification dark-field imaging microscope (Orenstein et al. 2020), contain plankton abundance in
182 terms of ROI/second (Regions Of Interest per second) and plankton size estimates in terms of bio-
183 area (mm²), minimum axis length (mm), and maximum axis length (mm). ROI/sec is a pseudo-unit
184 derived from the statistical output of each detected and processed object by the imaging microscope
185 that converts ROIs into image mosaics. In general, this unit is analogous to object counts for both
186 unicellular and colonial phytoplankton and zooplankton individuals (Merz et al. 2021). To be
187 comparable with model outputs, these data were converted into abundances (individual L⁻¹) based on
188 laboratory calibration functions (Merz et al. 2021). After that, the abundances were converted into
189 community biomass concentration (μM P), based on the size data of phytoplankton and zooplankton.
190 These conversions are explained in the following sub-sections and detailed equations are given in
191 Table S1.

192
193 **S2.1 Plankton abundances**

194 To obtain plankton abundances, we adopted the two calibrated functions derived by Merz et al. (2021)
195 and presented in Fig. S1. One equation was derived for the magnifier 5p0x (for smaller objects,
196 between 10 and 150 μm), whereas another was for the magnifier 0p5x (for larger objects, between
197 100 μm and 1 cm). We applied the equations to the plankton data according to the magnifier that
198 produced that data (Table S1). The phytoplankton species identified by the 5p0x magnifier are centric
199 diatom, cryptophyceae, rhodomonas, and chlorophytes, whereas those identified by the 0p5x
200 magnifier are dinobryon and uroglena. The zooplankton species captured from 5p0x and 0p5x are,
201 respectively, ciliates and daphnia.

202

203

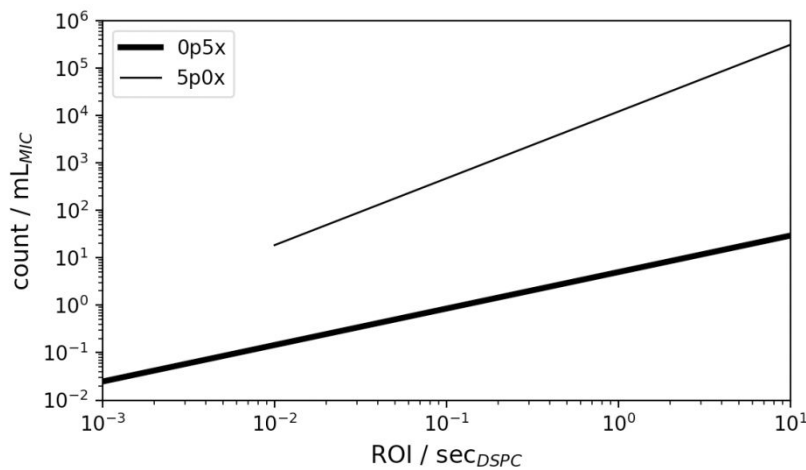


Fig. S1 Laboratory calibration of density estimated by microscopic count (count/mL) against density estimated by the plankton camera (ROI/sec). The functions are replotted from Merz et al. (2021): 5p0x ($r^2 = 0.889$) and 0p5x ($r^2 = 0.726$), Table S1.

S2.2 Phosphorus-equivalent phytoplankton biovolume

We transformed the bio-area (mm^2) estimates of phytoplankton to biovolume based on the relationship established by Ryabov et al. (2021), followed by calculating the phosphorus content using an allometric scaling function (Fig. S2) established by Hillebrand et al. (2022). The equations are shown in Table S1.

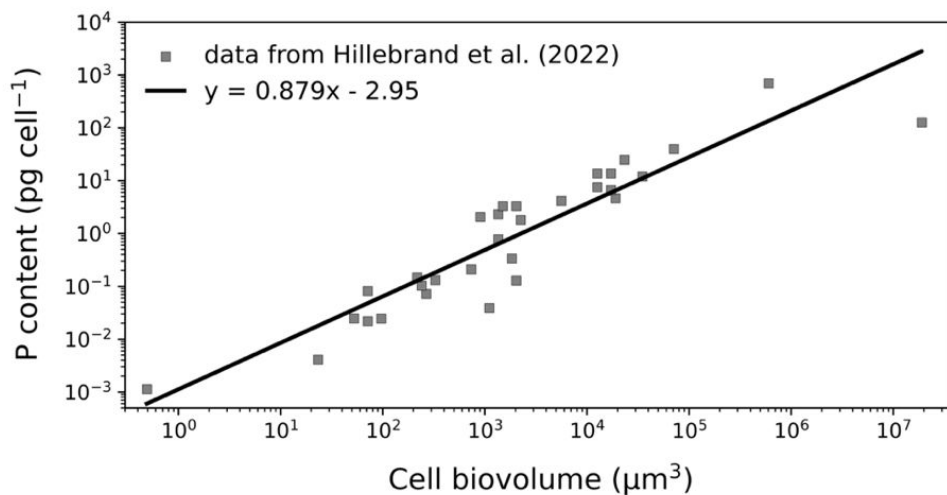


Fig. S2 The scaling function of phosphate content per cell in relation to phytoplankton biovolume (μm^3). The data (grey squares) are from Hillebrand et al. (2022) and the function (solid dark line) is a linear regression on the original data ($r^2 = 0.924$).

S2.3 Phosphorus-equivalent zooplankton biovolume

S2.3.1 Ciliates and Daphnia

222 The biovolume of ciliates was approximated assuming that the general shape of ciliates is a standard
223 ellipsoid, following Hillebrand et al. (1999). We use the major and minor axis length data of the
224 ciliates to represent the principal diameters of the ellipsoidal ciliate and assume that the third
225 dimension of the ellipsoidal ciliate is equal to the minor axis length (Table S1). We obtain the carbon
226 content per ciliate individual in relation to size according to the size-scaling relationship of Putt and
227 Stoecker (1989) and the stoichiometric transformation of carbon to phosphorus is based on a range
228 within 80-120 measured for a freshwater ciliates *Euplotes sp.* (Golz et al. 2015). We used the major
229 axis length data of daphnia to estimate their carbon content based on a body length to weight
230 relationship in Schalau et al. (2008) and considered a correction factor of 0.75 from the major axis
231 length to body length. The stoichiometric conversion of carbon to phosphorus is based on a ranges
232 from 80 to 385 in previous modelling studies (reviewed in Khattak et al. 2018), Table S1. The detailed
233 conversion is shown in Table S1, and the values of the conversion coefficients are given in Table S2.
234 Those values were manually optimized by comparing and matching the plankton abundances
235 obtained from the observations with the abundances produced by the model.
236

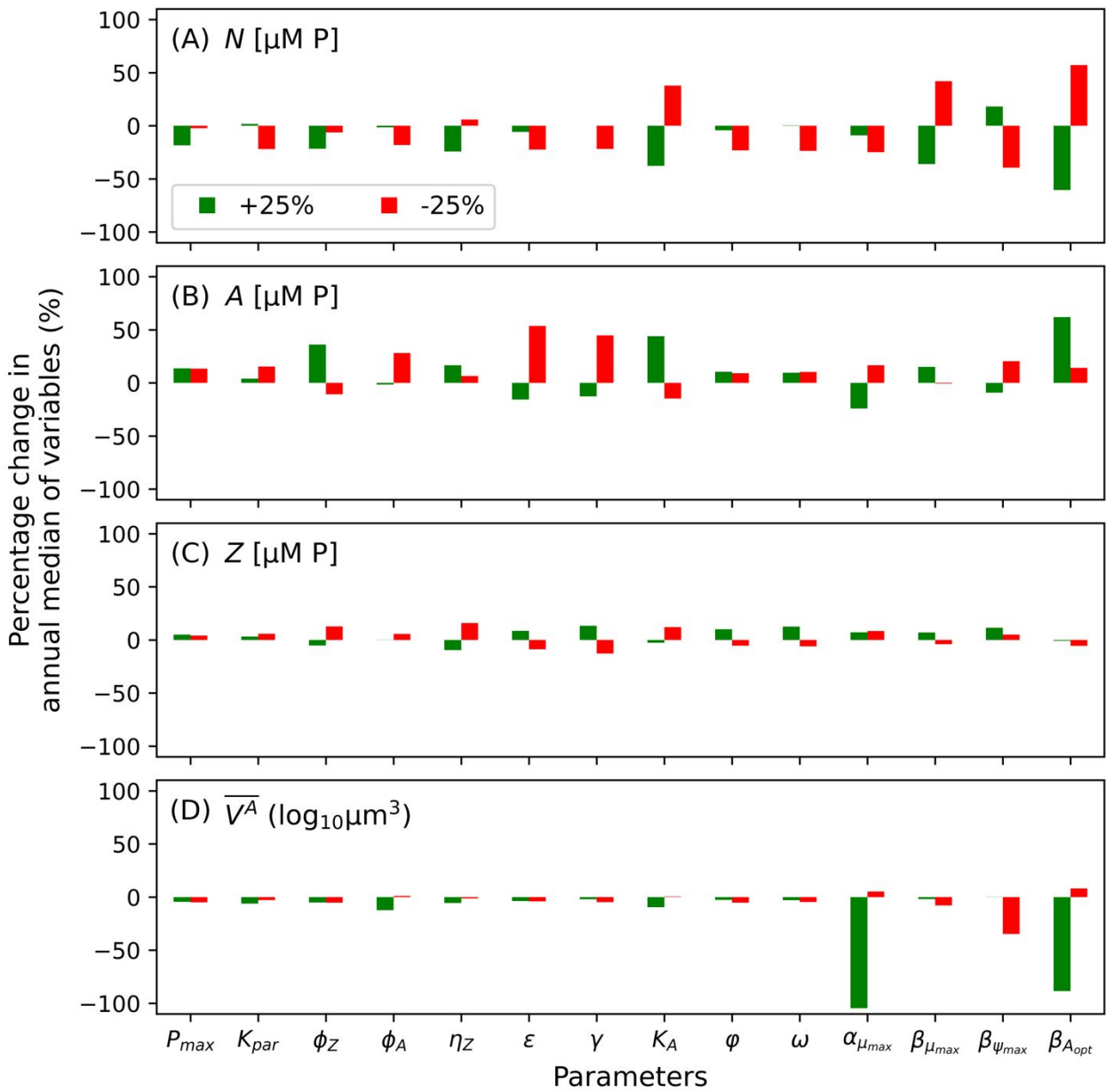
237 **Supporting Information S4 – Sensitivity analysis on standard model parameters**

238 We conducted a sensitivity analysis to evaluate the uncertainty of those parameters that could not be
 239 directly estimated from Greifensee data (Table 1). A total of 14 parameters were tested, except the
 240 thermal traits that were tested in a different sensitivity analysis (Fig. 4) and the parameters that were
 241 already calibrated by matching model results to observations (Fig. S5). We varied the 14 parameters
 242 by $\pm 25\%$ from standard value (Table 1) and quantified the percentage change on the annual median
 243 for four model variables, nutrient concentrations, community-integrated phytoplankton and
 244 zooplankton biomass, and community-mean phytoplankton biovolume. This sensitivity test is
 245 simulated for the standard model under the default physical forcing.

246
 247 The result shows that nutrient concentration and phytoplankton biomass are generally
 248 sensitive to the parameter tested (Fig. S3A, B). Zooplankton biomass and community-mean
 249 phytoplankton biovolume are generally not very sensitive to the parameter tested (Fig. S3C, D).
 250 Nutrient concentrations are most sensitive (affect nutrient values by max. $\pm 60\%$) to the slope and
 251 intercept of allometric relationships for phytoplankton growth ($\alpha_{\mu_{max}}$, $\beta_{\mu_{max}}$), zooplankton grazing
 252 ($\beta_{\psi_{max}}$), and optimal phytoplankton size for zooplankton ($\beta_{A_{opt}}$), Fig. S3A. This suggests that
 253 nutrient concentration is sensitive to growth and grazing parameters that vary with size-dependencies.
 254 Phytoplankton concentrations are more sensitive (altered by max. $\pm 60\%$) to parameters that control
 255 grazing, including zooplankton mortality ($\square_Z$), fraction of grazed material that is not ingested, i.e.,
 256 sloppy feeding (ϵ), fraction of grazed material that is assimilated into zooplankton growth (γ), half-
 257 saturation constant for grazing (K_A), and the intercept of allometric relationship of optimal
 258 phytoplankton size for zooplankton grazing ($\beta_{A_{opt}}$), Fig. S3B. Other parameters produced weaker
 259 effects to the annual median value of nutrient and phytoplankton biomass, i.e., within $\pm 30\%$ (Table
 260 S4).

261
 262 Our result shows moderate changes in zooplankton (within about $\pm 15\%$), Table S4. The
 263 community-mean phytoplankton biovolume is insensitive to most parameters (within $\pm 10\%$), except
 264 to the $+25\%$ slope of allometric relationships for maximum phytoplankton growth ($\alpha_{\mu_{max}}$) and the
 265 $+25\%$ intercept of allometric relationships of optimal phytoplankton size for zooplankton grazing
 266 ($\beta_{A_{opt}}$), Fig. S3D. Increasing these two parameters favours small phytoplankton by decreasing the
 267 growth ability of larger phytoplankton and by weakening grazing on small phytoplankton (grazing

268 shifts to target larger phytoplankton). Both parameters produce a strong decline in mean
269 phytoplankton biovolume (Fig. S3D).
270
271



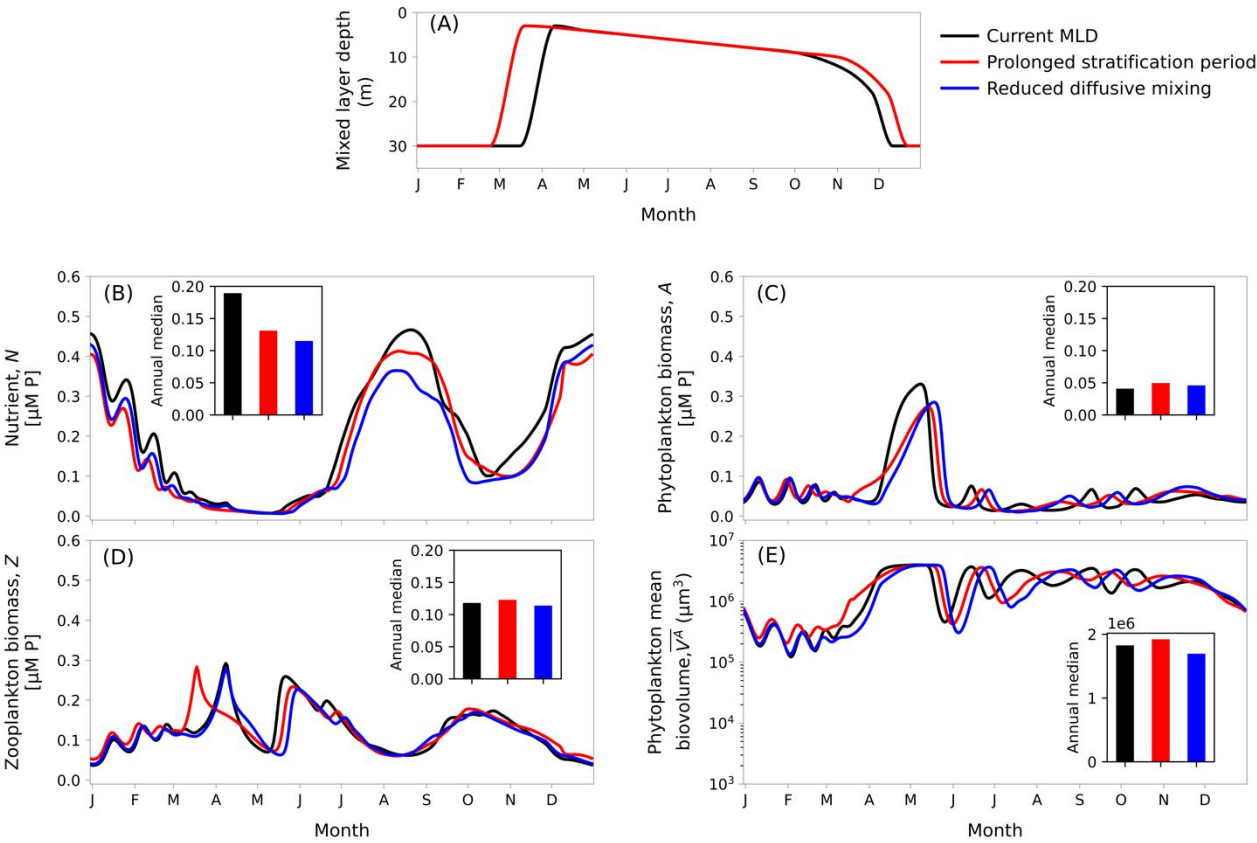
272
273 **Fig. S3** Results for the sensitivity analysis on fourteen model parameters. These are
274 parameters that were neither examined in sensitivity analyses nor calibrated with observations. The
275 result shows changes in annual median of four model variables, (A) nutrient concentration in the
276 epilimnion, (B) phytoplankton biomass, (C) zooplankton biomass, and (D) phytoplankton mean
277 biovolume, by +25% (green bars) and -25% (red bars) defaulted parameter values (Table 1). The
278 percentage change values are reported in Table S4.

Supporting Information S5 – Sensitivity test on alterations of lake stratification

In a warming world, higher temperature and longer stratification periods are factors (Woolway et al. 2020, 2021) that can affect nutrient transport (Behrenfeld et al. 2006), plankton biomass, and plankton phenology (Winder and Sommer 2012). Earlier onsets of stratification and longer periods of stratification are expected to occur in multiple temperate lakes across the Northern Hemisphere (Winder and Schindler 2004; Shatwell et al. 2019; Woolway et al. 2021).

Since our ecological projection did not consider any changes in the lake stratification, we performed a sensitivity test varying the lake mixing dynamics under the most severe climate change scenario, RCP 8.5. Lake stratification in our model is governed by two features, variations of the mixed layer depth (MLD) in combination with a cross-thermocline diffusive mixing, captured by a constant parameter (ω). Based on the estimation of the median change in stratification period by Woolway et al. (2021), we created a prolonged lake stratification that starts 22 days earlier and ends 11 days later (Fig. S4A). This leads to a longer summer stratification, by 33 days. In another case, we reduced the diffusive mixing coefficient by 50%. This parameter determines the strength of diffusive mixing between hypolimnion and epilimnion.

The model output shows that, these variations produce effects to all variables. The more prominent effects include a lower level of nutrient concentration throughout the year (Fig. S4B) and an earlier zooplankton spring peak under a prolonged stratification (Fig. S4D). A weaker diffusive mixing produces a lower level of nutrient accumulation (30% reduction in annual median) in summer compared to the standard mixing dynamics (Fig. S4B, inset). This is likely due to the reduced amount of nutrient recharged from the hypolimnion. Additionally, in our model, an earlier lake stratification leads to earlier development of zooplankton peak in spring. That is because the zooplankton peak is sensitive to the start of stratification (since the shallowing of mixed layer depth increases biomass concentration, Eq. S1), but they are not affected by changes in diffusive mixing. The alteration in lake stratification produces only minor changes to the biomass dynamics and community mean biovolume of phytoplankton (Fig. S4C, E).



309

310 **Fig. S4** Results of sensitivity test on alterations of lake stratification compared to the standard
311 mixing regime under the most severe climate change scenario (RCP 8.5, water temperature increases
312 by 4.2 °C throughout the year). (A) Standard mixed layer depth (black), future mixed layer depth
313 produced by (1) a prolonged stratification period (red) and (2) a weakened diffusive mixing between
314 hypolimnion and epilimnion (blue). The results show, for the three mixing regimes, temporal
315 dynamics of (B) nutrient concentration in the epilimnion, (C) phytoplankton biomass, (D)
316 zooplankton biomass, and (E) phytoplankton mean biovolume. The insets show the annual median
317 value for different alterations. The model is run to a steady annual cycle over a period of 10 years
318 with repeated annual forcing, the results shown are those of the 10th year.

319

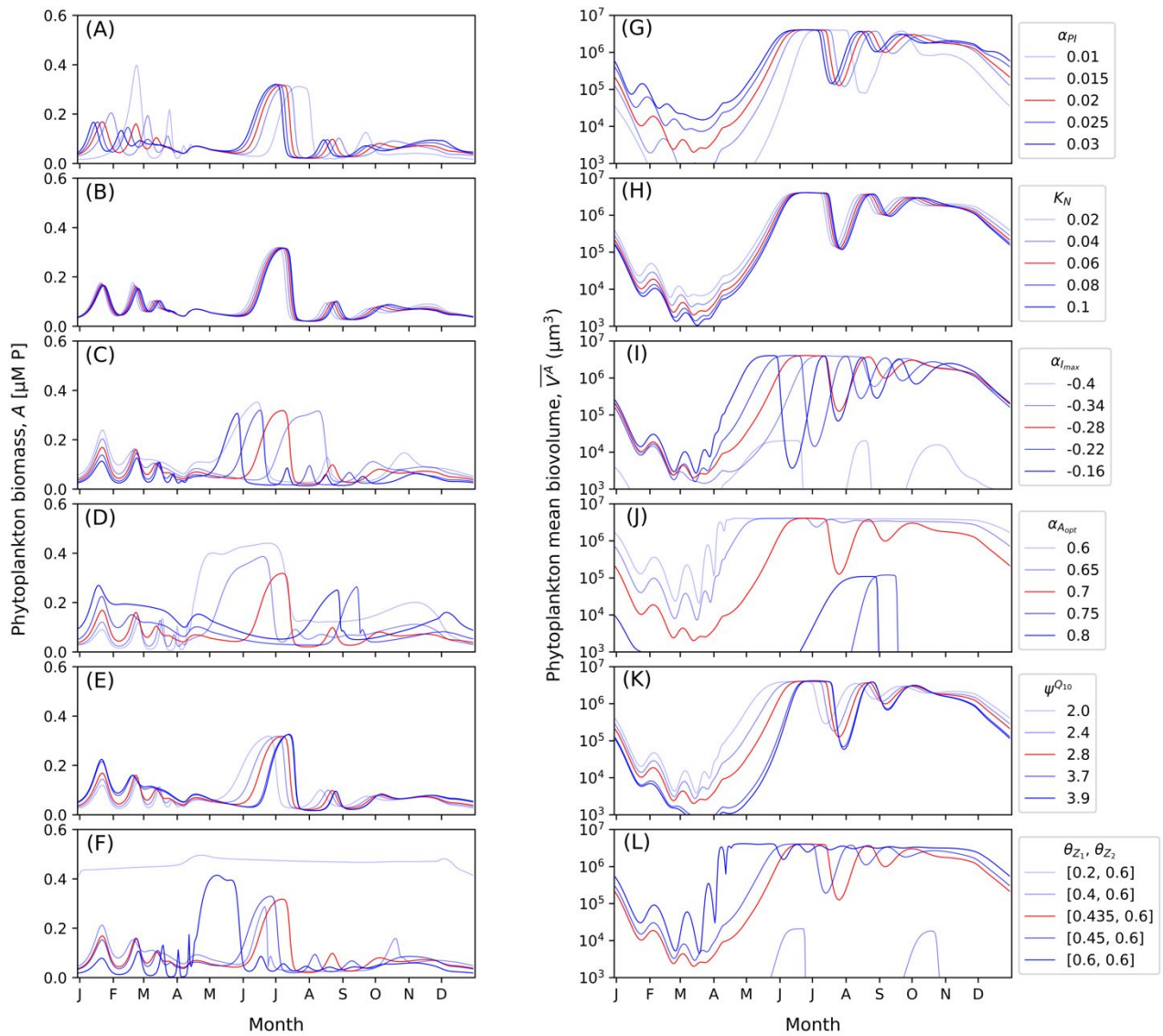
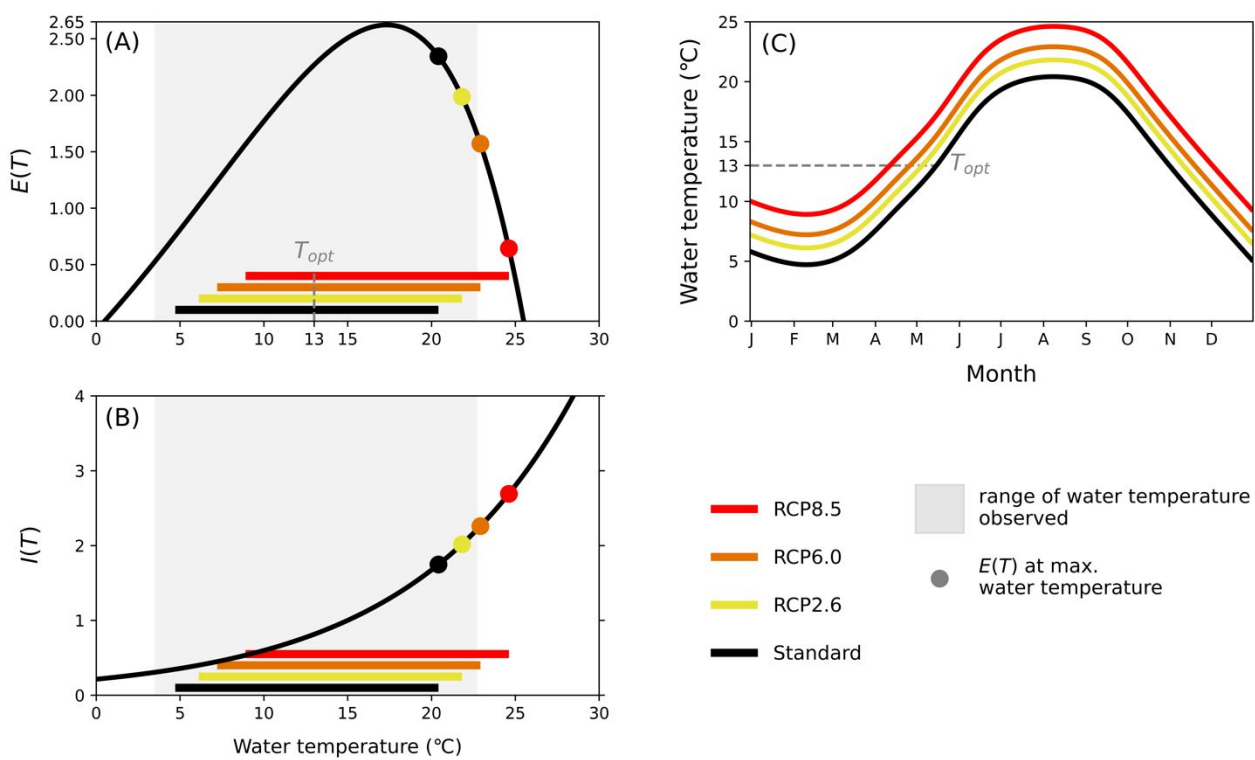


Fig. S5 Selected results for the calibrated parameters. (A, G) initial slope of P-I curve, α_{PI} ; (B, H) half saturation constant for nutrient, K_N ; (C, I) slope of the allometric relationship for maximum ingestion rate, $\alpha_{I_{max}}$; (D, J) slope of the allometric relationship for optimal phytoplankton size, $\alpha_{A_{opt}}$; (E, K) Q_{10} coefficient for maximum ingestion rate, $\psi^{Q_{10}}$; and (F, L) size-grazing breadth for Z_1 and Z_2 , $\theta_j^{Z_1}$ & $\theta_j^{Z_2}$. The results for (A-E) phytoplankton biomass concentrations, A , and (F-J) phytoplankton community mean biovolume, $\bar{V}^A$. The red line indicates the value used for the standard model run.



332

333 **Fig. S6** Temperature dependence curves for (A) phytoplankton growth, $E(T)$, and (B)
334 zooplankton grazing, $I(T)$, for different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0
335 (orange), and RCP 8.5 (red). (C) Optimal temperature for growth, T_{opt} . The standard range of water
336 temperature is represented by the black line. The climate change scenarios simulate an increase in the
337 standard range of water temperature by, respectively, 1.4 °C (RCP 2.6, yellow bar), 2.5 °C (RCP 6.0,
338 orange bar), and 4.2 °C (RCP 8.5, red bar). The circles (A, B) indicate the level of temperature
339 dependence at the hottest day of the year from different scenarios and show that phytoplankton growth
340 is becoming increasingly suboptimal while zooplankton grazing intensifies.

341

342

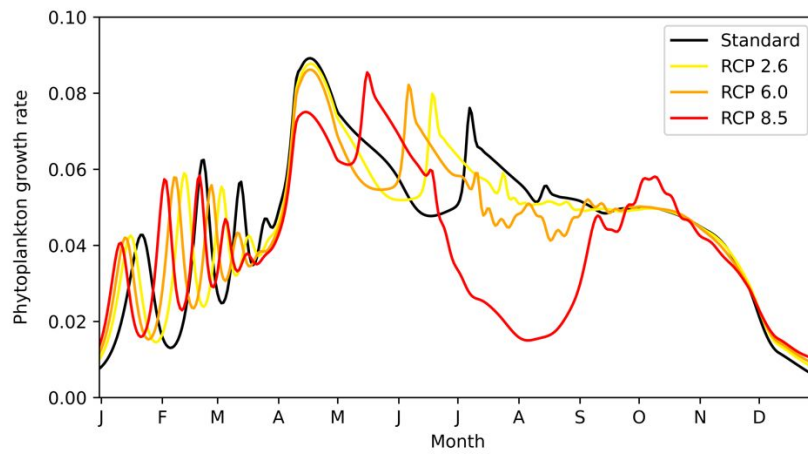


Fig. S7 Phytoplankton growth rates, $\sum \mu_i$ (Eq. S8), in different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0 (orange), and RCP 8.5 (red). It shows a pronounced decline in summer phytoplankton growth under RCP 8.5.

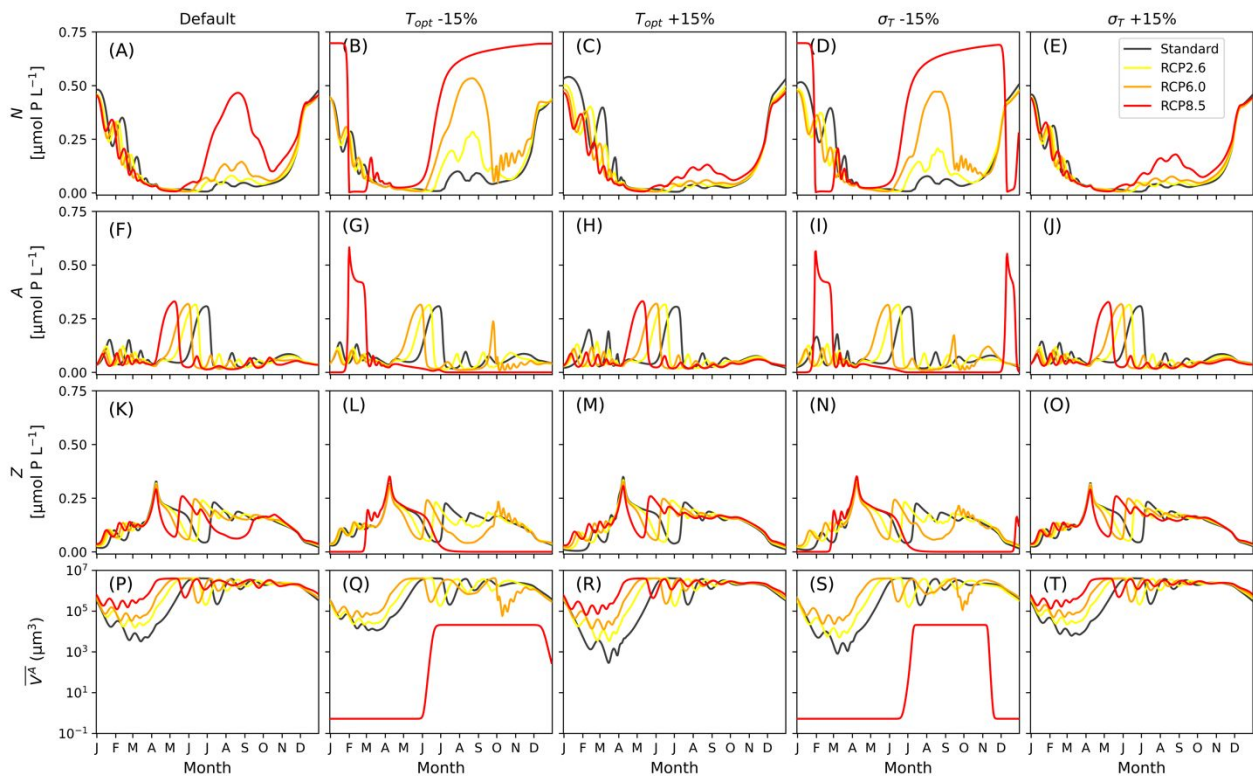


Fig. S8 Results of the sensitivity analysis ($\pm 15\%$) related to the two thermal traits, T_{opt} and σ_T . The result shows the temporal dynamics of four model variables: nutrient concentration, N (A-E); phytoplankton biomass concentration, A (F-J); zooplankton biomass concentration, Z (K-O); and phytoplankton community mean biovolume, $\bar{V}^A$ (P-T), under the different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0 (orange), and RCP 8.5 (red).

For Review Only

Table S1 Transformation of model and data output with physical, stoichiometric, and size-scaling coefficient. The references for the equations are provided below the equations and those for specific coefficient are given in Table S2

		Solar irradiance (W m ⁻²)	Abundance data		Phytoplankton size		Ciliates Minimum and maximum axis length (mm)	Ciliates carbon content (pg C ind ⁻¹)	Daphnia carbon content (µg C ind ⁻¹)
			ROI/sec (<100 µm)	ROI/sec (>100 µm)	Bio-area, A (µm ²)	Modeled ESD (µm)			
Physical	Net Short-wave solar irradiance i.e. light usable for photo- synthesis (W m ⁻²)	0.45 (Table S2)							
	Photosynthetically Active radiance (E m ⁻² d ⁻¹)	$\frac{5 \cdot 86400}{10^6}$ (Table S2)							
Abundance	Phytoplankton (ind L ⁻¹)		$10^{\frac{2.9}{0.71}} \cdot (10^{\text{ROI/sec}})^{\frac{1}{0.71}}$ Merz et al. (2021)						
	Zooplankton (ind L ⁻¹)		$10^{\frac{0.91}{1.3}} \cdot (10^{\text{ROI/sec}})^{\frac{1}{1.3}}$ Merz et al. (2021)						
Phytoplankton	Biovolume, V (µm ³)				$A^{\frac{1}{(0.68 \cdot 0.25)}}$ Ryabov et al. (2021)	$\frac{4}{3} \cdot \pi \cdot \left(\frac{ESD}{2}\right)^3$ Hillebrand et al. (1999)			
Zooplankton	Ciliates biovolume, V (µm ²)						$\frac{\pi}{6} \cdot a \cdot b \cdot h \cdot 10^3$ Hillebrand et al. (1999)	$10^{0.12} \cdot V^{0.19}$ Putt and Stoecker (1989)	
	Daphnia maximum axis length, L (mm)								$1.6 \cdot L^3$ Schalau et al. (2008)
Stoichiometry	Phosphate content per individual cell (pg P ind ⁻¹)				$10^{-2.95} \cdot V^{0.88}$ (Hillebrand et al., 2022)				
	Phosphorus- equivalent biomass concentration (µmol P ind ⁻¹)				$\frac{1}{\text{mw P}} \cdot 10^{-6}$ (Table S2)			$\frac{1}{\text{mw C}} \cdot \frac{1}{\text{C:P}_{\text{Ci}}} \cdot 10^{-6}$ (Table S2)	$\frac{1}{\text{mw C}} \cdot \frac{1}{\text{C:P}_{\text{Da}}}$ (Table S2)

359 **Table S2** Parameters for physical and stoichiometric conversion and their descriptions, standard values,
360 parameter range, units, and sources for the parameter or parameter range. Those that are provided with a range
361 was calibrated by manually comparing the range of biomass values obtained from the data with those from the
362 model.

Parameter	Description	Standard Value	Parameter range	Units	Source for parameter
-	Solar irradiance to net short-wave solar irradiance	0.45	0.41-0.5	W m ⁻²	Howell et al. (1983); Tsubo and Walker (2005)
-	Conversion of energy flow density to photon flux density	5	4-5	μE m ⁻² s ⁻¹ (mol P) ⁻¹	Lampert and Sommer (2007)
mw P	Molar weight of phosphorus	31.97	-	μg P	Standard unit
mw C	Molar weight of carbon	12.01	-	μg C	Standard unit
C:P _{Ci}	C:P ratio for ciliates	100	80-120	mol C (mol P) ⁻¹	Golz et al. (2015)
C:P _{Da}	C:P ratio for daphnia	350	80-385	mol C (mol P) ⁻¹	Khattak et al. (2018)

363

364

365

Table S3 Information for the increase in lake water surface temperature (LWST) for future projections (i.e. year 2099 or 2100) and the corresponding sources.

RCP scenarios	Value used for projections	Range of changes in LWST (°C)	Location	References
RCP 2.6	+1.4°C	1.36	Lake Erken, Sweden	Ayala et al. (2020) ¹
(low-emission scenario)		1.4	10 Polish lakes	Czernecki and Ptak (2018) ²
		0.09 decade ⁻¹	Rappbode Reservoir, Germany	Mi et al. (2020) ³
		0.51 century ⁻¹	Lake Kozjak, Croatia	Šarović and Klaić (2023) ⁴
RCP 6.0	+2.5°C	2.31	Lake Erken, Sweden	Ayala et al. (2020)
(medium-high-emission scenario)		1.8 – 2.7	The Great Lake region	Xue et al. (2022) ⁵
		0.32 decade ⁻¹	Rappbode Reservoir, Germany	Mi et al. (2020)
		1.41 century ⁻¹	Lake Kozjak, Croatia	Šarović and Klaić (2023)
RCP 8.5	+4.2°C	4.2	10 Polish lakes	Czernecki and Ptak (2018)
(high-emission scenario)		4.1 – 5.0	The Great Lake region	Xue et al. (2022)
		0.5 decade ⁻¹	Rappbode Reservoir, Germany	Mi et al. (2020)
		4.51 century ⁻¹	Lake Kozjak, Croatia	Šarović and Klaić (2023)

¹ Ayala et al. (2020) adopts the one-dimensional hydrodynamic model, GOTM, to simulate the lake surface water temperature in future scenarios that are projected by the GCMs from ISIMIP scheme. Their long-term simulations were forced with daily or hourly meteorological data.

² Czernecki and Ptak (2017) simulates future LWST using empirical-statistical downscaling modelling techniques, based on simulations of 33 atmosphere-ocean coupled GCMs obtained from the CMIP5 project.

³ Mi et al. (2020) uses a well-developed hydrodynamic model and consider a set of reanalyzed daily meteorological input (bias-corrected for ISIMIP).

⁴ Šarović et al. (2023) uses a one-dimension energy budget model to simulate changes in mean lake water temperature, based on projected meteorological data (i.e. CORDEX) taken from the World Climate Research Programme.

⁵ Xue et al. (2022) couples a regional climate model with a three-dimensional hydrodynamic lake model, GLARM, using GCMs obtained from the CMIP5 project.

381 **Table S4** Percentage change on the annual median value of four model variables: nutrient concentration in
382 the epilimnion, N , phytoplankton biomass, P , zooplankton biomass, Z , and phytoplankton mean biovolume, $\overline{V^A}$,
383 under $\pm 25\%$ change in fourteen model parameters for the standard model run.

Sensitivity of variables	N [$\mu\text{M P}$]		P [$\mu\text{M P}$]		Z [$\mu\text{M P}$]		$\overline{V^A}$ ($\log_{10} \mu\text{m}^3$)	
% Change in parameter values	+25%	-25%	+25%	-25%	+25%	-25%	+25%	-25%
P_{max}	-18.31	-2.31	13.68	13.38	5.06	4.1	-4.48	-4.93
K_{par}	1.74	-21.85	4.05	15.44	3.15	5.81	-6.11	-2.85
ϕ_A	-21.68	-6.32	36.17	-10.73	-5.16	12.7	-5.12	-5.25
ϕ_Z	-1.53	-18.06	-1.45	28.14	0.18	5.61	-12.33	0.97
η_Z	-24.25	5.87	16.54	6.52	-9.53	15.89	-5.49	-1.28
ε	-5.83	-22.43	-15.71	53.66	8.6	-8.74	-3.68	-3.96
γ	0.11	-21.7	-12.7	44.72	13.35	-12.78	-1.98	-4.79
K_A	-37.68	37.81	43.86	-14.7	-2.56	12.19	-9.61	0.37
φ	-4.31	-23.1	10.41	9.03	10.06	-5.42	-2.6	-5.31
ω	0.35	-23.68	9.53	10.33	12.62	-5.99	-2.95	-4.62
$\alpha_{\mu_{max}}$	-9.03	-24.79	-24.05	16.65	7.25	8.42	-104.65	5.13
$\beta_{\mu_{max}}$	-36.06	42.03	15.11	-0.66	6.99	-4.0	-1.87	-7.88
$\beta_{\psi_{max}}$	18.1	-39.41	-9.16	20.39	11.55	5.1	0.09	-34.71
$\beta_{A_{opt}}$	-60.51	57.14	61.83	14.11	-0.77	-5.48	-88.48	8.12

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1 *Research article*

2 **Future inorganic nutrient and plankton dynamics in a temperate lake**

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For Review Only

Abstract

Lakes are experiencing high temperatures and frequent phytoplankton blooms. These events threaten the health of lake ecosystems. Phytoplankton respond to lake warming in various ways, making it challenging to predict changes in community composition. Although understanding the effects of lake warming on phytoplankton communities is crucial for lake management, these effects are not clear and have been a matter of debate ~~for the last two decades~~ due to contrasting observations ~~patterns shown by empirical data~~. Here we use a data-~~driven~~informed, trait-based model to disentangle and project, based on future warming scenarios, the effects of temperature on the dynamics of inorganic nutrients and ~~lake~~-plankton. ~~We~~Assuming identical thermal characteristics among phytoplankton size classes, we find that nutrient concentrations and phytoplankton mean cell size increase with rising lake temperature. This is driven by the interplay between reduced growth and nutrient uptake by all phytoplankton under sub-optimal temperatures, resulting in higher summer nutrients and intensified zooplankton grazing on smaller phytoplankton cells. The small phytoplankton typically occurring during winter and spring is replaced by large phytoplankton. In addition, peaks of phytoplankton biomass, typically occurring in early summer, emerge earlier, in early spring. The earlier occurrence of these peaks is caused by a shift in the optimal temperature for phytoplankton growth from spring to late winter. ~~We also find a higher than normal accumulation of nutrients in summer due to reduced phytoplankton growth under sub-optimal temperatures. Our results indicate that, in a high temperature world, small phytoplankton typically occurring during winter and spring is replaced by large phytoplankton. This shift in size composition is driven by the interplay between temperature-mediated size-dependent competition for nutrients by phytoplankton and size-dependent grazing by zooplankton. Our work~~ Although, as in all studies of this type, our results are based on various modelling approximations, they highlight~~s~~ the importance of combining observations and trait-based

47 modelling for ~~providing~~gaining insights into the future compositions of lake phytoplankton and
48 establishes a foundation for examining ~~ecosystem functions in temperate lakes~~lake ecosystems
49 in a warming world.

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For Review Only

Introduction

~~Lakes~~Most lakes on our planet are under threat by climate change and, specifically, by increasing temperatures (~~O'Reilly et al. 2015; Dokulil et al. 2021~~). (O'Reilly et al. 2015; Woolway et al. 2022). The increase in lake water surface temperature (~~LWST~~) ~~leads to different physical responses, including alterations in physical mixing and disruptions of water cycles (Winder and Schindler 2004; Woolway et al. 2020).~~ These environmental effects can cascade on lake communities, altering their dynamics and compositions. Consequences may include ~~the dampening of trophic transfer efficiencies, with serious impacts on the stability of,~~ hereafter water temperature, couples with changes in nutrient conditions in the lake to alter the interactions and compositions of lake communities. This may dampen trophic transfer efficiencies, with serious impacts on the stability and productivity of lake food webs (Strecker et al. 2004; Barneche et al. 2021). Increasing occurrences of anomalous algal blooms (dominated, for example, by cyanobacteria) are observed in eutrophic lakes worldwide, especially in connection to rising temperatures (~~Paerl and Huisman 2008; Moss 2011; Ho et al. 2019~~) (Paerl and Huisman 2008; Ho et al. 2019). ~~The increasing occurrence of these bloom events not only degrades water quality, but also reduces the biodiversity and productivity of lakes (Heino et al. 2009; Woolway et al. 2022).~~ Severe algal blooms can deplete oxygen levels in lakes, causing large-scale fish kills. These events pose a threat to inland fish catches in large lake ecosystems, which are important sources of proteins for a portion of human societies (~~Lynch et al. 2017~~). Hence, global warming is a growing concern for lake ecologists and Nevertheless, the mechanistic reasons leading to these blooms are still not clear. In a number of temperate lakes, contrary to expectations (e.g., Pomati et al. 2020), a clear multi-decadal warming trend and a reduction in nutrient levels were linked to the dominance of large phytoplankton cells. Hence, a better understanding of the thermal responses of plankton communities is key to managing lake ecosystems in a warming world.

The effect of temperature on lake phytoplankton communities can be of two types: (1) direct effects can alter metabolic processes of cells and (2) indirect effects can reshape the interactions of biotic and abiotic components (Zohary et al. 2021). Elevated temperatures can indirectly affect phytoplankton nutrient uptake by varying autotrophic metabolic rates and nutrient recycling by varying remineralization rates (Vanni 2002; Dory et al. 2024). ~~Also, higher temperatures lead to stronger lake stratification, which in turn has an impact on competition for nutrients and light with consequences on the composition and dynamics of phytoplankton communities (Tilzer and Goldman 1978; Donis et al. 2021).~~ Additionally, temperature affects ~~the sinking velocities of phytoplankton communities by varying water density and viscosity (Zohary et al. 2017) and by influencing zooplankton grazing (Sommer and Lewandowska 2011)~~ the composition of phytoplankton communities indirectly by enhancing zooplankton grazing, which can favour grazer-resistant phytoplankton species (Sommer and Lewandowska 2011; Lüring 2021). The synergistic ~~interactions of~~ direct and indirect effects of temperature ~~effects can~~ have complex repercussions on phytoplankton communities. These repercussions are frequently observed but they remain difficult to disentangle and predict.

Warming-induced changes in the seasonal succession of phytoplankton have been modelled and observed in ~~many~~ different European lakes over the last two decades (reviewed in Winder and Sommer 2012; Vadadi-Fülöp and Hufnagel 2014). ~~These phenological shifts produce impacts on~~ Phenological shifts in planktonic communities impact key lake ecosystem functions such as productivity, ~~stability~~, diversity, and trophic transfer efficiency (Strecker et al. 2004; Woolway et al. 2022). Minimal predator-prey models proposed that earlier onsets can be caused by warming-induced shifts or decoupling of plankton dynamics (Scheffer et al. 2001; De Senerpont Domis et al. 2007a). A competition model based on several phytoplankton species found that warming-induced shifts observed in phytoplankton succession were driven

by species-specific responses to temperature change (De Senerpont Domis et al. 2007b).

Another way temperature can influence key ecosystem functions is by altering the size composition of phytoplankton, often with opposing effects, especially in the context of temperate lakes. Some mesocosm experiments found that warming favors small phytoplankton cells (Yvon-Durocher et al. 2011; Rasconi et al. 2015), whereas others observed a shift towards large cells (Rüger and Sommer 2012; Yvon-Durocher et al. 2015). Long-term field studies also showed similar contrasting results: high water temperatures fostered both large (Pomati et al. 2020; Dory et al. 2024) and small phytoplankton (Zohary et al. 2017).

The major effects of warming on phytoplankton sizes appear to be driven by temperature-mediated trophic interactions and resource availability (Murphy et al. 2020; Zohary et al. 2021). However, these two drivers are confounded in natural environments. ~~Given these results, it remains essential to isolate and quantify the different effects of temperature if we want~~ and that limits our ability to make sound predictions on the compositions and dynamics of planktonic communities in lake ecosystems. Mathematical models are useful tools for disentangling the effects of temperature on lake plankton. ~~Minimal predator-prey models proposed that earlier onsets or disappearances of the clear-water phase can be caused by warming-induced shifts or decoupling of plankton dynamics (Scheffer et al. 2001; De Senerpont Domis et al. 2007a). A competition model based on several phytoplankton species found that warming-induced shifts observed in the community dynamics of phytoplankton were driven by species-specific responses to temperature changes (De Senerpont Domis et al. 2007b). A diatom-daphnia interaction model applied to a shallow lake suggested that changes in nutrient loading are crucial to understanding anomalies in the timing of phytoplankton blooms under future warming conditions (Huber et al. 2008).~~ Recent modelling investigations ~~on the effects of warming~~ included more comprehensive ecological interactions between plankton communities and the environment. By considering seasonal variations in light,

~~LWST~~[water temperature](#), mixed layer depth, and nutrients in a plankton ecosystem model, Cagle and Roelke (2021) and To et al. (2024) found that grazing characteristics of zooplankton regulate ~~the timing of spring blooms and~~ the succession of different phytoplankton species [and their community composition](#) in temperate lakes. However, ~~similar to that existing for marine zooplankton (Serra Pompei et al. 2020), a more flexible and detailed model framework is needed to examine the explicit changes in the size compositions of lake phytoplankton communities under increasing temperature. such modelling investigations have not been implemented in combination with a high degree of biotic and abiotic data inputs. A data-informed model that is verifiable by in-situ observations can strengthen our understanding on the effects that lake warming can have on composition of phytoplankton communities.~~

Building on our previous NPZD modelling study (To et al. 2024), we present here an updated data-informed trait-based model to explain plankton dynamics in a specific temperate lake and to explore lake ecosystem responses under different warming scenarios. [Greifensee, a medium-size eutrophic lake near Zurich \(Switzerland\), experienced multi-decadal warming and re-oligotrophic trends \(Pomati et al. 2020; Merz et al. 2023\). The composition of the phytoplankton community in the lake was found to be sensitive to long-term changes in thermal and nutrient conditions \(Bürge et al. 2003; Monchamp et al. 2017\), making Greifensee a natural laboratory for studying the impacts of changing temperature on abundance and community composition of lake plankton.](#) This new model [version](#) includes additional physiological details on the thermal dependence of the plankton communities (presented in Methods) and assumes that temperature ~~has a direct impact on~~[impacts](#) phytoplankton growth and zooplankton grazing through ~~data-driven allometric~~[data-derived](#) relationships. The physical processes in the model are driven by observed environmental data and the model outputs are calibrated against [averaged](#) nutrient concentration, phytoplankton and zooplankton biomasses, and size data ~~all collected at Greifensee, an eutrophic lake nearby Zurich (Switzerland).~~ The calibrated model

is then run into the future to predict the evolution of the ecosystem under projected climate change scenarios. We considered three Representative Concentration Pathways (RCPs), RCP 2.6, RCP 6.0, and RCP 8.5, reflecting low, medium, and high emission scenarios, respectively. We aim to understand the ~~direct~~ effect of increased ~~LWST~~ [water temperature](#) on ~~the~~ future nutrient concentrations, ~~the~~ plankton biomasses, ~~the~~ size composition of the phytoplankton community, and ~~the~~ timing of the peaks in phytoplankton biomass.

Methods

Size-based plankton ecosystem model

We consider a size-based plankton model that builds on a previous NPZD-based modelling study (~~e.g.~~ To et al. 2024) ~~and describes a realistic lake set-up~~ [but that captures now a real lake, Greifensee](#). The model consists of: (1) an inorganic nutrient, N (i.e., phosphate, PO_4^{3-}), (2) a community of phytoplankton A_i , subdivided into ten size classes ($i = 1, \dots, 10$), covering the phytoplankton size ~~distribution~~ [range](#) typically observed at Greifensee, (3) a community of herbivorous zooplankton Z_j subdivided into two size classes ($j = 1, 2$), and (4) a detritus pool, D (Fig. 1). The phytoplankton community exploits nutrients and light for growth, while water temperature affects the growth of phytoplankton and the grazing of zooplankton. ~~As~~ [Since](#) phosphate is [considered](#) the most limiting inorganic nutrient for most lakes (Schindler 1978), ~~thus~~ all state variables are expressed in units of $\mu\text{M P}$. [A detailed model description, including equations and formulation of processes, is reported in Supporting Information S1. In the following, we present the novel features of the model, which include the temperature dependence of phytoplankton growth and zooplankton grazing, the model calibration procedure, and the ecological projections under the different warming scenarios.](#)

Physical configuration

The lake ecosystem model operates in a zero-dimensional physical configuration that consists of two layers stacked on top of each other, reflecting the so-called slab physics approximation (e.g. Fasham et al. 1990; To et al. 2024), the upper epilimnion and the bottom hypolimnion (Fig. 1). The two layers are separated by a seasonally varying mixed layer depth (MLD), defined by the depth of the epilimnion. The epilimnion is biologically active and is assumed to be well-mixed so that all model properties, physical and biological, are considered homogeneously distributed within its volume. The hypolimnion is biologically inactive but rich in nutrients. In the hypolimnion, a fixed amount of nutrient, N_0 , replenishes the inorganic nutrient pool in the epilimnion, N , through physical mixing. Changes in motile (zooplankton) and non-motile (nutrient, detritus, and phytoplankton) model variables, induced by physical processes, can occur due to (1) exchange of fluxes driven by diffusive mixing between the epilimnion and hypolimnion and (2) variations in the MLD. The variation in zooplankton biomass due to changes in the MLD is described by the following term (Evans and Parslow 1985):

$$\lambda_z = \frac{h}{MLD(t)}, \quad (1)$$

with $h = \frac{dMLD(t)}{dt}$. Here we assume that zooplankton can maintain their position within the upper epilimnion. The shallowing or deepening of the MLD will, respectively, increase or decrease their concentrations. A shallowing MLD leaves the concentrations of non-motile variables (i.e. nutrient, phytoplankton, and detritus) unchanged because their reduction in number due to detrainment is compensated by the reduction in volume of the epilimnion. A deepening MLD reduces the concentration of the non-motile variables. The non-motile variables are also subject to diffusive mixing. The variations in biomasses of non-motile variables due to physical processes can be written as follows (Evans and Parslow 1985):

$$\lambda = \frac{\omega + h^+(t)}{MLD(t)}, \quad (2)$$

with ω being the diffusive mixing constant and

$$-h^+(t) = \max(0, h). \quad (3)$$

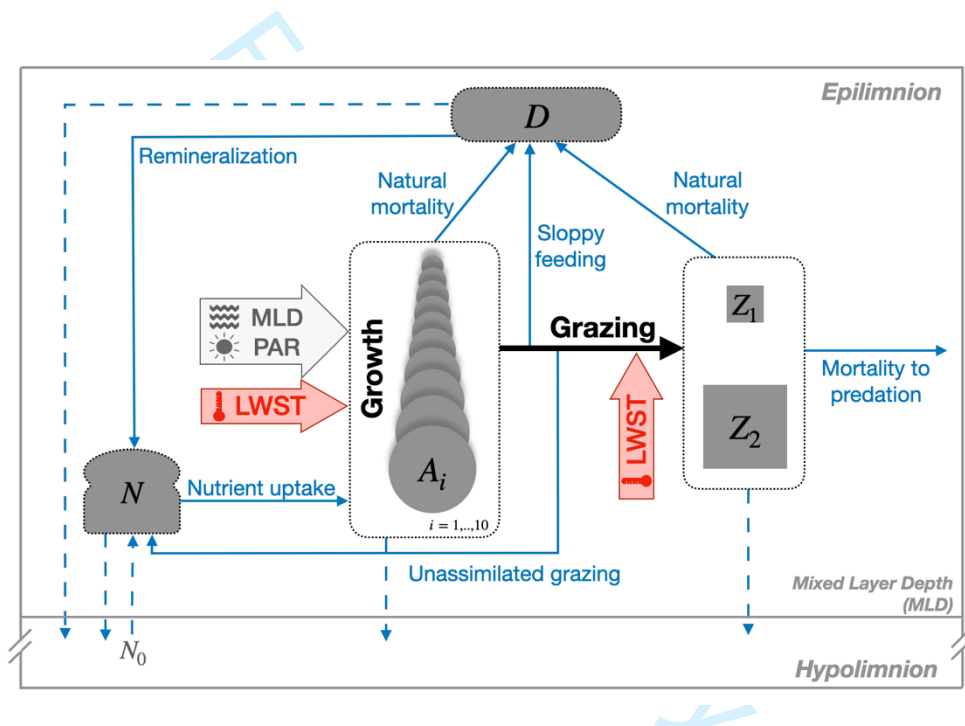


Fig. 1 Schematic of the size-based planktonic model. This model is based on our previous work (To et al. 2024). The new features of this model are highlighted in red. These are the temperature dependences of phytoplankton growth and zooplankton grazing, which are quantified through a unimodal curve and a Q_{10} exponential curve, respectively. The model assumes a zero-dimensional spatial configuration, the so-called slab physics approximation. The epilimnion contains a plankton ecosystem consisting of four components, inorganic nutrient, N , phytoplankton, A_i , zooplankton, Z_j , and detritus, D . Phytoplankton, A_i , is subdivided into i size classes (with $i = 1, \dots, 10$) and zooplankton, Z_j (with $j = 1, 2$), is subdivided into 2 size classes, Z_1 and Z_2 . Phytoplankton growth and zooplankton grazing are size-dependent. [The depth of the epilimnion reflects the Mixed Layer Depth \(MLD\).](#) The blue arrows indicate ecological and biogeochemical processes

(solid lines) and physical processes (dashed lines). Environmental forcing to this model is provided by in-situ observations of photosynthetically active radiations (PAR), mixed layer depth (MLD – an indicator of lake mixing), and ~~lake surface~~ water temperature (LWST) ~~from made at Greifensee, Switzerland.~~ To approximate real-world grazing, the grazed material is split into (1) a ~~portion~~ portion that is not ingested (due to sloppy feeding) and is immediately lost into detritus, (2) a portion that is ~~actually~~ ingested and assimilated by zooplankton, and (3) an additional portion that ~~after being ingested is not assimilated and quickly excreted into nutrient~~ is not assimilated after being ingested and is excreted and quickly remineralized into inorganic nutrients. The loss of zooplankton to higher order predation is density-dependent and formulated as a quadratic function. We assume that the loss of nutrients from the epilimnion is recharged through changes in MLD by a constant nutrient source, N_0 , from the hypolimnion. ~~N_0 is calibrated from observations and is also prescribed as forcing to the model.~~ We determined the value N_0 based on average winter observations, when the entire water column is well mixed and nutrient concentrations are homogeneously distributed throughout it. The detailed model configuration and all equations describing the processes shown in the schematic are reported in the Supporting Information S1.

Biological components

The model consists of a set of 14 ordinary differential equations:

$$\frac{dN}{dt} = - \sum_i \mu_i A_i + (1-\gamma) \varepsilon \sum_i \sum_j G_{ij} Z_j + \phi D + \lambda(N_0 - N) \quad (4)$$

$$\frac{dA_i}{dt} = (\mu_i A_i - \phi_A - \lambda) A_i - \sum_i \sum_j G_{ij} Z_j \quad (5)$$

$$\frac{dZ_j}{dt} = \left(c\gamma \sum_i \sum_j G_{ij} Z_j - \phi_Z - \lambda_Z - \eta_Z Z_j \right) Z_j \quad (6)$$

$$\frac{dD}{dt} = \sum_i \phi_p A_i + \sum_j \phi_z Z_j + (1 - e) \sum_i \sum_j G_{ij} Z_j - (\phi + \lambda) D \quad (7)$$

with $i = 1, \dots, 10$ and $j = 1, 2$. All the terms in the equations are described in Table 1 and in the following.

Table 1. Model equations

Description	Equations	Eq.
Phytoplankton growth	$\mu_i = \mu_{max_i}(S_i^A) \cdot E(T) \cdot NU(N) \cdot H(I_z)$	(8)
Phytoplankton maximum growth rate	$\mu_{max_i}(S_i^A) = \beta_{max} \cdot \left(\frac{S_i^A}{1 \mu\text{m}} \right)^{\alpha_{max}}$	(9)
Temperature dependence	$E(T) = e^{0.063T} \cdot \left[1 - \left(\frac{T - T_{opt}}{\sigma_T} \right)^2 \right]$	(10)
Nutrient uptake	$U(N) = \frac{N}{K_N + N}$	(11)
Light limitation	$H(I_z) = \frac{1}{MLD} \int_{z=0}^{z=MLD} P(I_z) dz$	(12)
	where $P(I_z) = P_{max} \frac{\alpha_{PF} I_z}{\sqrt{P_{max}^2 + (\alpha_{PF} I_z)^2}}$	(12a)
	and $I_z = I_0 \cdot e^{-K_{par} z}$	(12b)

Description	Equations	Eq.
	$H(I_z) = \frac{P_{max}}{K_{par}MLD} \log \left(\frac{\alpha_{p_i} I_0 + \sqrt{P_{max}^2 + (\alpha_{p_i} I_0)^2}}{-\alpha_{p_i} I_z + \sqrt{P_{max}^2 + (\alpha_{p_i} I_z)^2}} \right)$	(12e)
Gross Grazing	$G_{ij} = \left(I_{max_j}(S_j^Z) \cdot I(T) \right) \cdot \frac{\delta_{ij}(S_i^A, S_j^Z) A_i}{K_A + \sum_i \delta_{ij}(S_i^A, S_j^Z) A_i}$	(13)
	$\text{where } I(T) = I_{Q_{10}}^{\frac{T-Q_{10}}{10}}$	(13a)
Zooplankton maximum ingestion rate	$I_{max_j}(S_j^Z) = \beta_{I_{max}} \cdot S_j^{Z \alpha_{I_{max}}}$	(14)
Grazing preference	$\delta_{ij}(S_i^A, S_j^Z) = e^{\left[\frac{\log_{10}(S_i^A) - \log_{10}(P_{opt_j}(S_j^Z))}{\theta_j} \right]^2}$	(15)
Optimal phytoplankton size	$A_{opt_j}(S_j^Z) = \beta_{A_{opt}} \cdot S_j^{Z \alpha_{A_{opt}}}$	(16)

The dynamics of nutrient (Eq. 4) is driven by nutrient lost to phytoplankton growth, $\sum_i \mu_i A_i$, and nutrient replenishment from (1) the portion of grazed materials that is not assimilated by zooplankton, $(1-\gamma)\epsilon \sum_i \sum_j G_{ij} Z_j$, (2) the remineralization of detritus, ϕD , and (3) the exchange between the epilimnion and hypolimnion via vertical mixing, $\lambda(N_b - N)$, where N_b is the nutrient concentration in the hypolimnion calibrated from observations and prescribed as forcing to the model.

The temporal dynamics of the i^{th} phytoplankton size class (Eq. 5) is regulated by phytoplankton growth, $\mu_i A_i$, and by several loss processes. The growth rate of phytoplankton is regulated by a size-dependent maximum growth rate term, $\mu_{\max_i}(S_i^A)$, with S_i^A representing the size of the i^{th} phytoplankton size class (Eq. 9 and Fig. 2A), and three environmental factors: temperature T (representing LWST, °C), nutrient N , and light (PAR). The temperature-regulated growth term, $E(T)$ (Eq. 10), following Norberg (2004), an additional feature considered in our model, is elaborated in Section ‘Temperature dependences’. The nutrient uptake term follows a Monod function (Monod 1949), $U(N)$ (Eq. 11), with N representing inorganic nutrient concentration in the lake (Eq. 4) and K_N the half-saturation constant for nutrient. The light-limited growth term, $H(I_z)$ (Eq. 12), depends on the daily MLD-averaged photosynthesis rate and is calculated by integrating the photosynthesis irradiance curve, $P(I_z)$, of Lewis and Smith (1983) through depth z (Eq. 12a). In this equation, α_{P_i} is the initial slope of the curve, $P_{\max}$ is the maximum photosynthetic rate, and I_z is a function estimating the level of irradiance, I , at depth z , based on Beer-Lambert’s law (Eq. 12b). K_{PAR} is the light attenuation coefficient due to water clarity and phytoplankton self-shading, and I_0 is the irradiance value at surface i.e. the observed PAR prescribed as forcing to the model. To optimize numerical performance, $H(I_z)$ is computed based on an analytical approximation from Anderson et al. (2015) (Eq. 12c), which integrates light limitation with depth. Phytoplankton is lost due to natural mortality, ϕ_A , mixing, λ , and zooplankton grazing $\sum_i \sum_j G_{ij} Z_j$.

The temporal dynamics for the j^{th} zooplankton size class (Eq. 6) is regulated by net grazing, $c\gamma \sum_i \sum_j G_{ij} Z_j$, which is the portion of grazed phytoplankton successfully ingested and assimilated. The portion of grazed phytoplankton that is not ingested due to sloppy feeding, $(1-\gamma)c \sum_i \sum_j G_{ij} Z_j$, is assumed to flow into the detritus pool. The portion that is ingested but not assimilated, $(1-\gamma)c \sum_i \sum_j G_{ij} Z_j$, is excreted and quickly remineralized into the nutrient

pool. The grazing of zooplankton is determined by the size-dependent maximum ingestion rate, $I_{max,j}(S_j^Z)$, with S_j^Z representing the size of the j^{th} zooplankton size class (Eq. 13 and Fig. 2C). An additional temperature dependence function, $I(T)$, with T representing LWST ($^{\circ}C$), in our model is further elaborated in Section ‘Temperature dependences’ (Eq. 13A). The second part of the grazing formulation is composed of a half-saturation constant for phytoplankton intake, K_A , and the grazing preference for a specific range of phytoplankton size classes, $\delta_{ij}(S_i^A, S_j^Z)$ (Eq. 15). The grazing preference is characterized by a Gaussian curve over a specific range of phytoplankton size classes. The median and the width of this Gaussian are determined by, respectively, the optimal phytoplankton size, $A_{opt,j}(S_j^Z)$, and the level of selectivity of the grazer θ_j (Fig. 2B). Grazers of different sizes are characterized by different optimal phytoplankton sizes, following an allometric scaling (Eq. 16) that is empirically derived from the data of Hansen et al. (1997), reflecting the fact that large zooplankton prefer large phytoplankton cells (Fig. 2D). The loss processes for zooplankton include natural mortality, ϕ_Z , mixing, λ_Z , and higher order predation, $\eta_Z Z_j$.

The temporal dynamics of detritus is described in Eq. 7. The detritus pool acts as a nutrient recycling station for remineralising dead plankton and non-ingested materials from grazing.

Temperature dependences

[The basic features of the model considered in this study were introduced by To et al. \(2024\) and are presented in detail in Supporting Information S1.](#) ~~Additional features of our model are~~ [The novel features include](#) the temperature dependences ~~assumed~~ for phytoplankton growth and zooplankton grazing. ~~For phytoplankton, the temperature-regulated growth term, $E(T)$,~~

with T representing LWST ($^{\circ}\text{C}$), produces an unimodal curve along a temperature gradient (Norberg 2004). The temperature-regulated growth of phytoplankton is described by the following function (Norberg 2004):

$$E(T) = e^{0.063T} \cdot \left[1 - \left(\frac{T - T_{opt}}{\sigma_T} \right)^2 \right], \quad (1)$$

with T representing temperature ($^{\circ}\text{C}$). T_{opt} and σ_T representing, respectively, thermal optima and thermal tolerance and are described, respectively, by median and width of the temperature dependence curve. The curve is set to reflect the observed range of LWST temperatures, between 3 and 23 $^{\circ}\text{C}$ (Eq. 10 and Fig. 2E). For zooplankton grazing, $I_{max,j}(S_j^Z)$ is regulated by a temperature dependence term, $I(T)$, through a Q_{10} function along a temperature gradient (Eq. 13A and Fig. 2F). The curve is scaled by a coefficient, $I^{Q_{10}}$, and $T_{ref}^{Q_{10}}$ (i.e. set at 15 $^{\circ}\text{C}$, Table 2) represents Zooplankton grazing is regulated by temperature through the following function (Fig. 2F):

$$\psi(T) = Q_{10}^{\frac{T - T_{ref}}{10}}. \quad (2)$$

The function scales with the Q_{10} coefficient, with T_{ref} representing the reference temperature, at which there is no factorial change in $I_{max,j}(S_j^Z)$ maximum ingestion rate of zooplankton occurs (set to 15 $^{\circ}\text{C}$, Table 1). The Q_{10} coefficient was determined from a range of values observed by Hansen et al. (1997) for the maximum ingestion rate of Flagellates, Ciliates, and Copepods (functional groups abundant at Greifensee). In this model, T_{opt} , σ_T , and $I^{Q_{10}}$ are considered constant. In our model, the thermal optima (T_{opt}) and the thermal tolerance (σ_T) for phytoplankton growth and the Q_{10} coefficient for zooplankton grazing are assumed constant.

This assumption implies that different phytoplankton and zooplankton size classes have identical thermal responses in relation to, respectively, growth and grazing. These temperature dependencies allow us to examine the ~~direct~~ effects of temperature on phytoplankton growth and zooplankton grazing ~~in detail~~ and to investigate possible changes ~~made~~ to lake ecosystem components when temperature increases.

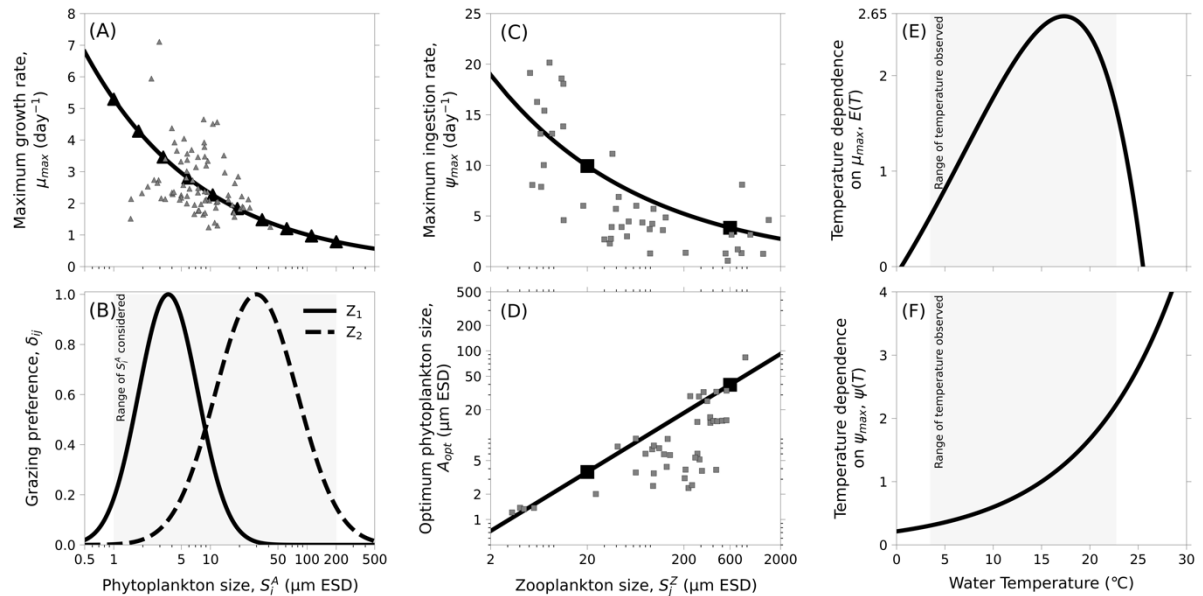
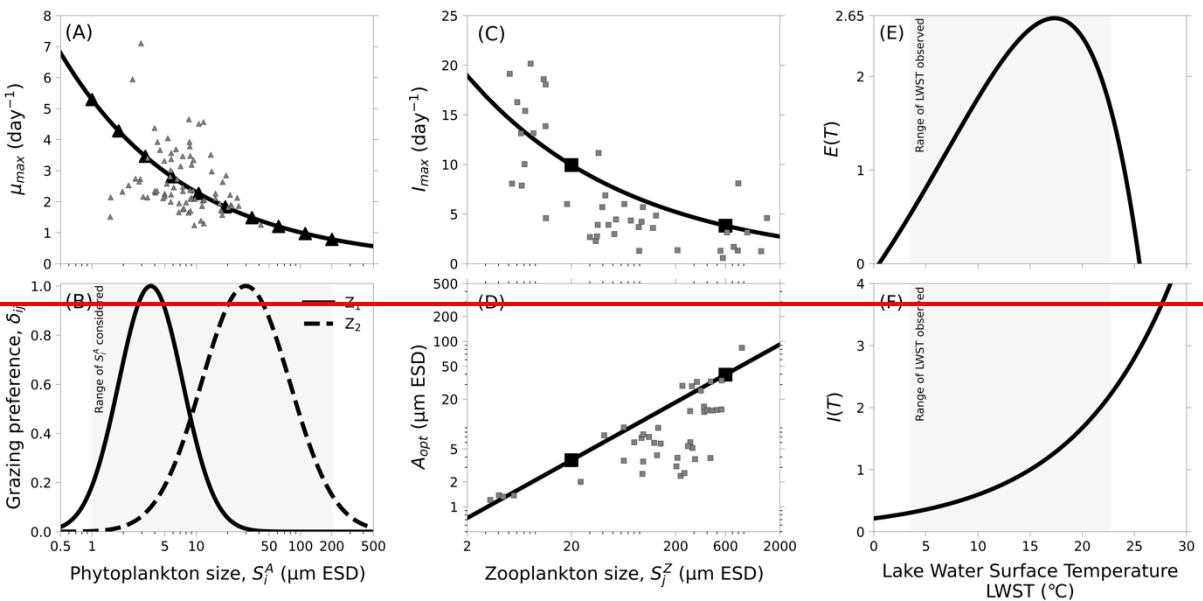


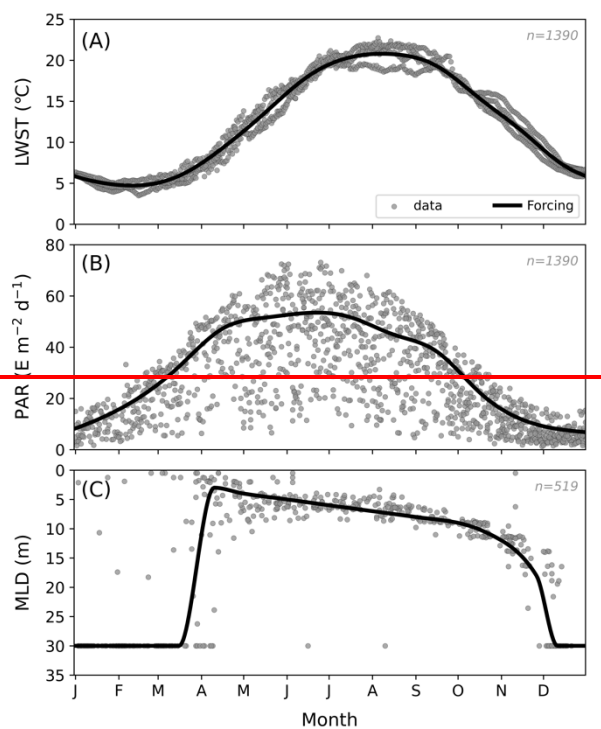
Fig. 2 Model assumptions related to phytoplankton growth and zooplankton grazing. (A) Allometric relationship defining maximum growth rates, μ_{max} , for a given phytoplankton size S_i^A . (B) ~~size~~Size-grazing breadth of the two zooplankton over a range of phytoplankton size classes. Allometric relationships defining (C) maximum ingestion rates, $I_{max}\psi_{max}$, and (D) optimal phytoplankton size, A_{opt} , for a given zooplankton size S_j^Z . The community-wide temperature dependence on (E) phytoplankton growth, $E(T)$, and (F) zooplankton grazing, $I\psi(T)$. The grey triangles and squares are the data points obtained from literature compilations by Edwards et al. (2012), (A); Hansen et al. (1994), (C); and Hansen et al. (1997), (D). Black triangles and squares represent the specific size values considered in this study for, respectively, phytoplankton and zooplankton. Shaded areas in panel B represent the range of phytoplankton sizes considered in this study, whereas in panels E and F represent the present observed range of ~~LWST~~water temperatures at Greifensee.

Model calibration

Environmental forcing

The model is forced with environmental variables observed at Greifensee, a medium-sized (about 8.45 km², with a maximum depth of 33 m) eutrophic lake near Zurich (Switzerland). The environmental variables are: ~~Lake Water Surface Temperature (LWST)~~water temperature, Photosynthetically Active Radiation (PAR), and Mixed Layer Depth (MLD). ~~The LWST data~~Water temperature is obtained from the CTD profiles as daily average epilimnetic temperature from March 2019 to December 2022. The PAR data is converted from the solar irradiance data collected daily by MeteoSwiss, from a weather station next to Greifensee, for the same period as the ~~LWST (Supp. Data S1)~~water temperature. We retrieved ~~MLD~~-data for mixed layer depth from Pomati et al. (2020), which documented ~~the~~ monthly MLD values at Greifensee from 1972 to 2015. The forcing for the model are daily values over a one-year period, ~~a daily average function for LWST, a 75 percentile function for PAR, and a monthly~~

average function for MLD, obtained from these multi-annual environmental data (Fig. 3). We used daily means from a three-year dataset (March 2019 to December 2022) for water temperature, 75-percentiles from the same three-year dataset for PAR, and daily values obtained by interpolating the monthly averages for MLD (Fig. 3). Details of the procedures followed for generating the forcing functions are reported in Supporting Information S1S2 and the environmental data are provided in Supplementary Data.



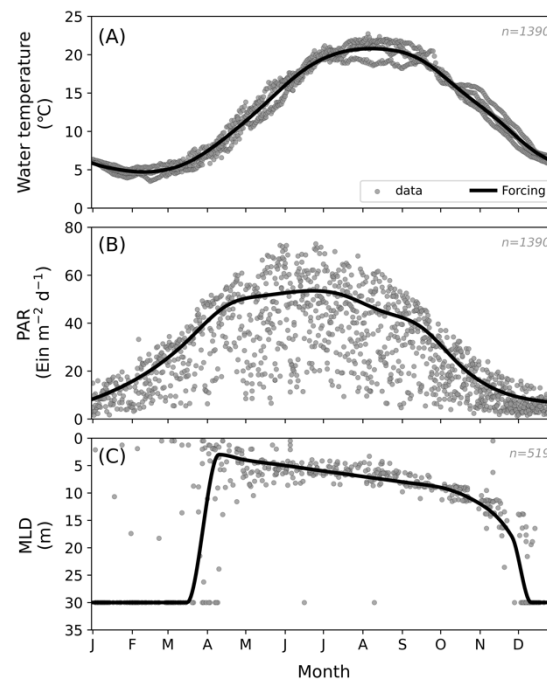


Fig. 3 Forcing functions used in our model. (A) Lake-Water Surface Temperature, LWST temperature ($n = 1390$). (B) Photosynthetically Active Radiation, PAR ($n = 1390$). (C) Mixed Layer Depth, MLD ($n = 519$). A more detailed description of the forcing functions is provided in Supporting Information S1S2.

Plankton abundance and size data and Data for model calibration

The model ~~is constrained~~ was calibrated by daily mean nutrient concentrations (phosphate), and phytoplankton and zooplankton abundance and size data, provided by the Swiss Federal Institute of Aquatic Science and Technology (Eawag). These data covered the period from March 2019 to December 2022. We selected the six most dominant groups of phytoplankton, namely, chlorophytes, rhodomonas, cryptophyceae, centric diatoms, dinobryon, and uroglena, and two dominant groups of herbivorous zooplankton, namely, ciliates and daphnia. The plankton data were collected at Greifensee using a novel dual-magnification dark-field imaging microscope deployed at 3 meters deep (Merz et al. 2021) ~~and cover the period from March 2019 to December 2022.~~ The measured abundance data were in units of ROI/sec

(Regions Of Interest per second). This pseudo-unit is a statistical output derived from each detected and processed object by the imaging microscope and is analogous to object counts for both unicellular and colonial phytoplankton and zooplankton individuals (Merz et al. 2021). We converted these data into plankton abundances (individual L^{-1}) based on laboratory-calibrated functions (Fig. S1, Table S1), derived by Merz et al. (2021). The plankton abundances were then converted into phosphorus-equivalent biomass concentrations, following previously reported scaling functions based on the size of phytoplankton and zooplankton (Supporting Information S3).

The size data were collected as bio-area (μm^2) and converted into biovolume, based on an empirical scaling law (Table S1). The median bio-area of the six phytoplankton species we selected ranged from 10^2 to $10^5 \mu m^2$. Our model captures cell size in terms of equivalent spherical diameter, ESD (μm), because it encompasses differences in size and geometric shape of plankton (Jennings and Paslow 1988). ~~We selected the six most dominant groups of phytoplankton, namely, chlorophytes, rhodomonas, cryptophyceae, centric diatoms, dinobryon, and uroglena, and two dominant groups of herbivorous zooplankton, namely, ciliates and daphnia.~~

~~The raw phytoplankton and zooplankton size data are collected as bio-area (μm^2), whereas size-based models typically capture cell size in terms of equivalent spherical diameter, ESD (μm). ESD is a common unit for planktonic size-based models (e.g. Acevedo-Trejos et al. 2015; To et al. 2024) because it encompasses differences in size and geometric shape of plankton (Jennings and Paslow 1988). However, for comparability purposes, we convert the model and lake size data into a unit of phytoplankton biovolume, V_t^A . The conversion from bio-area to biovolume is based on an empirical scaling law of bio-area to biovolume (Eq. S3 and Table S1). The median bio-area of the six phytoplankton species we selected ranges from 10^2~~

to $10^5 \mu\text{m}^2$. ~~Therefore, in the model, we considered~~ Therefore, in the model, we consider a size spectrum ranging from 1 to 200 μm ESD for the phytoplankton community, encompassing 10 size classes separated by logarithmic intervals. For zooplankton, we considered two size classes, Z_1 and Z_2 , of 20 and 600 μm , respectively. We start our simulation with an initial concentration of $6.25 \times 10^{-4} \mu\text{M}$ P for all planktonic organisms. We analyzed ~~plankton sizes in terms of phytoplankton mean biovolume, $\overline{V^A}$, given by:~~ the phytoplankton size composition in macroecological terms (e.g., Acevedo-Trejos et al. 2018; To et al. 2024), by considering a community-weighted mean cell biovolume, $\overline{V^A}$:

$$\overline{V^P V^A} = \frac{\sum_i^n A_i V_i^A}{\sum_i^n A_i} \quad (17)$$

where n denotes the total number of size classes, and A_i and V_i^A denote, respectively, total biomass and biovolume of the i^{th} class of phytoplankton.

~~—— The data collected by the imaging microscope and reflecting plankton abundances are in units of ROI/sec (Regions Of Interest per second). This pseudo-unit is a statistical output derived from each detected and processed object by the imaging microscope and is analogous to object counts for both unicellular and colonial phytoplankton and zooplankton individuals (Merz et al. 2021). To be comparable with model outputs, these data need to be transformed into abundances (individual L^{-1}) based on laboratory calibration functions, which were derived by Merz et al. (2021) and are reported in Fig. S2. The plankton abundances are then transformed into phosphorus-based biomass concentrations, following previously reported scaling functions based on the size of phytoplankton and zooplankton. The details of these transformations are reported in Supporting Information S2. We additionally optimized a number of unit conversion factors, such as the carbon to phosphate ratio, by comparing and~~

~~matching the plankton abundances obtained from the observations with the abundances produced by the model.~~

All model parameters ~~are~~were taken from literature (Table 21). Parameters that regulate nutrient uptake and light harvesting of phytoplankton and grazing of zooplankton ~~are~~were fine-tuned manually (i.e., α_{P_I} , K_n , ~~α_{max}~~ $\alpha_{\psi_{\text{max}}}$, $\alpha_{A_{\text{opt}}}$, $\theta_j^{Z_1}$ and $\theta_j^{Z_2}$; within ranges of values reported in the literature, see Table 21), in order to find best fits (based on visual inspection between model results and observations (Fig. S5). This parameter calibration ~~produces~~produced what we call a ‘standard model run’, i.e. a model run that yields the best match with observations. The time step of our model is one day. To calibrate parameters, we compared the model results against three consecutive years of nutrient and biological data, using daily averages over a one-year period (details can be found in Supporting Information S3). The model results, therefore, reflect the daily dynamics of the lake for a “typical” (i.e., average) year under current temperature conditions. Additionally, we undertook a sensitivity analysis to evaluate the effects of parameter uncertainty on model results. We addressed those parameters ($n=14$) that could not be directly constrained by the data specifically available for Greifensee and those that were not calibrated (Supporting Information S4). Since we are interested in the average state of the lake over one year, the model is run to steady-state for 10 years ~~with~~, using the same annual forcing for every year, and considering only the final year ~~is considered~~ for analysis. ~~The observations used for the parameter calibration are the median daily values over 3 years of data for nutrient concentrations, phytoplankton and zooplankton biomasses, and phytoplankton mean biovolume~~ Inter-annual variability was not explored as it was beyond the scope of our study.

Table 21. List of model parameters ~~along with their~~ corresponding symbols, standard values, parameter range, units, and literature sources. When ~~given~~ we report a parameter range, the standard value of the parameter ~~is~~ was derived from model calibration.

Parameter	Symbols	Standard Value*	Parameter range	Units	Sources for parameter/parameter range
Initial slope of P-I curve	α_{P_I}	0.02	0.02 - 0.1	$\text{Ein m}^{-2} \text{ d}^{-1}$	Anderson et al. (2015)
Maximum photosynthesis rate	P_{max}	1.1	-	d^{-1}	To et al. (2024)
Light attenuation coefficient	K_{par}	0.1	0.08 – 0.5	m^{-1}	Chen et al. (2002); Genkai-Kato et al. (2012)
Thermal tolerance	σ_T	12.5	-	$^{\circ}\text{C}$	Greifensee data
Thermal optima	T_{opt}	13	-	$^{\circ}\text{C}$	Greifensee data
Half-saturation constant for nutrient (phosphorus)	K_n	0.06	0.06 -0.1	$\mu\text{M P}$	Data in Edwards et al. (2012) <u>Edwards et al. (2012)</u>
<u>Slope of the allometric relationship for maximum growth rate, $\mu_{max_i}(S_i^A)$</u>	$\alpha_{\mu_{max}}$	<u>-0.36</u>	=	=	<u>Edwards et al. (2012)</u>
<u>Intercept of the allometric relationship for maximum growth rate, $\mu_{max_i}(S_i^A)$</u>	$\beta_{\mu_{max}}$	<u>$10^{0.69}$</u>	=	<u>d^{-1}</u>	<u>Edwards et al. (2012)</u>

Parameter	Symbols	Standard Value ^a	Parameter range	Units	Sources for parameter/ parameter range
Cross-thermocline diffusive mixing coefficient	ω	0.1	-	m day ⁻¹	Fasham et al. (1990)
Natural mortality - Phytoplankton	ϕ_A	0.1	-	d ⁻¹	this study
Natural mortality – Zooplankton	ϕ_Z	0.1	-	d ⁻¹	To et al. (2024)
Higher-order mortality rate	η_Z	0.34	-	d ⁻¹	Oschlies & Schartau (2005)
Fraction of grazing not egested	ε	0.69	-	-	Fasham et al. (1990)
Fraction of grazing assimilated to growth	γ	0.75	-	-	Oschlies & Schartau (2005)
Remineralization rate	φ	0.15	-	d ⁻¹	Kerimoglu et al. (2017)
Half-saturation constant to phytoplankton cells	K_A	0.19	-	μM P	Kerimoglu et al. (2017)
Size-grazing breadth - Z_1, Z_2	$\theta_j^{Z_1}, \theta_j^{Z_2}$	0.435, 0.6	0.2-0.8	μm ESD	Banas (2011)
Slope of the allometric relationship for I_{max_j} maximum ingestion rate, $\psi_{max_j}(S_j^Z)$	$\alpha_{I_{max}}$ $\alpha_{\psi_{ma}}$	-0.28	-0.25 - -0.4	-	Hansen et al. (1997)

Parameter	Symbols	Standard Value ^a	Parameter range	Units	Sources for parameter/ parameter range
Intercept of the allometric relationship for $I_{max,j}$ the maximum ingestion rate, $\psi_{max,j}(S_j^Z)$	$\beta_{I_{max}} \psi_{ma.}$	23	22-26	d ⁻¹	Hansen et al. (1997)
Q ₁₀ factor coefficient for $I_{max,j}(S_j^Z)$ the maximum ingestion rate	Q_{10}	2.8	2.0 - 3.9	-	Data in Hansen et al. (1997)
Reference temperature for the Q ₁₀ factor coefficient	$T_{ref}^{Q_{10}} T_{ref}$	15	-	°C	Hansen et al. (1997); Kerimoglu et al. (2017)
Slope of the allometric relationship for optimal phytoplankton size, $A_{opt,j}(S_j^Z)$	$\alpha_{A_{opt}}$	0.7	-	-	Hansen et al. (1997)
Intercept of the allometric relationship for optimal phytoplankton size, $A_{opt,j}(S_j^Z)$	$\beta_{A_{opt}}$	0.45	-	µm ESD	Hansen et al. (1997)
Cell sizes – phytoplankton	S_t^A	1-200	-	µm ESD	Greifensee data
Body sizes – zooplankton	S_j^Z	20 (Z ₁) 600 (Z ₂)	-	µm ESD	Greifensee data
Initial conditions for nutrient	N_0	0.75	-	µM P	Greifensee data
Initial conditions for phytoplankton, zooplankton, and detritus	A_0, Z_0, D_0	6.25e-4	-	µM P	this study

Ecological projections based on climate change scenarios

The model parameterization obtained from [the community-integrated outputs of](#) the standard model run produces model results that reflect current average annual conditions (in terms of nutrient concentrations, ~~plankton~~[total phytoplankton and zooplankton](#) biomasses and [aggregated phytoplankton](#) size structures) at Greifensee. With this parameter configuration, we forced the model with new temperatures (~~LWSTs~~)[based on, representing](#) three climate change scenarios,[based on three Representative Concentration Pathways \(RCPs\), Table S3,](#) expected by the end of this century (i.e., year 2099) in the regions [covering Greifensee, between](#) 40° N and 60° N, ~~covering Greifensee. The LWSTs scenarios are based on Representative Concentration Pathways (RCPs) produced by the Intergovernmental Panel on Climate Change (IPCC), Table S2. We considered RCP 2.6, RCP 6.0, and RCP 8.5, reflecting low, medium, and high emission scenarios, respectively. These scenarios produce an average annual increase in LWST of~~ [We considered average and seasonally uniform increase in water temperature by](#) 1.4 °C (RCP 2.6), 2.5 °C (RCP 6.0), and 4.2 °C (RCP 8.5). [However, our future projections did not include physical feedback that could be induced by increasing temperatures, like changes in lake stratification \(but see Supporting Information S5 for a sensitivity run exploring the impact of this omission on our results under scenario RCP 8.5\). Since we are not interested in the effects of transient thermal anomalies or in interannual or decadal warming trends, Also for these climate change scenarios,](#) we run the model ~~for these climate change investigations~~ to steady-state for 10 years, using the same annual forcing for every year, and [considering](#) only the final year ~~is considered for results and for~~ analyses.~~For the sake of simplicity, we considered uniform increases in LWSTs throughout the year.~~ In our analyses, annual seasons

are defined as follows: winter, from December to February; spring, from March to May; summer, from June to August; and autumn, from September to November.

Sensitivity analyses on phytoplankton thermal traits

~~Given the uncertainties involved in modelling studies like this, for~~ For each temperature scenario, we undertook a series of sensitivity analyses on ~~the thermal traits of the phytoplankton community, namely~~ the thermal optima, T_{opt} , and the thermal tolerance, σ_T , ~~of the phytoplankton community.~~ These sensitivity analyses examine the effects on the model results ~~of changes by (annual medians) under variations of~~ $\pm 15\%$ ~~for both T_{opt} and σ_T from their respective standard values of 13°C (i.e. 11.0°C and 15.0°C) and 12.5°C (i.e. 10.6°C and 14.4°C), respectively~~ (Fig. 4). These thermal traits play a key role in regulating phytoplankton responses to temperature change but are difficult to measure in natural systems, and their values can vary within ~~and between~~ species ~~and among communities~~ (Boyd et al. 2013; Thomas et al. 2016). ~~In the model, small thermal trait values produce poor phytoplankton growth (see Fig. 4).~~ The range of variability we considered for these sensitivity analyses covers the values of T_{opt} ~~thermal optima~~ and σ_T ~~thermal tolerance~~ observed in freshwater species (Thomas et al. 2016).

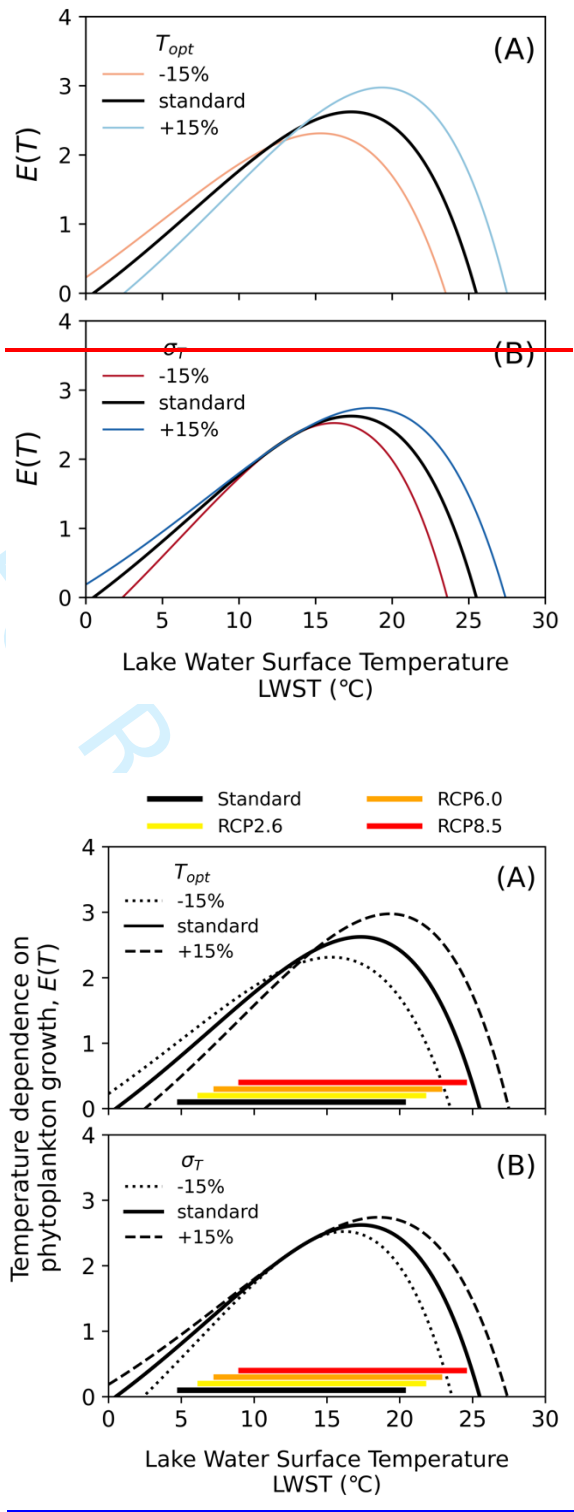


Fig. 4 Temperature dependence curves tested in the sensitivity analyses. These curves result from changes by $\pm 15\%$ of (A) thermal optima, T_{opt} , and (B) thermal tolerance, σ_T , from their standard values. The standard values for T_{opt} and σ_T are, respectively, 13 °C and 12.5 °C (black curves). The colored bars

represent the annual ranges of water temperatures for various warming scenarios: dark (current average state); yellow (RCP 2.6); orange (RCP 6.0); and red (RCP 8.5).

Results

Current status in the seasonal dynamics of ~~major variables~~ lake ecosystem components

For the model calibration, we focused ~~our attention~~ on the seasonal dynamics of four major variables: (1) nutrient concentration, (2) phytoplankton biomass, (3) zooplankton biomass, and (4) phytoplankton mean biovolume (the latter three are community-integrated). The simulated inorganic nutrient dynamics capture the average annual cycle ~~shown in the~~ of epilimnetic phosphorus ~~data~~ observed at Greifensee (Fig. 5A). The nutrient concentration is characterized by a steady decline, starting at the beginning of the year and lasting for about three months, it remains low throughout summer and for part of autumn, and it restocks by the end of the year (Fig. 5A).

The simulated plankton dynamics also capture the observed community-integrated biomass ~~levels~~ qualitatively well (Fig. 5B). The model captures the biomass accumulation ~~events by~~ of phytoplankton during spring, summer, and autumn, and reproduces correctly the low phytoplankton biomass levels after the summer and autumn peaks (Fig. 5B). ~~More specifically~~ However, the model ~~produced~~ produces a first winter peak that is not ~~reflected in~~ shown by the data (Fig. 5B) ~~(driven by a high efficiency of light harvesting i.e., high α_P , Fig. S5A), underestimated, it underestimates~~ the spring biomasses, and ~~overestimated it~~ overestimates the summer biomasses. The simulated zooplankton biomass matches well with the autumn and winter observations, despite a slight overestimation for the spring peak ~~biomass data~~ (Fig. 5C). ~~Consistently to the data, the model shows a peak in zooplankton biomass in~~

~~spring, although with a slight overestimation.~~ A mismatch is noticeable in June and July. ~~In this period,~~ when the simulated zooplankton biomass declines due to the dominance of large and less grazable phytoplankton (Fig. 5C and 5D). ~~This decline is immediately followed by a sharp increase that forms a second peak in zooplankton biomass, whereas the observations show a decrease in zooplankton biomass (Fig. 5C).~~ In summary, despite being an approximation of the real lake in terms of physics and ecology, the model captures several general features of the phytoplankton and zooplankton seasonal cycles; namely the spring, summer, and autumn biomass accumulations of phytoplankton and the spring biomass peaks of zooplankton.

A good match is also obtained between the simulated and observed seasonal patterns in the community-mean biovolume of phytoplankton. In general, both model and data show small mean biovolume in late winter and beginning of spring (Fig. ~~4D~~5D), when smaller cells outcompete larger cells due to weaker grazing on smaller cells (Fig. ~~4E~~5C). Relatively larger phytoplankton cells are present during summer and autumn (Fig. ~~4D~~5D), when zooplankton grazing on smaller cells increases and the larger, less grazable cells gain dominance. Additionally, we find that the phytoplankton size composition is strongly controlled by grazing characteristics, namely, the allometric scaling of grazing (i.e., maximum ingestion rate and optimal phytoplankton size) and the size-grazing breadth of zooplankton (Fig. ~~S5H-J~~S3A, S5I, S5J, S5L).

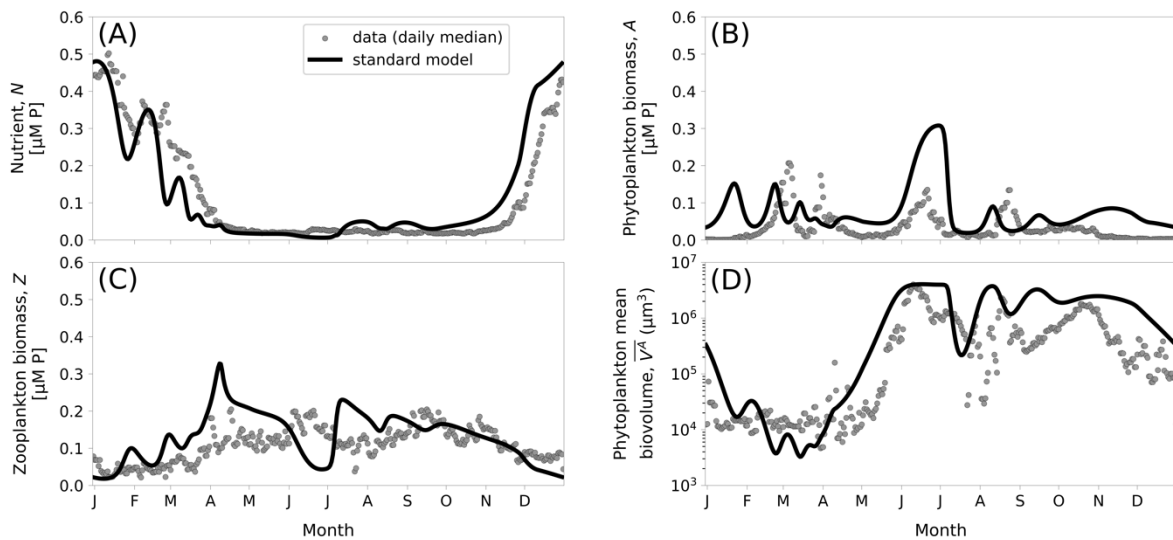


Fig. 5 Results of the standard run showing temporal dynamics of (A) nutrient concentration, (B) phytoplankton community biomass concentration, (C) zooplankton community biomass concentration, and (D) phytoplankton mean biovolume. The grey dots represent the observational data collected [in the epilimnion](#) by the underwater imaging device (see details about data transformation in Supporting Information [S2S3](#)) and the black lines represent the model outputs. The model is run to steady-state over 10 years and we show here the results of the 10th year. The observations are median daily values over 3 years of data (from March 2019 to December 2022).

Expected future seasonal dynamics of major variables

Alterations in nutrient dynamics

We find seasonal alterations in the nutrient dynamics under the [RCPwarming](#) scenarios, with the strongest alteration occurring under RCP 8.5 ([LWSTs](#)[water temperature](#) higher by 4.2 °C with respect to current average). The most prominent change in the nutrient concentration occurs in summer and autumn, i.e. from July to October, when there are excessive nutrient accumulations in the lake, [and accumulation intensifies with the warming scenarios](#) (Fig. 6B).

~~As lake warming intensifies, more nutrients accumulate over the summer months.~~ Under RCP

8.5, the nutrient accumulation in summer reaches an annual maximum, a pattern that contrasts sharply with the current status (standard run) when the highest nutrient level occurs in winter (Fig. 6B). This summer nutrient peak coincides with a decline in phytoplankton biomass, caused by the unfavorable temperature conditions for growth experienced by the system under RCP 8.5 (Fig. S6A and S7). Warming also causes an increase in the overall annual nutrient concentration in the epilimnion. Under RCP 8.5, the annual median nutrient concentration is about four times higher than ~~the median concentration that~~ under the current status, i.e. 0.05 μM P (current status) versus 0.2 μM P (RCP 8.5), Fig. 6C. In summary, our model projections suggest that as the lake warms up, nutrient accumulation increases in the summer months and the lake becomes more eutrophic.

Alterations in plankton phenology

As the lake warms up, the model shows lower winter peaks and earlier summer peaks of phytoplankton (Fig. 6D). The timing of the summer peaks depends on the timing of thermal optima, which shifts with the warming scenarios (Fig. S6C). Under RCP 8.5, the summer phytoplankton peak advances by about two months, from July to May. The earlier arrival of the summer peaks shortens the corresponding period of low phytoplankton biomass after the spring peaks. We also notice an increased number of minor phytoplankton peaks during the year as warming intensifies, with small reductions in annual median phytoplankton biomass concentrations (Fig. 6E). In summary, the increase in ~~LWST~~water temperature advances the timing of the major phytoplankton peaks and increases the number of minor peaks occurring throughout the year.

The model results also show phenological shifts for zooplankton. The timing of the first zooplankton peak in April remains the same in all climate scenarios (~~Fig. 6F~~), but the ~~period of sustained high zooplankton biomass during the first peak shortens and that the~~ second peak

556 moves earlier with the severity of the RCP scenarios [\(Fig. 6F\)](#). Consequently, the period
557 between the first and the second zooplankton peaks shortens by two months under RCP 8.5
558 ~~(Fig. 6F)~~. Also, the zooplankton biomass in late summer decreases noticeably under RCP 8.5,
559 compared to other scenarios, due to ~~a~~[the](#) decline in phytoplankton growth (Fig. S7). The annual
560 median zooplankton biomass is lower in RCP 8.5 compared to other scenarios (Fig. 6G). In
561 summary, major alterations in the phenology of zooplankton include a reduction in the period
562 between the first and the second peak and a reduction in the summer biomass.

563 *Alterations in phytoplankton mean biovolume*

564 An ~~increase in LWST~~[elevated water temperature](#) increases the mean biovolume of
565 phytoplankton and reduces the range of size variations during the year. In particular, as the lake
566 warms up, relatively smaller cells ~~are~~ replaced by larger cells in winter and [in](#) spring (Fig. 6H).
567 Under RCP 8.5, the dominance of relatively larger phytoplankton appears already in April, two
568 months earlier than in the current status. In summary, a warmer lake fosters a narrower size
569 diversity of phytoplankton and reduces seasonal variations in mean biovolumes (Fig. 6I).

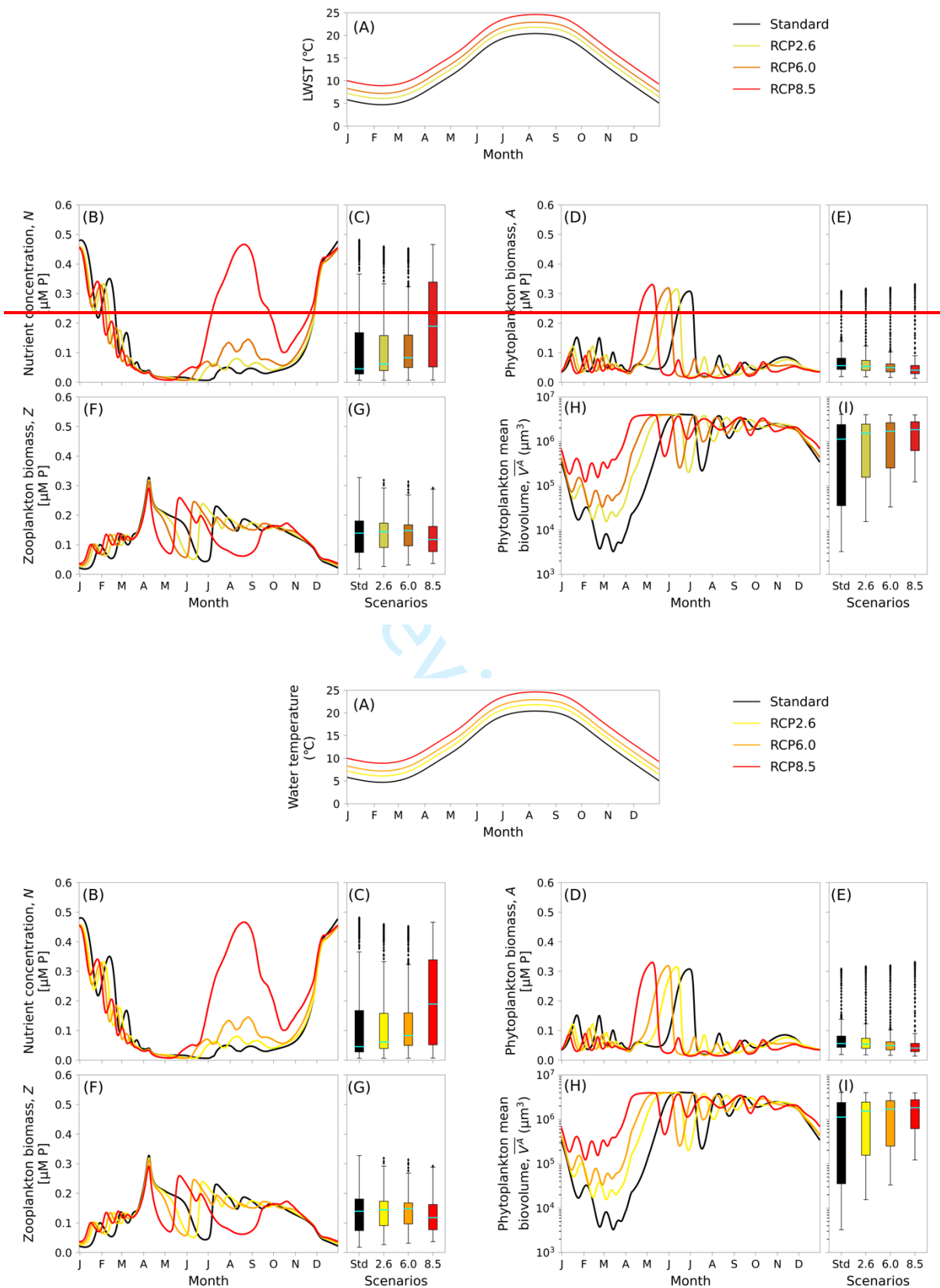


Fig. 6 Results of climate change scenarios compared to the current average status of the lake (standard run, black lines and Fig. 5). The climate change scenarios, RCP 2.6, RCP 6.0, and RCP 8.5 simulate an average

and seasonally uniform increase in ~~LWST~~water temperature by, respectively, 1.4 °C, 2.5 °C, and 4.2 °C (A). The results show temporal dynamics of (B) nutrient concentration in the epilimnion, (D) phytoplankton biomass, (F) zooplankton biomass, and (H) phytoplankton mean biovolume. Boxplots (panels C, E, G, and I) show the overall seasonal variability (width) and median (cyan lines) of the corresponding variables. The model is run to a steady annual cycle over a period of 10 years with repeated annual forcing, the results shown are those of the 10th year.

Sensitivity analysis of future scenarios

We examined the sensitivity of the model results by varying the values of the thermal traits, thermal optima (T_{opt}) and thermal tolerance (σ_T), by $\pm 15\%$ from ~~their~~ standard values. The results show that, under future warming scenarios, the nutrient concentration, among other variables, is most sensitive to changes in the thermal traits-. A reduction in either of the thermal traits leads to a higher median nutrient concentration, whereas an increase in either of the thermal traits leads to a lower median nutrient concentration (Fig. 7). These changes are exacerbated by the severity of the warming scenarios. Under RCP 8.5 (Fig. 7C), the median nutrient concentration increases by at least 200% when the phytoplankton community is characterized by ~~either a lower T_{opt} or a lower σ_T (Fig. 7C)~~ lower values of thermal traits (i.e., lower tolerance to higher temperatures, Fig. 4). Such higher nutrient concentrations are caused by reduced growth of phytoplankton, from spring to autumn, due to a less suitable temperature for growth (Fig. ~~S8S8G, S8I~~). Additionally, lower T_{opt} or σ_T leads to pronounced reductions in phytoplankton and zooplankton biomasses and in phytoplankton mean biovolume, particularly under RCP 8.5 (Fig. 7C). A higher T_{opt} (or σ_T) does not induce as many variations to the model variables; even under RCP 8.5, the greatest relative change is less than 50%. That

is because higher thermal trait values produce a more heat-tolerant phytoplankton community (Fig. 4).

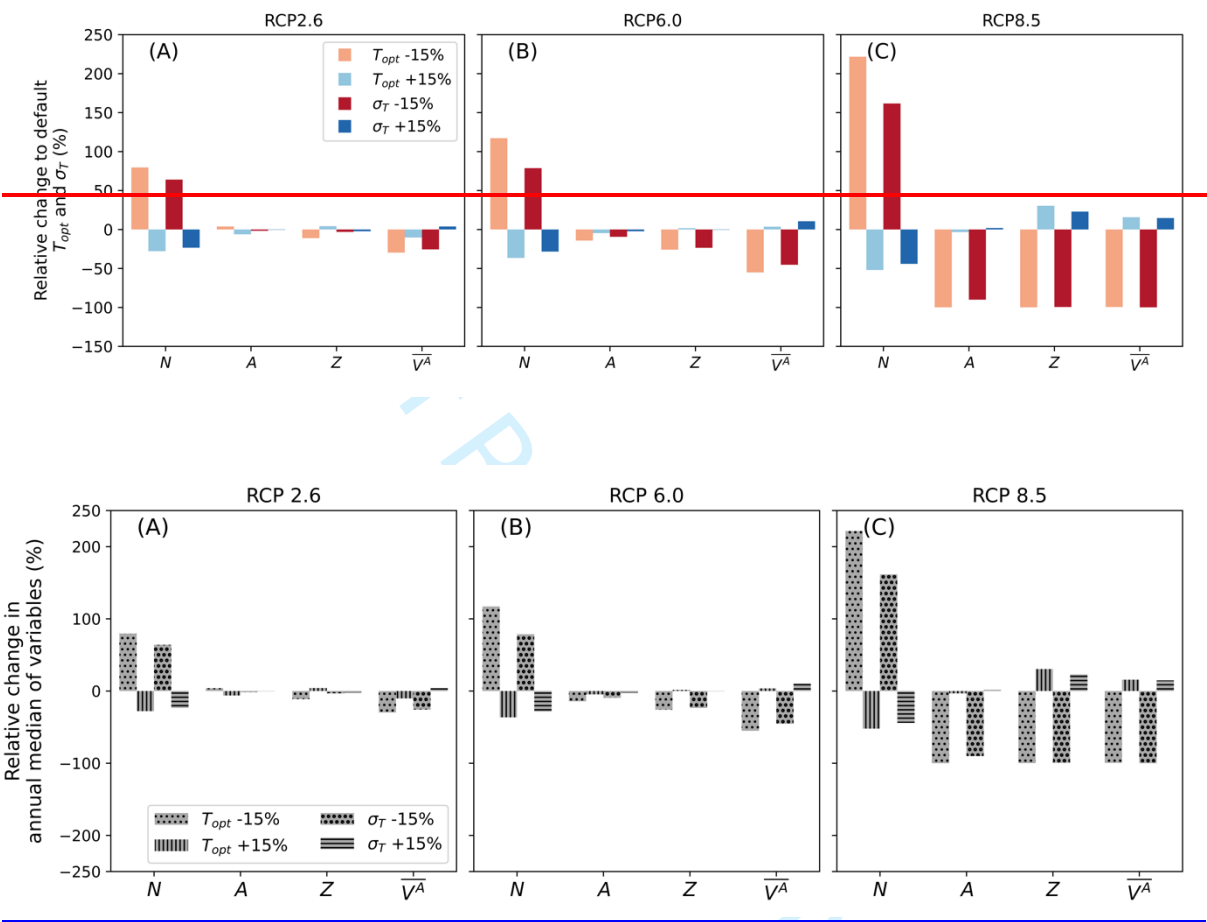


Fig. 7 Sensitivity of model variables (annual medians) caused by $\pm 15\%$ changes from standard values of T_{opt} or σ_T , thermal optima (T_{opt}) or thermal tolerance (σ_T), for the different climate change scenarios: (A) RCP 2.6, (B) RCP 6.0, and (C) RCP 8.5. The model variables are nutrient concentration, N ; phytoplankton biomass, A ; zooplankton biomass, Z ; and phytoplankton mean biovolume, $\overline{V^A}$.

Discussion

Mounting evidence over the last two decades shows rising lake temperatures on a planetary scale, with European temperate lakes expected to experience the fastest ~~increase in LWST~~ ~~(Woolway and Maberly 2020)~~ warming trend (Woolway et al. 2022). The consequences of rising temperatures on the physiology and ecology of plankton are far from clear. In our model, the growth of phytoplankton depends, among other factors, on a thermal dependence curve characterized by data-constrained (specifically, Greifensee data) optimal temperature for growth and tolerance range; and the grazing of zooplankton depends on a Q_{10} function constrained by literature data. Under future climate change scenarios, our model predicts increased epilimnetic nutrient concentrations, advancements in phytoplankton biomass peaks, and dominance of relatively larger phytoplankton. We also find that a phytoplankton community with lower thermal optima or lower thermal tolerances leads to even higher accumulation of nutrient concentrations due to the less favourable temperature ranges of phytoplankton growth and thus of lower nutrient uptake.

Current seasonal patterns of nutrients and phytoplankton concentrations

With respect to the current average state of Greifensee, the seasonal pattern of phosphorus dynamics produced by the model matches well with the observational data and reflects the typical nutrient dynamics of temperate lakes, which are characterized by high nutrient concentrations in winter and low concentrations in summer (Sommer et al. 2012). In temperate lakes, relatively abundant phosphorus at the beginning and at the end of the year coincides with scarce light and less favourable temperatures for growth, resulting in lower growth and uptake rates (Stachr and Sand-Jensen 2007; Sommer et al. 2012). The June-July peaks in the observed phytoplankton and zooplankton biomasses are temporally coincident, whereas the model results show a slight shift, with zooplankton peaking slightly later than phytoplankton. This small delay, typical of predator-prey models, is an expected feature of our model, given its

general resource-prey-predator structure. We find that the annual variations in the mean phytoplankton biovolume are characterized by relatively higher abundances of smaller phytoplankton in spring, higher abundances of larger phytoplankton in summer, and higher abundances of intermediate-size phytoplankton in autumn. These patterns are consistent with the general phytoplankton dynamics ~~described for eutrophic systems~~ expected to occur when interactions between inorganic nutrient limitation and grazing pressure are primary driving factors of lake ecosystems (Vanni and Temte 1990; Sommer et al. 2012). Our model further supports findings from previous size-based models (Banas 2011; Taniguchi et al. 2023; To et al. 2024) showing that the biomass of phytoplankton is regulated by variations in resources like light and nutrients, whereas the annual variations in the phytoplankton size composition are strongly controlled by grazing characteristics like the size-feeding breadth of zooplankton. ~~Previous size-based models demonstrated how the size-feeding breadth of zooplankton regulates the size compositions of phytoplankton communities (Fuchs and Franks 2010; Banas 2011; Taniguchi et al. 2023; To et al. 2024).~~ By using appropriate zooplankton size-feeding characteristics under the influence of major physical factors in lakes, we capture and elucidate the phenology of phytoplankton biomass and size composition at Greifensee.

Future increased phosphorus concentration driven by phenological shift of phytoplankton

The ~~model~~ ecological projections under future ~~LWSTs~~ warming scenarios show that epilimnetic phosphorus concentrations increase with the increasing severity of global warming. The ~~increased~~ higher phosphorus concentration is driven by a shifted phytoplankton phenology, characterized by milder accumulations of phytoplankton biomass in winter and earlier accumulations in spring, causing a reduction of phytoplankton biomass in summer due to sub-optimal temperature conditions for phytoplankton growth. ~~The low phytoplankton biomass in summer results from sub-optimal conditions for phytoplankton growth under the warming~~

~~scenarios.~~ The sub-optimal condition for growth also results from ~~This depends on~~ a mismatch between optimal temperature and light conditions, as the optimal temperature for growth occurs earlier than optimal light conditions. This mismatch retains unused phosphorus in the summer epilimnion because of reduced phosphorus uptake by the phytoplankton community. This result is based on the assumption that different phytoplankton size classes have identical thermal characteristics (Fig. 2). In nature, species sensitivities to temperature may vary and different phytoplankton can exhibit different growth and nutrient uptake abilities along thermal gradients (Thomas et al. 2016; Bestion et al. 2018). This is an aspect that could generate feedbacks on the nutrient dynamics that are not captured by our model. Also, higher temperatures lead to stronger lake stratification, which in turn has an impact on competition for nutrients and light with consequences on the composition and dynamics of phytoplankton communities (Tilzer and Goldman 1978; Donis et al. 2021). The stronger and longer stratification events in a warmer lake (Butcher et al. 2015; Woolway et al. 2021) can further disrupt the nutrient exchanges between epilimnion and hypolimnion and, thus, influence nutrient dynamics (Supporting Information S5, Fig. S4B). ~~Previous studies highlighted the synergistic impact of lake warming and anthropogenic eutrophication exacerbating phytoplankton blooms and reducing lake water quality (Moss 2011; Paerl et al. 2016). Our findings show, additionally, that lake warming increases nutrient concentrations via shifts in phytoplankton phenology, leading to reduced nutrient transfer efficiencies. The expected stronger and longer stratification events in a warmer lake (Woolway et al. 2020) can disrupt the nutrient exchanges between epilimnion and hypolimnion and can exacerbate nutrient accumulations in lakes. A nutrient-enriched epilimnion can alter lake internal nutrient loadings and phytoplankton responses to warming (Huber et al. 2008). Recent studies highlighted internal nutrient loadings as a key factor causing unexpected cyanobacterial blooms in lakes (Yindong et al. 2021; Kong et al. 2023).~~ Taken together, these results indicate that accounting

for the impacts of climate on nutrient regimes ~~and their internal feedback is crucial for designing effective lake management strategies~~ would require detailed considerations in relation to physical and biological feedbacks. Our results should be considered as a first order approximation with respect to the impacts that global warming can have on nutrient dynamics and plankton phenology.

Earlier accumulation of phytoplankton biomass driven by thermal optima

Our climate change scenarios produce earlier accumulations of summer phytoplankton biomass ~~as the timing of thermal optima advances with warming scenarios.~~ Earlier formation of spring blooms in temperate lakes was detected by previous studies based on simulations and observations. For example, a dynamic diatom-daphnia model showed that, under lake warming, a high phosphorus supply causes earlier diatom spring blooms ~~under lake warming~~ (Huber et al. 2008). ~~Large-scale simulation models predicted an earlier onset of spring blooms or chlorophyll peaks by about three weeks under a 4 °C increase in LWST (Gronechi et al. 2023; Mesman et al. 2023). Using phenological metrics, decadal observations of large lakes in the USA and UK also detected earlier onsets of phytoplankton peak biomass by around three weeks (Winder and Schindler 2004; Thackeray et al. 2013). Although our projections are largely consistent with these studies (earlier occurrences of phytoplankton peaks in summer by 18 to 53 days, Table S3).~~ A depth-resolved hydrodynamic model coupled to a simple phytoplankton growth model showed that earlier spring blooms can be caused by changes in lake stratification patterns (Peeters et al. 2007). In our model, the earlier peaks of large phytoplankton are produced by a warmer winter that advances the timing of optimal temperature for the growth of all phytoplankton size classes but intensifies the grazing of zooplankton on small phytoplankton size classes. Thermal characteristics, like optimal temperature for growth, are important factors for driving seasonal succession in culture experiments comprising marine

diatoms (Anderson and Rynearson 2020). Although our projections are largely consistent with these studies, a long-term observation conducted in a large Norwegian lake revealed delayed phytoplankton peaks (Moe et al. 2022). Their observation considered taxonomic details of the phytoplankton community and showed that the phenological changes of phytoplankton are contingent to the different succession patterns exhibited by individual taxa. ~~Our results also illustrate the role played by the composition of the phytoplankton community in terms of biovolume. In our model, the early formation of biomass peaks by large phytoplankton is driven by a warmer winter that advances the timing of optimal temperature for growth for all phytoplankton size classes and intensifies the grazing of zooplankton over small phytoplankton size classes.~~ Therefore, taxonomic details related to the different thermal responses in a community are important for understanding specific changes in the phenological patterns of phytoplankton (Thackeray et al. 2008; Segura et al. 2018). This shows the importance of considering size or taxonomic differences in thermal traits for capturing specific phenological trends.

Changes in phytoplankton mean biovolume ~~in a warmer lake~~ under increasing temperature

We find an increase in mean phytoplankton biovolume and a decrease in the range of variability with increasing average temperature. These trends are caused by the disappearance of smaller phytoplankton during spring and the earlier appearance of larger phytoplankton during summer. Our results are driven by ~~the~~ thermally-regulated bottom-up (nutrient uptake by phytoplankton) and top-down (zooplankton grazing) processes. As temperature increases, the optimal temperature for phytoplankton growth occurs earlier and the summer temperatures ~~are~~ become sub-optimal for growth; (Fig. S6C); whereas zooplankton grazing, which targets smaller phytoplankton, intensifies (Fig. S6B). These aspects interact not only to favor phytoplankton cells with intermediate biovolumes over cells with small biovolumes, but also

to foster the earlier appearance of large phytoplankton. Intensified grazing on smaller phytoplankton was observed in warmer freshwater environments (Winder and Sommer 2012). ~~Time-series analyses of dataset~~ Analyses of time-series from eight Swiss lakes revealed a rising proportion of larger phytoplankton cells in association with higher temperatures, higher zooplankton abundances, and lower nutrient levels (Pomati et al. 2020). This multi-decadal dataset showed that larger phytoplankton are found in summer, coinciding with higher grazing pressures compared to winter, suggesting ~~a~~ the presence of a grazing-mediated positive correlation between larger phytoplankton and higher temperatures. However, ~~an eight-year observation conducted in~~ lakes where zooplankton grazing is weak, there can be different results. In Lake Kinneret, ~~— (Israel, revealed that),~~ the prevalence of small phytoplankton in stratified summer waters ~~results~~ (based on an eight-year observation) resulted from a strong selection by sinking processes on the better-suspended small phytoplankton, ~~as zooplankton grazing plays a minor role in that lake~~ (Zohary et al. 2017). Sinking was not included in our model because Greifensee is characterized by a relatively shallow average depth (18 meters), but this loss process can contribute to varying the size composition of phytoplankton communities (Zohary et al. 2017). In summary, the physical and biological properties of a lake are important for understanding temperature-induced changes in the community composition of phytoplankton. Our findings here emphasize the role played by intensified grazing on the community compositions of phytoplankton in a warmer ~~lake~~ Greifensee.

Thermal traits affect future nutrient concentrations

The responses of phytoplankton to increasing temperatures are reflected in traits like optimal temperature for growth and thermal tolerance. Our sensitivity analyses suggest that, the future projections of nutrient dynamics in the lake depend highly on the optimal temperature for growth and thermal tolerance of phytoplankton, which control their growth and uptake

efficiencies under different temperature conditions. Previous modelling studies also showed that the thermal traits of phytoplankton regulate ~~their~~ community responses. ~~A global circulation~~ An ecosystem model resolving biogeochemical processes, for example, showed that alterations ~~in~~ of the Q_{10} factor ~~for phytoplankton growth~~ can markedly change the community ~~compositions in a future ocean~~ composition of phytoplankton under the most severe climate change scenario (Anderson et al. 2024). An NPZ model driven by seasonal variations of temperature ~~investigated the importance of species-specific thermal traits and~~ showed that tackling species differences, especially in terms of thermal optima, is ~~crucial~~ important for estimating the thermal sensitivities of phytoplankton (Chen 2022). ~~Our study does not resolve the thermal sensitivity of different phytoplankton species. In nature, thermal responses of phytoplankton vary among species and this variability can regulate succession patterns and community compositions (Moe et al. 2022). The optimal temperature~~ In addition, optimal temperatures for phytoplankton growth may co-vary with nutrient and light availabilities (Edwards et al. 2016; Bestion et al. 2018) in ways that are not captured by our model. We suggest that measuring size-specific thermal trait values would be a crucial asset for modelling studies such as this one. ~~H~~ These data would facilitate the formalization of a robust functional relationship between thermal traits and plankton sizes, a fundamental assumption for investigating the effects of temperature on plankton communities and the corresponding ecosystem feedback under climate change scenarios.

Limitations and future directions

Our study investigates the impact of increased ~~LWST~~ water temperature in a specific lake system, relying on specific, data-~~driven~~ derived assumptions regarding temperature-dependent phytoplankton growth and zooplankton grazing. ~~Alternative mechanisms of temperature-dependent growth and grazing were examined in other modelling studies (e.g. Uszko et al.~~

~~2017). Other factors not considered here could influence the planktonic ecosystem under global warming. Our results are based on a relatively simplified food web model. For example, our simplistic representation of higher trophic levels, such as those of carnivores, which we formalized as a higher-order mortality term for herbivores, might confound some predatory or cascading effects on the plankton community. Fish predation can impact on zooplankton abundances and their size compositions (Jeppesen et al. 2000). Plankton phenology may also be influenced by the food web structure (Matsuzaki et al. 2021). The minimal size composition (two size classes) we considered for the zooplankton community could underrepresent the diversity of the size-feeding milieu typically experienced by phytoplankton communities (Fuchs and Franks 2010). The underrepresentation of the zooplankton community could lead to an overestimation of the phytoplankton biomass, as shown in one of our previous modelling studies (To et al. 2024). Despite the unusual high resolution of the planktonic data (in terms of temporal intervals and sizes classes) available for Greifensee, a three-year dataset might not be sufficient to capture the range of thermal responses that the plankton community may exhibit in a lake. Species- or size-specific thermal characteristics may explain the slight mismatches between our standard model results and the plankton biomass observed at Greifensee. Although the general trends shown by the data are sufficiently well captured, we believe that resolving the thermal characteristics of different plankton species with appropriate data can enhance the usefulness and robustness of modelling studies like this and can optimize the forecasting for planktonic communities.~~

Our results are based on a relatively idealized lake physics, which does not specify vertical stratification and lateral transport processes. We consider that this choice is appropriate for Greifensee, a relatively shallow lake that is typically fully mixed in winter and stratified into two distinct and homogeneously mixed layers in summer (Mieleitner et al. 2008; Huntscha et al. 2018). Many studies (e.g., Livingstone 2003; Rempfer et al. 2010; Butcher et al. 2015; Woolway et al. 2021) also show that, in a warming world, lakes will

experience weaker mixing, with higher temperatures potentially structuring the water column into two stable layers, thus, making our simplified physical scheme an appropriate first-order approximation approach. Also, our results are based on a specific set of model parameters that are informed by lake data. This reduces the level of uncertainty in these parameters, especially those that regulate phytoplankton nutrient uptake and zooplankton grazing (Fig. S3), which are the major determinants of our results. Additionally, previous data suggested temperature dependences that were not considered in this study like phytoplankton respiration (Robarts and Zohary 1987) and nutrient remineralization rates (Boscolo-Galazzo et al. 2018). The absence of size-specific temperature dependencies in our model prevents niche differentiation with respect to thermal traits (Chen 2022). This limits the model ability to maintain all 10 phytoplankton size classes over time due to competitive exclusion (Hardin 1960), an aspect common to plankton models (e.g., To et al. 2025) but that does not have an equivalent in nature (the “paradox of the plankton”, Hutchinson 1961). Thus, some of the size classes simulated by our model disappear during the course of the simulations. This, however, does not preclude the model from capturing, both in magnitude and dynamics, nutrient and plankton concentrations, as well as mean phytoplankton biovolume.

~~Responses of plankton to changes in lake temperature could be improved by obtaining, through new observational campaigns and laboratory experiments, specific size-dependencies on the thermal traits.~~ Assessing how these limitations impact the community compositions of phytoplankton would be an interesting avenue for future research. Data-driven size-scaling relationships about optimal temperature for growth or thermal tolerance, which we consider crucial to phytoplankton physiology, remain scant ~~and relatively undefined~~. Resolving these scaling relationships can optimize the mechanistic investigations of size-based models under future scenarios (Chen 2022; Litchman 2023), particularly those in relation to the synergistic effects of temperature, nutrients, and zooplankton grazing on phytoplankton communities

~~(Pomati et al. 2020; Dory et al. 2024). The synergistic effects of temperature, nutrients, and zooplankton grazing on phytoplankton communities are still unclear. Major factors regulating the complicated interactions between temperature and nutrients on phytoplankton communities are still not fully understood, despite being critical for lake management under global warming (Moss 2011; Paerl et al. 2016). Future modelling studies could try to elucidate such relationships via hypothesis testing. Laboratory experiments and observations on warming and eutrophication could prove particularly useful in defining realistic hypotheses. The saturating relationship between thermal tolerance and nutrient conditions observed in incubation experiments suggests that this could be a fruitful research direction (Bestion et al. 2018)~~
examine the impacts of temperature on planktonic interactions more realistically by considering size-specific thermal responses (e.g., Chen 2022; Karakuş et al. 2022) or other size-related characteristics, like a more diverse zooplankton community in relation to body size. One could also consider additional temperature dependences on, for example, remineralization processes (Boscolo-Galazzo et al. 2018). Assuming that metabolic and biochemical rates would be enhanced by increasing temperatures, these dependences would produce more accurate dynamics of inorganic nutrients and detritus. Spatially explicit hydrodynamic modelling studies could investigate the extent to which our first-order slab physics approximation is appropriate to capture the future ecological regimes of lakes like Greifensee (e.g., Huntscha et al. 2018). Studying inter-annual variabilities would be another obvious next step when looking at future projections of lake ecosystem.

Conclusions

By combining a broad range of field observations and ~~data-driven~~informed trait-based modelling, our study provides insights into the future dynamics of nutrient, plankton biomasses and community compositions of phytoplankton in a temperate lake under global warming

scenarios. Our model predicts three potential consequences in a warmer lake, namely, increased epilimnetic nutrient retention, advancement of phytoplankton blooms, and increased mean phytoplankton biovolume, the latter being brought about by a replacement of small phytoplankton cells with large ones. Acknowledging a number of simplifications and limitations, this study represents a first-order approximation of the ecological responses that Greifensee may exhibit in a warmer world. Most of the model parameters and all environmental forcing are site-specific, thus the validity of our results is limited to Greifensee or to lake systems with characteristics similar to Greifensee. Future ~~testing~~ investigations of these phenomena, based on alternative sets of assumptions or expanded datasets, would be a ~~valuable~~ valuable effort to verify the likelihood of their occurrence ~~also in lakes with physical and biological characteristics different than those of Greifensee~~. Our approach emphasizes the multifaceted responses of planktonic communities to global warming, highlighting both challenges and opportunities involved in understanding and managing a lake ecosystem in a changing world. We thus argue that long-term monitoring programs, like those undertaken at Greifensee, in combination with modelling studies based on temperature dependences on the physiology and ecology of plankton can improve our predictions and enhance our capacity to design effective lake management strategies. Finally, investigating how the changes predicted by our model could impact lake ecosystem functions would help the scientific community to assess the significance that these studies have for lake management and conservation strategies in a warming world.

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1171 **Data Availability Statement**

1172 The lake data and model outputs used for this study are available on Zenodo (doi:
1173 10.5281/zenodo.11390345). The code to reproduce the analyses of this study is available on
1174 Github (<https://github.com/systemsecologygroup/TempSizeMod>).

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1176 **Conflict of Interest**

1177 None declared.

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